DevOrSysAdminScripts

Laurent Frédéric Bernard François Lyaudet
https://lyaudet.eu/laurent/
Laurent.Lyaudet@gmail.com

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Current number of commits: 1022
Current git SHA1: 3c1fbe3e69f5d123919339c7341622e64e77e144
Code lines: 14995 total lines, 13712 not empty lines, 1283 empty lines.
Buildtarget: 0 A4

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1 Files trees

```
.
├── archive_libre_office_documents.exec.php
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_dependencies_notes.exec.php
│   ├── build_dependencies_notes.libr.php
│   ├── build_file_from_printable_file__standard.exec.sh
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spelling_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── phpmd_baseline.xslt
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── dependencies_data.json
│   ├── dependencies_notes.txt
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_names_listing.txt
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpcs_ruleset.xml
│   ├── phpmd_baseline.xml
│   ├── phpstan_baseline.neon
│   ├── phpstan_baseline.neon.tpl
│   ├── phpstan_config.neon
│   ├── phpstan_config.neon.tpl
│   ├── pylintrc.toml
```

```
│   ├── repository_data.txt
│   ├── repository_data.txt.tpl
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts-A4.pdf
├── DevOrSysAdminScripts-Book.pdf
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── DevOrSysAdminScripts-USLetter.pdf
├── files.libr.php
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── .gitignore
├── IMAP_transfert.exec.py
├── LiensUtiles.txt
├── miscellaneous
│   ├──
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── funny.sh
│   ├── grep_benchmark.sh
│   ├── spampty_data.txt
│   ├── string_escaping.html
│   ├── string_escaping.xhtml
│   └──
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── README.html
├── README.md
├── README.md.tpl
├── string_escaping.libr.php
├── unicode_exceptions.libr.php
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 90 files

[ 12K 2026-11-11 15:51]  ./
├── [1.8K 2026-08-14 17:47]  archive_libre_office_documents.exec.php*
├── [4.0K 2026-11-11 15:51]  build_and_checks_dependencies/
│   ├── [4.4K 2026-05-12 19:33]  build_and_checks.exec.sh.tpl*
│   ├── [1.5K 2026-08-14 17:46]  build_dependencies_notes.exec.php*
│   ├── [ 10K 2026-08-14 15:46]  build_dependencies_notes.libr.php
│   ├── [2.6K 2026-08-13 12:28]
│   │ build_file_from_printable_file__standard.exec.sh*
│   ├── [4.2K 2026-08-12 19:31]  build_md_from_printable_md.exec.sh*
│   ├── [3.1K 2026-05-12 19:34]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.3K 2026-05-12 19:37]  check_shell_scripts_indentation.libr.sh
│   ├── [4.0K 2026-08-12 19:14]  check_URLs.libr.sh
│   ├── [ 43K 2026-11-11 15:41]  common_build_and_checks.exec.sh*
│   ├── [ 38K 2026-09-18 22:14]  common_build_and_checks.exec.sh.tpl
│   ├── [ 19K 2026-05-12 19:37]  common_options.libr.sh
│   ├── [6.2K 2026-05-12 19:37]  comparisons.libr.sh
│   ├── [ 23K 2026-11-11 15:40]  create_PDF.exec.sh*
```

```
│   ├── [3.7K 2026-05-12 19:32]  generate_from_template.libr.sh
│   ├── [2.1K 2026-08-12 19:16]  get_common_text_glob_patterns.libr.sh
│   ├── [3.3K 2026-05-12 19:37]  grammar_and_spelling_check.libr.sh
│   ├── [4.0K 2026-08-13 13:00]  licenses_templates/
│   │   ├── [ 16K 2026-08-13 13:00]  build_licenses_templates.exec.sh*
│   │   ├── [ 720 2026-05-12 19:16]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 699 2026-05-12 19:17]  license_file_header_GPLv3+.tpl
│   │   └── [ 720 2026-05-12 19:18]  license_file_header_LGPLv3+.tpl
│   ├── [6.0K 2026-05-13 17:48]  lines_counts.libr.sh
│   ├── [ 15K 2026-05-13 15:25]  lines_filters.libr.sh
│   ├── [3.0K 2026-05-12 19:37]  lines_maps.libr.sh
│   ├── [3.3K 2026-05-12 19:37]  overwrite_if_not_equal.libr.sh
│   ├── [1.6K 2026-08-14 14:40]  phpmd_baseline.xslt
│   ├── [2.4K 2026-05-12 19:37]  python_black_complement.libr.sh
│   ├── [3.1K 2026-05-13 15:29]  python_isort_complement.libr.sh
│   ├── [2.9K 2026-05-13 15:08]  repository_data.libr.sh
│   ├── [3.9K 2026-05-12 19:37]  shell_checks_complement.libr.sh
│   ├── [2.8K 2026-08-14 17:45]  split_score.exec.php*
│   ├── [5.9K 2026-08-14 14:56]  split_score.libr.php
│   ├── [ 47K 2026-05-13 16:32]  strings_functions.libr.sh
│   ├── [4.8K 2026-08-12 19:25]  too_long_code_lines.libr.sh
│   ├── [4.5K 2026-08-14 14:33]  update_common_build_and_checks.exec.sh*
│   └── [6.0K 2026-05-12 19:37]
    update_or_check_files_names_listing.exec.sh*
├── [4.6K 2026-11-11 15:41]  build_and_checks.exec.sh*
├── [4.0K 2026-11-11 15:51]  build_and_checks_variables/
│   ├── [  15 2026-01-05 20:00]  bandit_baseline.json
│   ├── [  25 2026-01-05 18:49]  bandit.ini
│   ├── [  30 2026-02-28 20:30]  black.toml
│   ├── [5.1K 2026-04-17 16:50]  dependencies_data.json
│   ├── [4.3K 2026-04-17 16:57]  dependencies_notes.txt
│   ├── [5.7K 2026-05-12 19:31]  DevOrSysAdminScripts.html.tpl
│   ├── [ 38K 2026-11-11 15:41]  DevOrSysAdminScripts.tex
│   ├── [4.8K 2026-11-11 15:50]  DevOrSysAdminScripts.tex.tpl
│   ├── [5.3K 2026-08-14 17:42]  files_names_listing.txt
│   ├── [  23 2026-03-24 19:33]  .gitignore
│   ├── [1.4K 2026-04-17 16:49]  licenses_notes.txt
│   ├── [ 16K 2026-08-14 17:54]  phpcs_ruleset.xml
│   ├── [5.2K 2026-08-14 14:53]  phpmd_baseline.xml
│   ├── [1.6K 2026-08-14 18:06]  phpstan_baseline.neon
│   ├── [1.7K 2026-08-14 18:06]  phpstan_baseline.neon.tpl
│   ├── [ 236 2026-08-13 12:29]  phpstan_config.neon
│   ├── [ 244 2026-08-12 19:23]  phpstan_config.neon.tpl
│   ├── [ 678 2026-02-24 22:38]  pylintrc.toml
│   ├── [5.8K 2026-11-11 14:52]  repository_data.txt
│   ├── [ 963 2026-11-11 15:43]  repository_data.txt.tpl
│   ├── [1.5K 2026-08-13 16:02]  ruff.toml
│   ├── [ 166 2026-03-24 19:33]  shellcheck.ini
│   └── [3.6K 2026-05-13 17:18]  shellcheck_notes.txt
├── [ 34K 2024-05-06 00:49]  COPYING
├── [7.5K 2025-12-27 21:48]  COPYING.LESSER
├── [468K 2026-11-11 15:42]  DevOrSysAdminScripts-A4.pdf
├── [468K 2026-11-11 15:42]  DevOrSysAdminScripts-Book.pdf
├── [575K 2026-11-11 15:41]  DevOrSysAdminScripts.html
├── [466K 2026-09-18 22:18]  DevOrSysAdminScripts.pdf
├── [468K 2026-11-11 15:42]  DevOrSysAdminScripts-USLetter.pdf
├── [ 21K 2026-08-14 18:01]  files.libr.php
├── [4.0K 2026-05-13 17:28]  git_helpers/
│   ├── [3.8K 2026-05-13 17:22]  git_checks.libr.sh
│   ├── [2.1K 2026-05-12 19:31]  post-commit*
│   ├── [ 15K 2026-05-13 17:28]  post_commit_renaming_comment.libr.sh
```

```
│   └── [1.8K 2026-05-12 19:31]  pre-commit*
├── [   8 2026-03-26 10:37]  .gitignore
├── [8.4K 2026-05-12 19:31]  IMAP_transfert.exec.py
├── [1.1K 2026-03-31 19:59]  LiensUtiles.txt
├── [ 24K 2026-08-13 18:22]  miscellaneous/
│   ├── [   0 2026-03-04 09:13]
│   │  a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │  an_see_that_it_works.txt
│   ├── [1.3K 2026-05-12 19:37]  funny.sh
│   ├── [2.4K 2026-08-13 18:22]  grep_benchmark.sh
│   ├── [  94 2026-03-03 19:56]  spampty_data.txt
│   ├── [ 461 2026-05-13 18:15]  string_escaping.html
│   ├── [ 653 2026-05-12 14:38]  string_escaping.xhtml
│   └── [   0 2026-03-04 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.1K 2026-05-12 12:17]  README.html
├── [4.3K 2026-03-22 21:29]  README.md
├── [4.3K 2026-03-22 21:29]  README.md.tpl
├── [7.5K 2026-08-14 17:03]  string_escaping.libr.php
├── [2.5K 2026-08-13 18:54]  unicode_exceptions.libr.php
├── [ 31K 2026-08-14 17:04]  unicode.libr.php
├── [6.4K 2026-05-13 18:46]  upgrade_build_and_checks.exec.sh*
└── [2.5K 2026-05-12 19:37]  wget_sha512.libr.sh

6 directories, 90 files
```

2 Notes on licenses and dependencies

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
  bandit;
- The following dependencies are BSD-3-Clause licensed:
  phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,

- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/,
https://tiswww.case.edu/php/chet/bash/bashtop.html,
https://savannah.gnu.org/projects/bash/.

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/,
https://savannah.gnu.org/projects/coreutils/.

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/,
https://github.com/uutils/coreutils.

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/,
https://savannah.gnu.org/projects/diffutils/.

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/,
https://savannah.gnu.org/projects/findutils/.

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/,
https://savannah.gnu.org/projects/grep/.

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/,
https://savannah.gnu.org/projects/sed/.

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/,
https://github.com/koalaman/shellcheck.

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/,
https://savannah.gnu.org/projects/wget/.

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/,
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/,
https://github.com/util-linux/util-linux official?.

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.
mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.
phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/,
https://docs.phpdoc.org/,
https://github.com/phpdocumentor/phpdocumentor.
pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).
ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.
xmllint (libxml2) can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home,
https://gitlab.gnome.org/GNOME/libxml2.
bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.
phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer,
https://packagist.org/packages/squizlabs/php_codesniffer.
phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.
php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.
tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.
pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.
python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.
If you see something that I missed regarding my dependencies, please tell me/email me :).

3 Listing of files

3.1 archive_libre_office_documents.exec.php

```
1 #!/usr/bin/php
2 <?php
3
4 /**
5 This file is part of DevOrSysAdminScripts.
```

```
6
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12 or (at your option) any later version.
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23 along with DevOrSysAdminScripts.
24 If not, see <https://www.gnu.org/licenses/>.
25
26 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
27
28 @category Executable
29 @package DevOrSysAdminScripts
30 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33 */
34
35 declare(strict_types=1);
36 declare(encoding='UTF-8');
37
38 require_once 'files.libr.php';
39
40 if(PHP_SAPI !== 'cli'){
41   fwrite(
42     STDERR,
43     "archive_libre_office_documents.exec.php must be run in a shell.\n",
44   );
45   die(1);
46 }
47
48 if(count($argv) < 3 || count($argv) > 4){
49   fwrite(
50     STDERR,
51     "split_score.exec.php must be given 2 or 3 arguments.\n",
52   );
53   die(1);
54 }
55
56 DOSASfiles\archive_directory_into_another(
57   '/home/'.$argv[1].'/.config/libreoffice/4/user/backup/',
58   '/home/'.$argv[2].'/Documents/LibreOfficeArchives/',
59   true,
60   true,
61   // Following argument is if you want same relative path and same content
62   // at the same time.
63   // This is in case you want to track when the file goes back to some
64   // old state.
65   isset($argv[3]) ? (bool) (int) $argv[3] : false,
66   true,
```

```
    true,
    '\\.~\\d+~',
);
?>
```

3.2 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
```

```
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir --parents "${LFBFL_subdir}/licenses_templates"
  local LFBFL_variables_directory
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  readonly LFBFL_variables_directory
  mkdir --parents -- "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  local LFBFL_script_download_URL
  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
  LFBFL_script_download_URL+="${LFBFL_common_file_name}"
  readonly LFBFL_script_download_URL
  declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
```

```
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.3 build_and_checks_dependencies/build_dependencies_notes.exec.php

```
#!/usr/bin/php
<?php

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@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once 'string_escaping.libr.php';
require_once 'build_dependencies_notes.libr.php';

if(PHP_SAPI !== 'cli'){
  fwrite(
    STDERR,
    "build_dependencies_notes.exec.php must be run in a shell.\n",
  );
  die(1);
}

```

```
if(count($argv) !== 2){
  fwrite(
    STDERR,
    "split_score.exec.php must be given exactly 1 argument.\n",
  );
  die(1);
}

\DOSAS_dependencies_notes\build_dependencies_notes($argv[1]);
?>
```

3.4 build_and_checks_dependencies/build_dependencies_notes.libr.php

```
<?php

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2026-03-26T11:59+01:00: This file was renamed from
"build_dependencies_notes.exec.php"
to
"build_dependencies_notes.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

// require_once 'string_escaping.libr.php';
namespace DOSAS_dependencies_notes;

use Exception;
use string_escaping;

/**
This function looks at the dependencies data found in a work directory,
```

```
49  and generate three files: a txt file, a TeX file (.sub),
50  and an HTML file (.sub) with the data in a nicer format.
51
52  @param string $s_work_directory The directory where the work must be done.
53
54  @throws \Exception When the content of one of the files needed as input
55                     cannot be loaded into a string.
56                     Or that some part of the structure from the JSON
57                     isn't valid.
58
59  @return void
60  */
61  function build_dependencies_notes(string $s_work_directory) : void {
62    $s_subdir1 = $s_work_directory.'/build_and_checks_variables';
63    $s_subdir2 = $s_subdir1.'/temp';
64    $s_data_file_path = $s_subdir1.'/dependencies_data.json';
65    $s_data = file_get_contents($s_data_file_path);
66    if($s_data === false){
67      throw new Exception(
68        'Unexpected failure when fetching content of dependencies_data.json'
69      );
70    }
71    $arr_data = json_decode($s_data, true);
72    if(!is_array($arr_data)){
73      throw new Exception(
74        "dependencies_data.json doesn't contain an array."
75      );
76    }
77    $s_data_file_path2 = $s_subdir1.'/repository_data.txt';
78    $s_data2 = file_get_contents($s_data_file_path2);
79    if($s_data2 === false){
80      throw new Exception(
81        'Unexpected failure when fetching content of repository_data.txt'
82      );
83    }
84    $i_max_line_length = 70;
85    foreach(explode("\n", $s_data2) as $s_line){
86      if(str_starts_with($s_line, 'max_line_length=')){
87        $i_max_line_length = (
88          (int) substr($s_line, strlen('max_line_length='))
89        );
90        break;
91      }
92    }
93
94    $s_result_content1 = '';
95    $s_result_content2 = '';
96    $s_result_content3 = '';
97
98    foreach($arr_data as $arr_dependency){
99      if(!is_array($arr_dependency)){
100       throw new Exception("Dependency isn't an array.");
101     }
102     if(!isset($arr_dependency['URLs'])){
103       throw new Exception("Dependency doesn't have an URLs array.");
104     }
105     if(!is_array($arr_dependency['URLs'])){
106       throw new Exception("URLs isn't an array.");
107     }
108     if(!isset($arr_dependency['dependency_name'])){
109       throw new Exception("Dependency doesn't have a name.");
```

```
110       }
111       if(!is_string($arr_dependency['dependency_name'])){
112         throw new Exception("Dependency name isn't a string.");
113       }
114
115       $s_URL_word = 'URL';
116       if(count($arr_dependency['URLs']) > 1){
117         $s_URL_word .= 's';
118       }
119
120       $s_result_content3 .= "<p>\n";
121
122       $s_dependency1 = $arr_dependency['dependency_name'];
123       $s_dependency2 = string_escaping\TeX\escape_text($s_dependency1);
124       $s_dependency3 = string_escaping\HTML\escape_text($s_dependency1);
125
126       $s_sentence_end = ' can be found at the following '.$s_URL_word.':';
127       $s_result_content1 .= $s_dependency1.$s_sentence_end."\n";
128       $s_result_content2 .= $s_dependency2.$s_sentence_end."\\newline\n";
129       $s_result_content3 .= $s_dependency3.$s_sentence_end."<br>\n";
130
131       for(
132         $i = 0, $i_max = count($arr_dependency['URLs']) - 1;
133         $i <= $i_max;
134         ++$i
135       ){
136         $mixed_URL = $arr_dependency['URLs'][$i];
137         $s_comment = '';
138         $s_URL = '';
139         if(is_array($mixed_URL)){
140           if(!isset($mixed_URL['URL'])){
141             throw new Exception("Dependency URL doesn't have an URL.");
142           }
143           if(!isset($mixed_URL['comment'])){
144             throw new Exception("Dependency URL doesn't have a comment.");
145           }
146           $s_URL = $mixed_URL['URL'];
147           $s_comment = $mixed_URL['comment'];
148         }
149         else{
150           $s_URL = $mixed_URL;
151         }
152         if(!is_string($s_URL)){
153           throw new Exception("URL isn't a string.");
154         }
155         if(!is_string($s_comment)){
156           throw new Exception("Comment isn't a string.");
157         }
158
159         $i_URL_length = strlen($s_URL);
160         $i_comment_length = strlen($s_comment);
161
162         $s_result_content1 .= $s_URL;
163         $i_previous_line_length = $i_URL_length;
164         if($i_comment_length > 0){
165           if(
166             $i_previous_line_length + $i_comment_length + 1
167             > $i_max_line_length
168           ){
169             $s_result_content1 .= "\n";
170           }
```

```
171           else{
172             $s_result_content1 .= ' ';
173           }
174           $s_result_content1 .= $s_comment;
175         }
176         $s_result_content1 .= "\n";
177
178         if($i_URL_length > 60){
179           $s_result_content2 .= (
180             "{\\small\n"
181             .'  \url{'.$s_URL."}\n"
182             .'}'
183           );
184           $i_previous_line_length = 1;
185         }
186         else{
187           $s_result_content2 .= '\url{'.$s_URL.'}';
188           $i_previous_line_length = $i_URL_length + 6;
189         }
190         if($i === $i_max){
191           ++$i_previous_line_length;
192         }
193         else{
194           $i_previous_line_length += 9;
195         }
196         $s_comment2 = string_escaping\TeX\escape_text($s_comment);
197         $i_comment_length2 = strlen($s_comment2);
198         if($i_comment_length2 > 0){
199           if(
200             $i_previous_line_length + $i_comment_length2 + 1
201             > $i_max_line_length
202           ){
203             $s_result_content2 .= "\n";
204           }
205           else{
206             $s_result_content2 .= ' ';
207           }
208           $s_result_content2 .= $s_comment2;
209         }
210         if($i === $i_max){
211           $s_result_content2 .= ".\n";
212         }
213         else{
214           $s_result_content2 .= ",\\newline\n";
215         }
216
217         $s_result_content3 .= (
218           "<a\n"
219           .'href="'.string_escaping\HTML\escape_attribute($s_URL)."\"\n"
220           ."><span style=\"font-family:monospace\"\n"
221           .'>'.string_escaping\HTML\escape_text($s_URL)."</span\n"
222           .'></a>'
223         );
224         if($s_comment !== ''){
225           $s_result_content3 .= (
226             ' '.string_escaping\HTML\escape_text($s_comment)
227           );
228         }
229         if($i === $i_max){
230           $s_result_content3 .= ".\n";
231         }
```

```
232         else{
233           $s_result_content3 .= ",<br>\n";
234         }
235       }//end for urls
236
237       $s_result_content1 .= "\n";
238       $s_result_content2 .= "\n";
239       $s_result_content3 .= "</p>\n\n";
240     }//end foreach($arr_data as $arr_dependency)
241
242     // Add a "footer".
243     $s_closing_sentence = (
244       "If you see something that I missed regarding my dependencies,\n"
245       ."please tell me/email me :).\n"
246     );
247     $s_result_content1 .= $s_closing_sentence;
248     $s_result_content2 .= $s_closing_sentence;
249     $s_result_content3 .= "<p>\n".$s_closing_sentence."</p>\n";
250
251     // We overwrite results if needed.
252     $s_result_path1 = $s_subdir1.'/dependencies_notes.txt';
253     $s_existing_result_content1 = '';
254     if(file_exists($s_result_path1)){
255       $s_existing_result_content1 = file_get_contents($s_result_path1);
256     }
257     if($s_result_content1 !== $s_existing_result_content1){
258       file_put_contents($s_result_path1, $s_result_content1);
259     }
260     /*
261     Example of generated content of dependencies_notes.txt:
262     ------------------------------------------------------------------------
263     more of util-linux can be found at the following URLs:
264     https://www.kernel.org/pub/linux/utils/util-linux/
265     https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
266     https://github.com/util-linux/util-linux official?
267     ------------------------------------------------------------------------
268     */
269     $s_result_path2 = $s_subdir2.'/dependencies_notes.tex.sub';
270     $s_existing_result_content2 = '';
271     if(file_exists($s_result_path2)){
272       $s_existing_result_content2 = file_get_contents($s_result_path2);
273     }
274     if($s_result_content2 !== $s_existing_result_content2){
275       file_put_contents($s_result_path2, $s_result_content2);
276     }
277     /*
278     Example of generated content of dependencies_notes.tex.sub:
279     ------------------------------------------------------------------------
280     more of util-linux can be found at the following URLs:\newline
281     \url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
282     {\small
283       \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
284     },\newline
285     \url{https://github.com/util-linux/util-linux} official?.
286     ------------------------------------------------------------------------
287     */
288     $s_result_path3 = $s_subdir2.'/dependencies_notes.html.sub';
289     $s_existing_result_content3 = '';
290     if(file_exists($s_result_path3)){
291       $s_existing_result_content3 = file_get_contents($s_result_path3);
292     }
```

```
  if($s_result_content3 !== $s_existing_result_content3){
    file_put_contents($s_result_path3, $s_result_content3);
  }
  /*
  Example of generated content of dependencies_notes.html.sub:
  <p>
  more of util-linux can be found at the following URLs:<br>
  <a
  href="https://www.kernel.org/pub/linux/utils/util-linux/"
  ><span style="font-family:monospace"
  >https://www.kernel.org/pub/linux/utils/util-linux/</span
  ></a>,<br>
  <a
  href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
  ><span style="font-family:monospace"
  >https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
  ></a>,<br>
  <a
  href="https://github.com/util-linux/util-linux"
  ><span style="font-family:monospace"
  >https://github.com/util-linux/util-linux</span
  ></a> official?.
  </p>
  */
}//end build_dependencies_notes()
?>
```

3.5 build_and_checks_dependencies/build_file_from_printable_file__standard.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_file_from_printable_file__standard(){
  # $1=base_name
```

```
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --remove-leading-whitespaces-after-escaped-line-return
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -i LFBFL_remove_leading_whitespaces_after_escaped_line_return=0
  get_some_flag LFBFL_remove_leading_whitespaces_after_escaped_line_return\
    --remove-leading-whitespaces-after-escaped-line-return 1 "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name
  get_some_option LFBFL_base_name --base-name '' '' 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    if [[ "${LFBFL_base_name}" == "" ]]; then
      return 2
    fi
    printf "Searching file to build from printable file: %s.\n"\
      "${LFBFL_base_name}"
  fi

  local LFBFL_s_expression='s/\\\n//Mg'
  if [[
    LFBFL_remove_leading_whitespaces_after_escaped_line_return -eq 1
  ]]; then
    LFBFL_s_expression='s/\\\n *//Mg'
  fi
  if [[ -f "${LFBFL_base_name}.tpl" ]]; then
    sed --regexp-extended --null-data --expression="${LFBFL_s_expression}"\
      -- "${LFBFL_base_name}.tpl"\
      > "${LFBFL_base_name}.temp"
    overwrite_if_not_equal "${LFBFL_base_name}" "${LFBFL_base_name}.temp"
  else
    printf "No file %s.tpl\n" "${LFBFL_base_name}"
    return 3
  fi

  if [[ ! -f "${LFBFL_base_name}" ]]; then
    printf "No file %s\n" "${LFBFL_base_name}"
    return 4
  fi
}

build_file_from_printable_file__standard "$@"
```

3.6 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
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```

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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name
  get_some_option LFBFL_base_name --base-name README README 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 && "${LFBFL_base_name}" != "README" ]]; then
    printf "Searching md file: %s.\n" "${LFBFL_base_name}"
  fi

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence, it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed --regexp-extended --null-data\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      --expression="${LFBFL_sed_expression}"\
      -- "${LFBFL_base_name}.md.tpl"\
```

```
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 2
  fi

  pandoc --from=markdown --to=html --standalone\
    --shift-heading-level-by=-1 --output="${LFBFL_base_name}.html.temp"\
    -- "${LFBFL_base_name}.md"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed --in-place --regexp-extended\
    --expression='s~(  <meta .*) />~\1>~g'\
    --expression='s~<html .*>~<html lang="en">~'\
    --expression='/    code\{white-space: pre-wrap;\}/d'\
    -- "${LFBFL_base_name}.html.temp"
  sed --in-place --regexp-extended --null-data\
    --expression='s~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
    --expression="${LFBFL_prewrap}"\
    -- "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

3.7 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # $1=file_path
  #
  # declare -r LFBFL_file_path=$(basename -- "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
  # ---------------------------------------------------------------
  # is of course slower than
  # ---------------------------------------------------------------
  # diff <(head --lines=1 -- "$1") <(printf '#!/usr/bin/env bash')
  # ---------------------------------------------------------------
  # is of course slower than
  # ---------------------------------------------------------------
  declare -r LFBFL_head=$(head --lines=1 -- "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
```

```
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_beginning "${LFBFL_file_path}"
  done
}
```

3.8 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  # $1=file_path
  declare -r LFBFL_NULL_PREVIOUS_LINE="--"
  declare -i LFBFL_i_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_i_line_length
  declare -i LFBFL_i_line_end_length
  declare -i LFBFL_i_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_i_offset
```

```
44    local LFBFL_substring1
45    local LFBFL_substring2
46    declare -r LFBFL_pattern='^ *-e '
47
48    # This grep would have been enough without some sed commands.
49    declare -r LFBFL_s_lines=$(
50      grep --before-context=1 --extended-regexp --line-number\
51        --regexp='^(  )* ([^ ]|$)' --with-filename -- "$1"
52    )
53    if [[ -z "${LFBFL_s_lines}" ]]; then
54      return
55    fi
56
57    declare -a LFBFL_arr_lines
58    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
59    readonly LFBFL_arr_lines
60
61    for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
62      if [[ "${LFBFL_some_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]]; then
63        # When switching between code fragments, we reinit.
64        # It cannot happen for other grep output lines with line-number.
65        LFBFL_previous_line="${LFBFL_NULL_PREVIOUS_LINE}"
66        continue
67      fi
68      # Note how it's funny that this loop always work because we check
69      # shell scripts beginnings in another script and thus we always
70      # have a line before.
71      if [[ "${LFBFL_previous_line}" == "${LFBFL_NULL_PREVIOUS_LINE}" ]];
72      then
73        LFBFL_previous_line="${LFBFL_some_line}"
74        continue
75      fi
76      # Remove the name of the file and then the line number.
77      LFBFL_line_end=${LFBFL_some_line#*:}
78      LFBFL_line_end=${LFBFL_line_end#*:}
79      # But for the previous line '-' or ':' is used instead of ':'
80      # and filenames with '-' are much more frequent than with ':'...
81      LFBFL_i_line_length=${#LFBFL_some_line}
82      LFBFL_i_line_end_length=${#LFBFL_line_end}
83      LFBFL_i_line_start_length=$((
84        LFBFL_i_line_length - LFBFL_i_line_end_length
85      ))
86      LFBFL_i_offset=$((LFBFL_i_line_start_length - 2))
87      # Checking if increasing number of digits in the line numbers
88      # (99 -> 100).
89      LFBFL_substring1=${LFBFL_previous_line:${LFBFL_i_offset}:1}
90      # Nice puzzle below :) Look for comments above :)
91      case ${LFBFL_substring1} in
92        -) ((LFBFL_i_offset += 1));;
93        :) ((LFBFL_i_offset += 1));;
94        [0-9]) ((LFBFL_i_offset += 2));;
95        *) printf ":-(\n";;
96      esac
97      LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_i_offset}}
98      if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
99        # Line starts with spacing and -e
100       # Keep max spaces
101       LFBFL_spacing=${LFBFL_line_end%%-*}
102       LFBFL_i_offset=${#LFBFL_spacing}
103       LFBFL_substring1=${LFBFL_line_end:${LFBFL_i_offset}:3}
104       LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_i_offset}:3}
```

```
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_i_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done

  if [[ LFBFL_i_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

3.9 build_and_checks_dependencies/check_URLs.libr.sh

```
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS

  declare -Ar LFBFL_substitutions=(
    ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
  )

  local LFBFL_pattern
  local LFBFL_base_file_name
  local LFBFL_substitution
  local LFBFL_substitution2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "check_URLs: Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\0'\
      | xargs -0 grep --with-filename -- 'http:'\
      | grep --invert-match "| xargs -0 grep --with-filename -- 'htt"\
```

```
      | grep --invert-match --perl-regexp\
        "(['\"])http(:[^'\"]*)\\1.*(['\"])https\\2\\3"\
      | grep --invert-match\
        --regexp='http://www.w3.org/1999/xhtml'\
        --regexp="http://www.w3.org/1999/XSL/Transform"\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd'\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd'\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd'\
        --regexp='http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd'
    # Second grep after xargs grep just above will remove false positives
    # from substitutions that fit on one line.

    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_main_html_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]] && printf "check_URLs: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
      [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
        || continue
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "check_URLs: Handling the file: %s.\n" "${LFBFL_file_path}"
      fi
      for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
        LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
        if grep --quiet "${LFBFL_substitution}" -- "${LFBFL_file_path}";
        then
          sed --in-place\
          --expression="s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
            -- "${LFBFL_file_path}"
        fi
      done
    done
  done
}
```

3.10 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
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```

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# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  declare -a LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    "--work-directory=${LFBFL_work_directory}"
  )

  source ./wget_sha512.libr.sh

  local LFBFL_subdir="build_and_checks_dependencies"
  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
  declare -r LFBFL_subdir3="build_and_checks_variables"
  local LFBFL_file_name
  local LFBFL_script_download_URL
  local LFBFL_file_path
  local LFBFL_correct_sha512

  wrapped_wget_sha512(){
    wget_sha512 "${LFBFL_file_path}"\
      "${LFBFL_script_download_URL}"\
      "${LFBFL_correct_sha512}"\
      "${LFBFL_verbose}"\
```

```
        "${LFBFL_check_download}"
    }

    LFBFL_file_name="build_dependencies_notes.exec.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='3d881a757c69fbcc02cbc5a568c508cfa6249c1330ea6'
    LFBFL_correct_sha512+='7eeca6d284950b3ca2ced7b4acce4142cdf90c5aaa88'
    LFBFL_correct_sha512+='5bb3ee77f3a32ae0d4b412173cec0d944278ab0'
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="build_dependencies_notes.libr.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='6203cb1c31b656004ec8467d4f84288f342fcc7f5c236'
    LFBFL_correct_sha512+='270e819b986fe967aa6abe101188a7fc1a55e2f4c9b5'
    LFBFL_correct_sha512+='9a6dde6c9aa125b385d0ce85e02b1de69ce1084'
    wrapped_wget_sha512

    LFBFL_file_name="build_file_from_printable_file__standard.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='9c5b11280019b4bbc679bddb8eb5c66d9ef715d04c8c1'
    LFBFL_correct_sha512+='9e2cbd55e08ad38169b82aa2de5fe59f46053552a358'
    LFBFL_correct_sha512+='d92f3554cfc8ce7d930055b400ec1e6ee5e3069'
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="build_md_from_printable_md.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='a39101cb7c0a7a743386b0c8745071cc15f8fd93be3ac'
    LFBFL_correct_sha512+='94bd5d69860836e4ae94b08a3219c2542f9d6501f0b4'
    LFBFL_correct_sha512+='d4b1bc40700afcb1b555bbbe25de6485f95c361'
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='0b3e861820e09148a141142c8bb5549e7fa29d906dda9'
    LFBFL_correct_sha512+='5e0fe44132610910687e6f767f1050646abe1ea262b8'
    LFBFL_correct_sha512+='37b3a5d5e79497d0cdc42e43d3c74967f266abb'
    wrapped_wget_sha512

    LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='3596fddc4b20ce353facac3e355e38fadb5593caee3c8'
    LFBFL_correct_sha512+='05464aedd513d34860a79b296efa648dae18717ae4c0'
    LFBFL_correct_sha512+='24a6df11eb0549c4797643427c49c4f80947840'
    wrapped_wget_sha512

    LFBFL_file_name="check_URLs.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='d3ddcd56520d1786cd5afb8fd0f7ee278ed7307a6f5d2'
    LFBFL_correct_sha512+='1f948d0f2f6f2ed5210072d923f2036643c7181012fc'
    LFBFL_correct_sha512+='15d7921f5617105012a14b8adc9acc76d2b50cb'
    wrapped_wget_sha512
```

```
140
141     LFBFL_file_name="common_options.libr.sh"
142     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
143     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
144     LFBFL_correct_sha512='3cfaf6cea99f67ade931cf21e9f49658c421e182dc4ac'
145     LFBFL_correct_sha512+='211be526f9d99b2e57a09844991cf0d2e6d5e49c19e2'
146     LFBFL_correct_sha512+='a63bf9e8f1e3fe999542e9a4e45e5196b375fa9'
147     wrapped_wget_sha512
148
149     LFBFL_file_name="comparisons.libr.sh"
150     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
151     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
152     LFBFL_correct_sha512='f146beca0d7b4046c140f3b2292cec4c31515cf8c7dbe'
153     LFBFL_correct_sha512+='7e9b07fc6dae32513aff6692c6286fa31055caff1a20'
154     LFBFL_correct_sha512+='9a6a7384568fea0b50b3cbb2608a3ed9e6e5125'
155     wrapped_wget_sha512
156
157     LFBFL_file_name="create_PDF.exec.sh"
158     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
159     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
160     LFBFL_correct_sha512='588f29f1de07d9980a14259a710cea5f5093102e0ff8c'
161     LFBFL_correct_sha512+='366e71c45bc383f5040954a858ec0c84ba712c51ee01'
162     LFBFL_correct_sha512+='cb2c5ced6354103dfc9ea7e735c70b205cafbe6'
163     wrapped_wget_sha512
164     chmod +x "./${LFBFL_file_path}"
165
166     LFBFL_file_name="generate_from_template.libr.sh"
167     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
168     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
169     LFBFL_correct_sha512='5f322f27bd20d7042c729f4cb5d8f14f587b5b415c77e'
170     LFBFL_correct_sha512+='4a224902d8d4cdeb1483639ae85d81873c9f20c175bb'
171     LFBFL_correct_sha512+='4de76a6eaa3dbcbf8a300368e65bbd5303afd3d'
172     wrapped_wget_sha512
173
174     LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
175     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
176     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
177     LFBFL_correct_sha512='0945c9896ce9d26637a8ba5c08a1fdad102b725de26ce'
178     LFBFL_correct_sha512+='088e519a3696925465007ad0b460ae7433174d8af848'
179     LFBFL_correct_sha512+='b77dfdff0da78534453e90303d3bd458bf378e9'
180     wrapped_wget_sha512
181
182     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
183     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
184     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
185     LFBFL_correct_sha512='88cc4f031c05d576a74ea7b14e79077f06c53b3b8a294'
186     LFBFL_correct_sha512+='e9b8ddcfe884db5e2cb5cebde09a8d9aea7980fb595c'
187     LFBFL_correct_sha512+='85c562462fb67e10503a471ee46da89c8f73b47'
188     wrapped_wget_sha512
189
190     # /licenses_templates/ ------------------------------------------
191     local LFBFL_dependencies_URL2
192     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
193     readonly LFBFL_dependencies_URL2
194     LFBFL_file_name="build_licenses_templates.exec.sh"
195     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
196     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
197     LFBFL_correct_sha512='9b06e0631a64496c0e412ef412fb676bff0ace10cdbf7'
198     LFBFL_correct_sha512+='a6b92b7cb087049f8d2ae38261b316f74eedb5a9d2e3'
199     LFBFL_correct_sha512+='4825396438729e70ce82e38df766df7fb416b5b'
200     wrapped_wget_sha512
```

```
201     chmod +x "./${LFBFL_file_path}"
202
203     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
204     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
205     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
206     LFBFL_correct_sha512='5dc0dc20914c57feee2e26c2d329f020f221357b11b0d'
207     LFBFL_correct_sha512+='0ff6add69941ad953f8356b68be32cd60df8c56e91a2'
208     LFBFL_correct_sha512+='716c9cbf70ab8698e74e6aacc9c5439a93cc2bc'
209     wrapped_wget_sha512
210
211     LFBFL_file_name="license_file_header_GPLv3+.tpl"
212     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
213     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
214     LFBFL_correct_sha512='2d4b31697e78ba36ee0e3bd2b5393a57e15b5f5e62b2e'
215     LFBFL_correct_sha512+='802b812dcb087588f63cb92a851b21180be24bf5dee4'
216     LFBFL_correct_sha512+='eff7927244cde817e964512eaa1fb51d9064c01'
217     wrapped_wget_sha512
218
219     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
220     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
221     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
222     LFBFL_correct_sha512='43976b6dc9a54fcea0440cfaf87530dade73efe22b5ae'
223     LFBFL_correct_sha512+='a968453c9c5ac1fdc4d3798fdd445fb38a53f997da82'
224     LFBFL_correct_sha512+='7c30be6b7ad42d5e56ae90aac5095b9526b203d'
225     wrapped_wget_sha512
226     # ----------------------------------------------------------------
227
228     LFBFL_file_name="lines_counts.libr.sh"
229     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
230     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
231     LFBFL_correct_sha512='d131e24ad596988a6953b9362b13e5d17330f3bd8013b'
232     LFBFL_correct_sha512+='03fc2acf74a6b931660a619149cce59d1fa3b075c11c'
233     LFBFL_correct_sha512+='0a5e9c85e847b437d7bc3b58f034c00df3c5702'
234     wrapped_wget_sha512
235
236     LFBFL_file_name="lines_filters.libr.sh"
237     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
238     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
239     LFBFL_correct_sha512='d7b9d7456d18965b7910516cf80f96b5a89b76c59f315'
240     LFBFL_correct_sha512+='f7558487f5f37afc8d0af13d4f11928f7c2bec8a03a1'
241     LFBFL_correct_sha512+='7b49a9a5f71945d2a6fff955ce98518f5305234'
242     wrapped_wget_sha512
243
244     LFBFL_file_name="lines_maps.libr.sh"
245     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
246     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
247     LFBFL_correct_sha512='ccb9115be582a5b39790725dc96a628beb21b0c9951e7'
248     LFBFL_correct_sha512+='03a7a7bc8ab2736e3940ed3df6a2fcb04e75350c10d7'
249     LFBFL_correct_sha512+='3e92bcfada424b4049cc6141d41bf7901c0525b'
250     wrapped_wget_sha512
251
252     LFBFL_file_name="overwrite_if_not_equal.libr.sh"
253     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
254     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
255     LFBFL_correct_sha512='378f03b8ba77430e05db4ef5d5926803b466a87e1a2ec'
256     LFBFL_correct_sha512+='eef0b51036f25460a93a018bd86d3f90146b4dfec9ab'
257     LFBFL_correct_sha512+='15c5d1d57a526296bbdc4a0c03650088f3e9653'
258     wrapped_wget_sha512
259
260     LFBFL_file_name="phpmd_baseline.xslt"
261     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
```

```
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='88f39ea6952acebdad60c267e0f49a0489b486f7ead53'
    LFBFL_correct_sha512+='4c6ccf32e9f2bfa86f92d772c2fa59fecef00d6c9e6c'
    LFBFL_correct_sha512+='4d31cf074dda4943070db069c64581a18bfe724'
    wrapped_wget_sha512

    LFBFL_file_name="python_black_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='4f08461329f1b78b58e54f675fb433aec363de2c96a91'
    LFBFL_correct_sha512+='34ba5e96f3c77b93c9d7d6e28771a0b886501c286457'
    LFBFL_correct_sha512+='a4a410389556f180f2dc8f66d7956bf5dd5acf2'
    wrapped_wget_sha512

    LFBFL_file_name="python_isort_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='f91f3236ff9720412b31ec9fec2a5a5b3ef2424931815'
    LFBFL_correct_sha512+='c0e6d1bd9821e86ef80a13a27a3f6d831327dc4b9698'
    LFBFL_correct_sha512+='5935a75793c87bdf735c4ae2a94327289d1c2ed'
    wrapped_wget_sha512

    LFBFL_file_name="repository_data.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='8148fae16b5853b5ed6ca141b73b32f6d8695d8d34a54'
    LFBFL_correct_sha512+='73ad8e8cc0dab61d92e87fd925b6f25c0632f70b2fee'
    LFBFL_correct_sha512+='78ee53b0e3af6aa25b3cc58d8769292c3e4defd'
    wrapped_wget_sha512

    LFBFL_file_name="shell_checks_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='143575d86b5fc0fd78e575b6e46054b0c8facaf2c6df6'
    LFBFL_correct_sha512+='07dc596b6db0494c2b88211d5e0e011dee9abe70a2fc'
    LFBFL_correct_sha512+='b8f2bc04eb5cd50b248b0afe423103ee1cd568b'
    wrapped_wget_sha512

    LFBFL_file_name="split_score.exec.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='2845905b970c2dae5cf4a072df3e00670607f576d88db'
    LFBFL_correct_sha512+='22176e57f47081d487eaf0d5d006950550932bae2686'
    LFBFL_correct_sha512+='9ad1933bb2adcaf2719469fad83ea55e53e154e'
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="split_score.libr.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='a66cf904ba51bf9180c887d67f81b8d64eb0ac9b95a3c'
    LFBFL_correct_sha512+='696e51e050743956029d2d247e2252e2763ceffbd201'
    LFBFL_correct_sha512+='459ac09ed5ed53e2af2436ed3610fe56a88d384'
    wrapped_wget_sha512

    LFBFL_file_name="strings_functions.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='526bdaeca9ce0732ad429c16410909f9f8014eef9e031'
    LFBFL_correct_sha512+='837c428786b19bc31fea3eff6c8d93eea260b9076b67'
    LFBFL_correct_sha512+='71d21220528e70aa08a1a2fb7579926db662ff2'
```

```
    wrapped_wget_sha512

    LFBFL_file_name="too_long_code_lines.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='0571332f70cd5700f35ba5eb804bbe20a58d01ca33635'
    LFBFL_correct_sha512+='c36f40a8aa24ddc7926ee66d6372191b91e6b2ffcac3'
    LFBFL_correct_sha512+='9d4e89ef8a1e0c32cbae6c2e1e69437a65d3efc'
    wrapped_wget_sha512

    LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    LFBFL_correct_sha512='188fa5dbcab63be8e0f5b3c8997e099c7259f7c804433'
    LFBFL_correct_sha512+='581c2c05a7d9c7bd3d82bc424e2021afa90cb9e9d6a5'
    LFBFL_correct_sha512+='f84f549ed692c18b6351983c31b875846d7e1af'
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    # shellcheck source=check_shell_scripts_beginnings.libr.sh
    source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
    # shellcheck source=check_shell_scripts_indentation.libr.sh
    source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
    # shellcheck source=check_URLs.libr.sh
    source "./${LFBFL_subdir}/check_URLs.libr.sh"
    # shellcheck source=common_options.libr.sh
    source "./${LFBFL_subdir}/common_options.libr.sh"
    # shellcheck source=comparisons.libr.sh
    source "./${LFBFL_subdir}/comparisons.libr.sh"
    # shellcheck source=generate_from_template.libr.sh
    source "./${LFBFL_subdir}/generate_from_template.libr.sh"
    # shellcheck source=get_common_text_glob_patterns.libr.sh
    source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
    # shellcheck source=grammar_and_spelling_check.libr.sh
    source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
    # shellcheck source=lines_counts.libr.sh
    source "./${LFBFL_subdir}/lines_counts.libr.sh"
    # shellcheck source=lines_filters.libr.sh
    source "./${LFBFL_subdir}/lines_filters.libr.sh"
    # shellcheck source=lines_maps.libr.sh
    source "./${LFBFL_subdir}/lines_maps.libr.sh"
    # shellcheck source=overwrite_if_not_equal.libr.sh
    source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
    # shellcheck source=python_black_complement.libr.sh
    source "./${LFBFL_subdir}/python_black_complement.libr.sh"
    # shellcheck source=python_isort_complement.libr.sh
    source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
    # shellcheck source=shell_checks_complement.libr.sh
    source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
    # shellcheck source=strings_functions.libr.sh
    source "./${LFBFL_subdir}/strings_functions.libr.sh"
    # shellcheck source=too_long_code_lines.libr.sh
    source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

    enhanced_set_bash_option extglob
    # shellcheck source=repository_data.libr.sh
    source "./${LFBFL_subdir}/repository_data.libr.sh"
    # shellcheck disable=SC2154,SC2309
    [[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
      && enhanced_unset_bash_option extglob

```

```
384   declare -a LFBFL_return_traps_stack
385   init_return_trap
386
387   enhanced_set_shell_option pipefail --trap-unset
388
389   declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
390   keep_options LFBFL_some_common_options2 --verbose
391
392   printf "Building license headers\n"
393   "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
394     "${LFBFL_some_common_options[@]}"
395
396   printf "Building README.md\n"
397   "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
398     --work-directory="${LFBFL_work_directory}"\
399     --base-name=README\
400     "${LFBFL_verbose}"
401
402   printf "Building other MarkDown files\n"
403   local LFBFL_some_directory
404   declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
405   local LFBFL_s_files_paths
406   LFBFL_s_files_paths=$(
407     find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
408     | relevant_find
409   )
410   declare -a LFBFL_arr_files_paths
411   if [[ -n "${LFBFL_s_files_paths}" ]]; then
412     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
413     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
414       [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
415       printf "Found template %s.\n" "${LFBFL_file_path}"
416       LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
417       LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
418       LFBFL_file_name=${LFBFL_file_name%.md.tpl}
419       "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
420         --work-directory="${LFBFL_some_directory}"\
421         --base-name="${LFBFL_file_name}"\
422         "${LFBFL_verbose}"
423     done
424   fi
425
426   printf "Building other standard files\n"
427   LFBFL_s_files_paths=$(
428     find "${LFBFL_work_directory}" -type f -name "*.neon.tpl"\
429     | relevant_find
430   )
431   if [[ -n "${LFBFL_s_files_paths}" ]]; then
432     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
433     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
434       printf "Found template %s.\n" "${LFBFL_file_path}"
435       LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
436       LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
437       LFBFL_file_name=${LFBFL_file_name%.tpl}
438       "./${LFBFL_subdir}/build_file_from_printable_file__standard.exec.sh"\
439         --work-directory="${LFBFL_some_directory}"\
440         --base-name="${LFBFL_file_name}"\
441         "${LFBFL_verbose}"\
442         --remove-leading-whitespaces-after-escaped-line-return
443     done
444   fi
```

```
445
446   printf "Building dependencies_notes\n"
447   # https://github.com/php/php-src/issues/21538
448   local LFBFL_annoying_warning
449   LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
450   LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
451   LFBFL_annoying_warning=" by settings in"
452   readonly LFBFL_annoying_warning
453   if [[ -e "${LFBFL_subdir3}/dependencies_data.json" ]]; then
454     php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
455       -- "${LFBFL_work_directory}"\
456       2>&1\
457       | grep --invert-match "${LFBFL_annoying_warning}"
458   elif [[ LFBFL_i_verbose -eq 1 ]]; then
459     printf "No file dependencies_data.json.\n"
460   fi
461
462   declare -i LFBFL_i_directory_changed
463   pushd_to_work_directory
464   LFBFL_i_directory_changed=$?
465   can_continue_after_enhanced_pushd || return 1
466
467   local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"
468
469   declare -i LFBFL_i_max_line_length
470   grep_variable "${LFBFL_data_file_path}" max_line_length\
471     --result-variable-prefix=LFBFL_i_
472
473   LFBFL_some_common_options+=(
474     "--max-line-length=${LFBFL_i_max_line_length}"
475   )
476   LFBFL_some_common_options2+=(
477     "--max-line-length=${LFBFL_i_max_line_length}"
478   )
479
480   declare -i LFBFL_i_upgrade_venvs=0
481   local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
482   readonly LFBFL_upgrade_venvs_ts_file
483   declare -i LFBFL_i_upgrade_venvs_ts
484   declare -i LFBFL_i_current_ts
485   local LFBFL_upgrade_venvs_answer
486
487   local LFBFL_upgrade_venvs_time_interval_in_seconds=""
488   get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
489     "${LFBFL_some_common_options2[@]}"
490
491   if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
492     LFBFL_i_upgrade_venvs_ts=$(
493       stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
494     )
495     LFBFL_i_current_ts=$(date +%s)
496     ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
497     ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
498     if [[ LFBFL_i_current_ts -gt 0 ]]; then
499       LFBFL_i_upgrade_venvs=1
500     fi
501   else
502     LFBFL_i_upgrade_venvs=1
503   fi
504   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
505     local LFBFL_upgrade_venvs=""
```

```
    grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

  printf "Analyzing too long lines\n"
  too_long_code_lines "${LFBFL_some_common_options2[@]}"

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└├│«»..."
  grep --binary-files=without-match --color=always --exclude-dir .git\
    --line-number --perl-regexp --recursive --\
    "[^${LFBFL_usual_characters}]" .\
    | relevant_grep

  printf "Grammar and spelling check\n"
  grammar_and_spelling_check "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  printf "Running shellcheck\n"
  if [[ -e "${LFBFL_subdir3}/shellcheck.ini" ]]; then
    LFBFL_s_files_paths=$(
      find . -type f -name "*.sh"\
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_paths}" ]]; then
      mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
      for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
        shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
          -- "${LFBFL_file_path}"
      done
    fi
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file shellcheck.ini.\n"
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options2[@]}"

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
    | relevant_grep

  printf -- "---Python---\n"

  maybe_deactivate(){
```

```
    if command -v deactivate
    then
      deactivate
    fi
  }

  source_without_return_trap(){
    declare -i LFBFL_i_no_return_traps=1
    # shellcheck disable=SC1090,SC1091
    source -- "$1"
  }

  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_path}" isort_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort --profile=black .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options2[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
    local LFBFL_update_max_length="s/^line-length = [0-9]*$"
    LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
    sed --regexp-extended --expression="${LFBFL_update_max_length}"\
      "${LFBFL_subdir3}/black.toml"\
      > "${LFBFL_subdir3}/black.toml.temp"
    overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
      "${LFBFL_subdir3}/black.toml.temp"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file black.toml.\n"
  fi

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_path}" black_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
    black --config="${LFBFL_subdir3}/black.toml" .
  fi
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
```

```
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options2[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_path}" mypy_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  local LFBFL_directory_path
  declare -i LFBFL_i_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
        mypy -- "${LFBFL_directory_path}"
      fi
      LFBFL_i_no_toml=0
    done
  fi
  if [[ LFBFL_i_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_path}" bandit_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  if [[ -e "${LFBFL_subdir3}/bandit.ini" ]]; then
    bandit --ini="${LFBFL_subdir3}/bandit.ini"\
      --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
      --recursive .
    # Saving new baseline in temp if necessary.
    bandit --ini="${LFBFL_subdir3}/bandit.ini"\
      --format=json\
      --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
```

```
      --recursive .
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file bandit.ini.\n"
  fi
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_path}" pylint_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  if [[ -e "${LFBFL_subdir3}/pylintrc.toml" ]]; then
    pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file pylintrc.toml.\n"
  fi
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_path}" ruff_venv\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    maybe_deactivate
    source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  if [[ -e "${LFBFL_subdir3}/ruff.toml" ]]; then
    ruff check --config="${LFBFL_subdir3}/ruff.toml"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file ruff.toml.\n"
  fi
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHP native linter\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.php"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      php --syntax-check -f="${LFBFL_file_path}"\
```

```
      2>&1\
      | grep --invert-match "${LFBFL_annoying_warning}"
  done
fi

printf "Running PHP_CodeSniffer\n"
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  composer global require squizlabs/php_codesniffer
fi
if [[ -e "${LFBFL_subdir3}/phpcs_ruleset.xml" ]]; then
  phpcs --report=code\
    --standard="${LFBFL_subdir3}/phpcs_ruleset.xml"
elif [[ LFBFL_i_verbose -eq 1 ]]; then
  printf "No file phpcs_ruleset.xml.\n"
fi

printf "Running PHPMD\n"
if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
  composer global require phpmd/phpmd
fi

declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_temp_phpmd_baseline

local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
LFBFL_phpmd_rulesets+="design,naming,unusedcode"
readonly LFBFL_phpmd_rulesets

phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Saving new baseline in temp if necessary.
rm "${LFBFL_temp_phpmd_baseline}"
phpmd --color --generate-baseline\
  --baseline-file="${LFBFL_temp_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Remove all superfluous whitespaces.
# sed --in-place --regexp-extended --null-data\
#   --expression='s/\n +//Mg'\
#   "${LFBFL_temp_phpmd_baseline}"
xsltproc --output "${LFBFL_temp_phpmd_baseline}"\
  build_and_checks_dependencies/phpmd_baseline.xslt\
  "${LFBFL_temp_phpmd_baseline}"
# sed --in-place --expression='s/" file="/"\n    file="/g'\
#   "${LFBFL_temp_phpmd_baseline}"
# sed --in-place --regexp-extended\
#   --expression='s~(file=".*)/>~\1\n  />~g'\
#   "${LFBFL_temp_phpmd_baseline}"
sed --in-place\
  --expression='s/\([^"]*\("[^"]*"\)*\) rule="/\1\n    rule="/'\
  --expression='s/\([^"]*\("[^"]*"\)*\) file="/\1\n    file="/'\
  --expression='s/\([^"]*\("[^"]*"\)*\) method="/\1\n    method="/'\
  --expression='s~/>~\n  />~'\
  "${LFBFL_temp_phpmd_baseline}"
diff --suppress-common-lines\
  "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
  | grep --invert-match\
    --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
# See miscellaneous/grep_benchmark.

```

```
811   printf "Running PHPSTAN\n"
812   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
813     composer global require phpstan/phpstan
814     composer global require phpstan/phpstan-strict-rules
815   fi
816
817   local LFBFL_temp_phpstan_baseline="${LFBFL_subdir3}/temp/"
818   LFBFL_temp_phpstan_baseline+="phpstan_baseline.neon"
819   readonly LFBFL_temp_phpstan_baseline
820   if [[ -e "${LFBFL_subdir3}/phpstan_config.neon" ]]; then
821     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"
822     # Saving new baseline in temp if necessary.
823     rm "${LFBFL_temp_phpstan_baseline}"
824     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"\
825       --generate-baseline="${LFBFL_temp_phpstan_baseline}"\
826       --allow-empty-baseline
827     sed --in-place --expression='s/\t/ /g' "${LFBFL_temp_phpstan_baseline}"
828     cat "${LFBFL_temp_phpstan_baseline}"
829   elif [[ LFBFL_i_verbose -eq 1 ]]; then
830     printf "No file phpstan_config.neon.\n"
831   fi
832
833   printf "Running phpDocumentor\n"
834   # shellcheck disable=SC2312
835   docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
836   printf -- "---PHP end---\n"
837
838   printf -- "---JS---\n"
839   local LFBFL_npm_lint_directories=""
840   grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
841     --result-variable-prefix=LFBFL_
842   if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
843     printf "Running ESLint\n"
844     local LFBFL_JS_directory
845     declare -a LFBFL_arr_paths
846     mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
847     for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
848       (cd -- "${LFBFL_JS_directory}" && npm run lint)
849     done
850   fi
851   printf -- "---JS end---\n"
852
853   [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory
854
855   printf "Checking listed files\n"
856   "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
857     "${LFBFL_some_common_options[@]}"
858
859   printf "Creating the PDF file of the listing of the source code\n"
860   "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"
861
862   pushd_to_work_directory --trap-popd
863   can_continue_after_enhanced_pushd || return 1
864
865   if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
866     "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
867   fi
868
869   printf "Running xmllint\n"
870   # In some cases, we need to handle a cache of common DTDs by hand.
871   # XHTML DTDs validation can yield errors:
```

```
    # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
    # If you use Ubuntu/Debian, install w3c-sgml-lib instead.
    LFBFL_file_name="xhtml1-strict.dtd"
    LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='fe11db06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
    LFBFL_correct_sha512+='d8352bf19947168c9bcf1ae212eca035824293abb1f9'
    LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
    wrapped_wget_sha512

    LFBFL_file_name="xhtml1-transitional.dtd"
    LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
    LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
    LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
    wrapped_wget_sha512

    LFBFL_file_name="xhtml1-frameset.dtd"
    LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
    LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
    LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
    wrapped_wget_sha512

    LFBFL_file_name="xhtml11.dtd"
    LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
    LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
    LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
    wrapped_wget_sha512

    LFBFL_file_name="xhtml-lat1.ent"
    LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
    LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
    LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
    wrapped_wget_sha512

    LFBFL_file_name="xhtml-special.ent"
    LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
    LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
    LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
    wrapped_wget_sha512

    LFBFL_file_name="xhtml-symbol.ent"
    LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
    LFBFL_script_download_URL="${LFBFL_a}"
    LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
    LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
```

```
933   LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
934   LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
935   wrapped_wget_sha512
936
937   declare -r LFBFL_s_files_for_xmllint=$(
938     find . -type f -iregex ".*\.\(xml\|xhtml\)"\
939     | relevant_find\
940     | sort
941   )
942   if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
943     LFBFL_xmllint_use_downloaded_DTDs=""
944     grep_variable "${LFBFL_data_file_path}" xmllint_use_downloaded_DTDs\
945       --result-variable-prefix=LFBFL_
946
947     declare -a LFBFL_arr_files_for_xmllint
948     mapfile -t LFBFL_arr_files_for_xmllint\
949       <<< "${LFBFL_s_files_for_xmllint}"
950     readonly LFBFL_arr_files_for_xmllint
951     local LFBFL_xmllint_files_without_DTD=""
952     grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
953       xmllint_files_without_DTD\
954       --result-variable-prefix=LFBFL_
955     declare -a LFBFL_arr_xmllint_files_without_DTD
956     if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
957       mapfile -t LFBFL_arr_xmllint_files_without_DTD\
958         <<< "${LFBFL_xmllint_files_without_DTD}"
959     fi
960     for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
961       [[ LFBFL_i_verbose -eq 1 ]]\
962         && printf "xmllint on %s\n" "${LFBFL_file_path}"
963       # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
964       # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
965       # xmllint --pedantic --noout -- "${LFBFL_file_path}"
966       xmllint --pedantic --noout "${LFBFL_file_path}"
967     done
968
969     local LFBFL_file_path2
970     for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
971       for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
972       do
973         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
974           LFBFL_file_path=""
975         fi
976       done
977       if [[ "${LFBFL_file_path}" == "" ]]; then
978         continue
979       fi
980       [[ LFBFL_i_verbose -eq 1 ]]\
981         && printf "xmllint on %s\n" "${LFBFL_file_path}"
982       if [[
983         "${LFBFL_file_path}" == *.xhtml
984         && "${LFBFL_xmllint_use_downloaded_DTDs}" == "true"
985       ]]; then
986         if grep --fixed-strings\
987           "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
988           "${LFBFL_file_path}"
989         then
990           xmllint --pedantic --noout\
991             --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
992             "${LFBFL_file_path}"
993         elif grep --fixed-strings\
```

```
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
          "${LFBFL_file_path}"
        then
          xmllint --pedantic --noout\
            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
            "${LFBFL_file_path}"
        elif grep --fixed-strings\
          "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
          "${LFBFL_file_path}"
        then
          xmllint --pedantic --noout\
            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
            "${LFBFL_file_path}"
        elif grep --fixed-strings\
          "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
          "${LFBFL_file_path}"
        then
          xmllint --pedantic --noout\
            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
            "${LFBFL_file_path}"
        else
          printf "Unknown XHTML DTD.\n"
        fi
      else
        xmllint --pedantic --noout --valid "${LFBFL_file_path}"
      fi
    done
  fi
  # ---------------------------------------------------------------------

  # ---------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
```

```
    # vnu has false positives for xhtml.
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_s_files_for_Nu=$(
      find . -type f -iregex "${LFBFL_find_regexp}"\
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
      declare -a LFBFL_arr_files_for_Nu
      mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
      readonly LFBFL_arr_files_for_Nu
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
    fi
  fi
  # ----------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.11 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
```

```
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
  declare -r LFBFL_dependencies_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  declare -a LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    "--work-directory=${LFBFL_work_directory}"
  )

  source ./wget_sha512.libr.sh

  local LFBFL_subdir="build_and_checks_dependencies"
  declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
  declare -r LFBFL_subdir3="build_and_checks_variables"
  local LFBFL_file_name
  local LFBFL_script_download_URL
  local LFBFL_file_path
  local LFBFL_correct_sha512

  wrapped_wget_sha512(){
    wget_sha512 "${LFBFL_file_path}"\
      "${LFBFL_script_download_URL}"\
      "${LFBFL_correct_sha512}"\
      "${LFBFL_verbose}"\
      "${LFBFL_check_download}"
  }

  LFBFL_file_name="build_dependencies_notes.exec.php"
  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_build_dependencies_notes.exec.php@
  wrapped_wget_sha512
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="build_dependencies_notes.libr.php"
```

```
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_build_dependencies_notes.libr.php@
    wrapped_wget_sha512

    LFBFL_file_name="build_file_from_printable_file__standard.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_build_file_from_printable_file__standard.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="build_md_from_printable_md.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_build_md_from_printable_md.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_check_shell_scripts_beginnings.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_check_shell_scripts_indentation.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="check_URLs.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_check_URLs.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="common_options.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_common_options.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="comparisons.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_comparisons.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="create_PDF.exec.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_create_PDF.exec.sh@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="generate_from_template.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_generate_from_template.libr.sh@
    wrapped_wget_sha512
```

```
151
152     LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
153     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
154     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
155     @sha512_get_common_text_glob_patterns.libr.sh@
156     wrapped_wget_sha512
157
158     LFBFL_file_name="grammar_and_spelling_check.libr.sh"
159     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
160     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
161     @sha512_grammar_and_spelling_check.libr.sh@
162     wrapped_wget_sha512
163
164     # /licenses_templates/ ---------------------------------------------
165     local LFBFL_dependencies_URL2
166     LFBFL_dependencies_URL2="${LFBFL_dependencies_URL}/licenses_templates"
167     readonly LFBFL_dependencies_URL2
168     LFBFL_file_name="build_licenses_templates.exec.sh"
169     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
170     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
171     @sha512_build_licenses_templates.exec.sh@
172     wrapped_wget_sha512
173     chmod +x "./${LFBFL_file_path}"
174
175     LFBFL_file_name="license_file_header_AGPLv3+.tpl"
176     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
177     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
178     @sha512_license_file_header_AGPLv3+.tpl@
179     wrapped_wget_sha512
180
181     LFBFL_file_name="license_file_header_GPLv3+.tpl"
182     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
183     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
184     @sha512_license_file_header_GPLv3+.tpl@
185     wrapped_wget_sha512
186
187     LFBFL_file_name="license_file_header_LGPLv3+.tpl"
188     LFBFL_script_download_URL="${LFBFL_dependencies_URL2}/${LFBFL_file_name}"
189     LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
190     @sha512_license_file_header_LGPLv3+.tpl@
191     wrapped_wget_sha512
192     # ------------------------------------------------------------------
193
194     LFBFL_file_name="lines_counts.libr.sh"
195     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
196     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
197     @sha512_lines_counts.libr.sh@
198     wrapped_wget_sha512
199
200     LFBFL_file_name="lines_filters.libr.sh"
201     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
202     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
203     @sha512_lines_filters.libr.sh@
204     wrapped_wget_sha512
205
206     LFBFL_file_name="lines_maps.libr.sh"
207     LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
208     LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
209     @sha512_lines_maps.libr.sh@
210     wrapped_wget_sha512
211
```

```
    LFBFL_file_name="overwrite_if_not_equal.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_overwrite_if_not_equal.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="phpmd_baseline.xslt"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_phpmd_baseline.xslt@
    wrapped_wget_sha512

    LFBFL_file_name="python_black_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_python_black_complement.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="python_isort_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_python_isort_complement.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="repository_data.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_repository_data.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="shell_checks_complement.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_shell_checks_complement.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="split_score.exec.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_split_score.exec.php@
    wrapped_wget_sha512
    chmod +x "./${LFBFL_file_path}"

    LFBFL_file_name="split_score.libr.php"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_split_score.libr.php@
    wrapped_wget_sha512

    LFBFL_file_name="strings_functions.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_strings_functions.libr.sh@
    wrapped_wget_sha512

    LFBFL_file_name="too_long_code_lines.libr.sh"
    LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
    @sha512_too_long_code_lines.libr.sh@
    wrapped_wget_sha512

```

```
LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_script_download_URL="${LFBFL_dependencies_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wrapped_wget_sha512
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spelling_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spelling_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

declare -a LFBFL_return_traps_stack
init_return_trap

enhanced_set_shell_option pipefail --trap-unset

declare -a LFBFL_some_common_options2=("${LFBFL_some_common_options[@]}")
keep_options LFBFL_some_common_options2 --verbose

printf "Building license headers\n"
"./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
  "${LFBFL_some_common_options[@]}"

```

```
334   printf "Building README.md\n"
335   "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
336     --work-directory="${LFBFL_work_directory}"\
337     --base-name=README\
338     "${LFBFL_verbose}"
339
340   printf "Building other MarkDown files\n"
341   local LFBFL_some_directory
342   declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
343   local LFBFL_s_files_paths
344   LFBFL_s_files_paths=$(
345     find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
346     | relevant_find
347   )
348   declare -a LFBFL_arr_files_paths
349   if [[ -n "${LFBFL_s_files_paths}" ]]; then
350     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
351     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
352       [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
353       printf "Found template %s.\n" "${LFBFL_file_path}"
354       LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
355       LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
356       LFBFL_file_name=${LFBFL_file_name%.md.tpl}
357       "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
358         --work-directory="${LFBFL_some_directory}"\
359         --base-name="${LFBFL_file_name}"\
360         "${LFBFL_verbose}"
361     done
362   fi
363
364   printf "Building other standard files\n"
365   LFBFL_s_files_paths=$(
366     find "${LFBFL_work_directory}" -type f -name "*.neon.tpl"\
367     | relevant_find
368   )
369   if [[ -n "${LFBFL_s_files_paths}" ]]; then
370     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
371     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
372       printf "Found template %s.\n" "${LFBFL_file_path}"
373       LFBFL_some_directory=$(dirname -- "${LFBFL_file_path}")
374       LFBFL_file_name=$(basename -- "${LFBFL_file_path}")
375       LFBFL_file_name=${LFBFL_file_name%.tpl}
376       "./${LFBFL_subdir}/build_file_from_printable_file__standard.exec.sh"\
377         --work-directory="${LFBFL_some_directory}"\
378         --base-name="${LFBFL_file_name}"\
379         "${LFBFL_verbose}"\
380         --remove-leading-whitespaces-after-escaped-line-return
381     done
382   fi
383
384   printf "Building dependencies_notes\n"
385   # https://github.com/php/php-src/issues/21538
386   local LFBFL_annoying_warning
387   LFBFL_annoying_warning="PHP Warning:  declare(encoding=...) ignored"
388   LFBFL_annoying_warning=" because Zend multibyte feature is turned off"
389   LFBFL_annoying_warning=" by settings in"
390   readonly LFBFL_annoying_warning
391   if [[ -e "${LFBFL_subdir3}/dependencies_data.json" ]]; then
392     php -f="./${LFBFL_subdir}/build_dependencies_notes.exec.php"\
393       -- "${LFBFL_work_directory}"\
394       2>&1\
```

```
      | grep --invert-match "${LFBFL_annoying_warning}"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file dependencies_data.json.\n"
  fi

  declare -i LFBFL_i_directory_changed
  pushd_to_work_directory
  LFBFL_i_directory_changed=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_path="${LFBFL_subdir3}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_path}" max_line_length\
    --result-variable-prefix=LFBFL_i_

  LFBFL_some_common_options+=(
    "--max-line-length=${LFBFL_i_max_line_length}"
  )
  LFBFL_some_common_options2+=(
    "--max-line-length=${LFBFL_i_max_line_length}"
  )

  declare -i LFBFL_i_upgrade_venvs=0
  local LFBFL_upgrade_venvs_ts_file="${LFBFL_subdir3}/upgrade_venvs_ts"
  readonly LFBFL_upgrade_venvs_ts_file
  declare -i LFBFL_i_upgrade_venvs_ts
  declare -i LFBFL_i_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_path}"\
    "${LFBFL_some_common_options2[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_i_upgrade_venvs_ts=$(
      stat --format %Y -- "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_upgrade_venvs_ts))
    ((LFBFL_i_current_ts -= LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      LFBFL_i_upgrade_venvs=1
    fi
  else
    LFBFL_i_upgrade_venvs=1
  fi
  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    local LFBFL_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_path}" upgrade_venvs\
      --result-variable-prefix=LFBFL_
    if [[ "${LFBFL_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_i_upgrade_venvs=0
      fi
    fi
  fi

```

```
456   printf "Analyzing too long lines\n"
457   too_long_code_lines "${LFBFL_some_common_options2[@]}"
458
459   printf "Analyzing URLs\n"
460   check_URLs "${LFBFL_some_common_options2[@]}" \
461     | relevant_grep
462
463   printf "Analyzing strange characters: hover over in doubt\n"
464   # shellcheck disable=SC1111
465   LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└┼│«»..."
466   grep --binary-files=without-match --color=always --exclude-dir .git\
467     --line-number --perl-regexp --recursive --\
468     "[^${LFBFL_usual_characters}]" .\
469     | relevant_grep
470
471   printf "Grammar and spelling check\n"
472   grammar_and_spelling_check "${LFBFL_data_file_path}"\
473     "${LFBFL_some_common_options2[@]}"
474
475   printf "Running shellcheck\n"
476   if [[ -e "${LFBFL_subdir3}/shellcheck.ini" ]]; then
477     LFBFL_s_files_paths=$(
478       find . -type f -name "*.sh"\
479       | relevant_find
480     )
481     if [[ -n "${LFBFL_s_files_paths}" ]]; then
482       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
483       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
484         shellcheck --rcfile="${LFBFL_subdir3}/shellcheck.ini"\
485           -- "${LFBFL_file_path}"
486       done
487     fi
488   elif [[ LFBFL_i_verbose -eq 1 ]]; then
489     printf "No file shellcheck.ini.\n"
490   fi
491   printf "Running shell_checks_complement\n"
492   shell_checks_complement "${LFBFL_some_common_options2[@]}"
493
494   printf "Analyzing shell scripts beginnings\n"
495   check_shell_scripts_beginnings "${LFBFL_some_common_options2[@]}" \
496     | relevant_grep
497
498   printf "Analyzing shell scripts indentation\n"
499   check_shell_scripts_indentation "${LFBFL_some_common_options2[@]}" \
500     | relevant_grep
501
502   printf -- "---Python---\n"
503
504   maybe_deactivate(){
505     if command -v deactivate
506     then
507       deactivate
508     fi
509   }
510
511   source_without_return_trap(){
512     declare -i LFBFL_i_no_return_traps=1
513     # shellcheck disable=SC1090,SC1091
514     source -- "$1"
515   }
516
```

```
517   printf "Running isort\n"
518   local LFBFL_isort_venv=""
519   grep_variable "${LFBFL_data_file_path}" isort_venv\
520     --result-variable-prefix=LFBFL_\
521     --replace-line-returns-by=""
522   if [[ -n "${LFBFL_isort_venv}" ]]; then
523     maybe_deactivate
524     source_without_return_trap "${LFBFL_isort_venv}/bin/activate"
525   fi
526   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
527     pip install --upgrade isort
528   fi
529   isort --profile=black .
530   if [[ -n "${LFBFL_isort_venv}" ]]; then
531     deactivate
532   fi
533   printf "Running python_isort_complement\n"
534   python_isort_complement "${LFBFL_some_common_options2[@]}"
535
536   printf "Running black\n"
537   # First, we update the configuration file with max_line_length.
538   if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
539     local LFBFL_update_max_length="s/^line-length = [0-9]*$"
540     LFBFL_update_max_length+="/line-length = ${LFBFL_i_max_line_length}/"
541     sed --regexp-extended --expression="${LFBFL_update_max_length}"\
542       "${LFBFL_subdir3}/black.toml"\
543       > "${LFBFL_subdir3}/black.toml.temp"
544     overwrite_if_not_equal "${LFBFL_subdir3}/black.toml"\
545       "${LFBFL_subdir3}/black.toml.temp"
546   elif [[ LFBFL_i_verbose -eq 1 ]]; then
547     printf "No file black.toml.\n"
548   fi
549
550   local LFBFL_black_venv=""
551   grep_variable "${LFBFL_data_file_path}" black_venv\
552     --result-variable-prefix=LFBFL_\
553     --replace-line-returns-by=""
554   if [[ -n "${LFBFL_black_venv}" ]]; then
555     maybe_deactivate
556     source_without_return_trap "${LFBFL_black_venv}/bin/activate"
557   fi
558   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
559     pip install --upgrade black
560   fi
561   if [[ -e "${LFBFL_subdir3}/black.toml" ]]; then
562     black --config="${LFBFL_subdir3}/black.toml" .
563   fi
564   if [[ -n "${LFBFL_black_venv}" ]]; then
565     deactivate
566   fi
567   printf "Running python_black_complement\n"
568   python_black_complement "${LFBFL_some_common_options2[@]}"
569
570   printf "Probing if mypy should be runned\n"
571   local LFBFL_mypy_venv=""
572   grep_variable "${LFBFL_data_file_path}" mypy_venv\
573     --result-variable-prefix=LFBFL_\
574     --replace-line-returns-by=""
575   if [[ -n "${LFBFL_mypy_venv}" ]]; then
576     maybe_deactivate
577     source_without_return_trap "${LFBFL_mypy_venv}/bin/activate"
```

```
578   fi
579   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
580     pip install --upgrade mypy
581   fi
582   local LFBFL_directory_path
583   declare -i LFBFL_i_no_toml=1
584   LFBFL_s_files_paths=$(
585     find . -type f -name "pyproject.toml"\
586     | relevant_find
587   )
588   if [[ -n "${LFBFL_s_files_paths}" ]]; then
589     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
590     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
591       if grep --quiet "Typing :: Typed" -- "${LFBFL_file_path}"; then
592         printf "Running mypy\n"
593         LFBFL_directory_path=$(dirname -- "${LFBFL_file_path}")
594         mypy -- "${LFBFL_directory_path}"
595       fi
596       LFBFL_i_no_toml=0
597     done
598   fi
599   if [[ LFBFL_i_no_toml -eq 1 ]]; then
600     printf "Running mypy\n"
601     mypy .
602   fi
603   if [[ -n "${LFBFL_mypy_venv}" ]]; then
604     deactivate
605   fi
606
607   printf "Running bandit\n"
608   LFBFL_bandit_venv=""
609   grep_variable "${LFBFL_data_file_path}" bandit_venv\
610     --result-variable-prefix=LFBFL_\
611     --replace-line-returns-by=""
612   if [[ -n "${LFBFL_bandit_venv}" ]]; then
613     maybe_deactivate
614     source_without_return_trap "${LFBFL_bandit_venv}/bin/activate"
615   fi
616   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
617     pip install --upgrade bandit
618   fi
619   if [[ -e "${LFBFL_subdir3}/bandit.ini" ]]; then
620     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
621       --baseline="${LFBFL_subdir3}/bandit_baseline.json"\
622       --recursive .
623     # Saving new baseline in temp if necessary.
624     bandit --ini="${LFBFL_subdir3}/bandit.ini"\
625       --format=json\
626       --output="${LFBFL_subdir3}/temp/bandit_baseline.json"\
627       --recursive .
628   elif [[ LFBFL_i_verbose -eq 1 ]]; then
629     printf "No file bandit.ini.\n"
630   fi
631   if [[ -n "${LFBFL_bandit_venv}" ]]; then
632     deactivate
633   fi
634
635   printf "Running pylint\n"
636   LFBFL_pylint_venv=""
637   grep_variable "${LFBFL_data_file_path}" pylint_venv\
638     --result-variable-prefix=LFBFL_\
```

```
639       --replace-line-returns-by=""
640     if [[ -n "${LFBFL_pylint_venv}" ]]; then
641       maybe_deactivate
642       source_without_return_trap "${LFBFL_pylint_venv}/bin/activate"
643     fi
644     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
645       pip install --upgrade pylint
646     fi
647     if [[ -e "${LFBFL_subdir3}/pylintrc.toml" ]]; then
648       pylint --rcfile="${LFBFL_subdir3}/pylintrc.toml" --recursive=y .
649     elif [[ LFBFL_i_verbose -eq 1 ]]; then
650       printf "No file pylintrc.toml.\n"
651     fi
652     if [[ -n "${LFBFL_pylint_venv}" ]]; then
653       deactivate
654     fi
655
656     printf "Running ruff\n"
657     LFBFL_ruff_venv=""
658     grep_variable "${LFBFL_data_file_path}" ruff_venv\
659       --result-variable-prefix=LFBFL_\
660       --replace-line-returns-by=""
661     if [[ -n "${LFBFL_ruff_venv}" ]]; then
662       maybe_deactivate
663       source_without_return_trap "${LFBFL_ruff_venv}/bin/activate"
664     fi
665     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
666       pip install --upgrade ruff
667     fi
668     if [[ -e "${LFBFL_subdir3}/ruff.toml" ]]; then
669       ruff check --config="${LFBFL_subdir3}/ruff.toml"
670     elif [[ LFBFL_i_verbose -eq 1 ]]; then
671       printf "No file ruff.toml.\n"
672     fi
673     if [[ -n "${LFBFL_ruff_venv}" ]]; then
674       deactivate
675     fi
676     printf -- "---Python end---\n"
677
678     printf -- "---PHP---\n"
679     printf "Running PHP native linter\n"
680     LFBFL_s_files_paths=$(
681       find . -type f -name "*.php"\
682       | relevant_find
683     )
684     if [[ -n "${LFBFL_s_files_paths}" ]]; then
685       mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
686       for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
687         php --syntax-check -f="${LFBFL_file_path}"\
688           2>&1\
689           | grep --invert-match "${LFBFL_annoying_warning}"
690       done
691     fi
692
693     printf "Running PHP_CodeSniffer\n"
694     if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
695       composer global require squizlabs/php_codesniffer
696     fi
697     if [[ -e "${LFBFL_subdir3}/phpcs_ruleset.xml" ]]; then
698       phpcs --report=code\
699         --standard="${LFBFL_subdir3}/phpcs_ruleset.xml"
```

```
700   elif [[ LFBFL_i_verbose -eq 1 ]]; then
701     printf "No file phpcs_ruleset.xml.\n"
702   fi
703
704   printf "Running PHPMD\n"
705   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
706     composer global require phpmd/phpmd
707   fi
708
709   declare -r LFBFL_phpmd_baseline="${LFBFL_subdir3}/phpmd_baseline.xml"
710   local LFBFL_temp_phpmd_baseline="${LFBFL_subdir3}/temp/"
711   LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
712   readonly LFBFL_temp_phpmd_baseline
713
714   local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
715   LFBFL_phpmd_rulesets+="design,naming,unusedcode"
716   readonly LFBFL_phpmd_rulesets
717
718   phpmd --color --baseline-file="${LFBFL_phpmd_baseline}"\
719     . text "${LFBFL_phpmd_rulesets}"
720   # Saving new baseline in temp if necessary.
721   rm "${LFBFL_temp_phpmd_baseline}"
722   phpmd --color --generate-baseline\
723     --baseline-file="${LFBFL_temp_phpmd_baseline}"\
724     . text "${LFBFL_phpmd_rulesets}"
725   # Remove all superfluous whitespaces.
726   # sed --in-place --regexp-extended --null-data\
727   #   --expression='s/\n +//Mg'\
728   #   "${LFBFL_temp_phpmd_baseline}"
729   xsltproc --output "${LFBFL_temp_phpmd_baseline}"\
730     build_and_checks_dependencies/phpmd_baseline.xslt\
731     "${LFBFL_temp_phpmd_baseline}"
732   # sed --in-place --expression='s/" file="/"\n    file="/g'\
733   #   "${LFBFL_temp_phpmd_baseline}"
734   # sed --in-place --regexp-extended\
735   #   --expression='s~(file=".*)/>~\1\n  />~g'\
736   #   "${LFBFL_temp_phpmd_baseline}"
737   sed --in-place\
738     --expression='s/\([^"]*\("[^"]*"\)*\) rule="/\1\n    rule="/'\
739     --expression='s/\([^"]*\("[^"]*"\)*\) file="/\1\n    file="/'\
740     --expression='s/\([^"]*\("[^"]*"\)*\) method="/\1\n    method="/'\
741     --expression='s~/>~\n  />~'\
742     "${LFBFL_temp_phpmd_baseline}"
743   diff --suppress-common-lines\
744     "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"\
745     | grep --invert-match\
746       --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
747   # See miscellaneous/grep_benchmark.
748
749   printf "Running PHPSTAN\n"
750   if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
751     composer global require phpstan/phpstan
752     composer global require phpstan/phpstan-strict-rules
753   fi
754
755   local LFBFL_temp_phpstan_baseline="${LFBFL_subdir3}/temp/"
756   LFBFL_temp_phpstan_baseline+="phpstan_baseline.neon"
757   readonly LFBFL_temp_phpstan_baseline
758   if [[ -e "${LFBFL_subdir3}/phpstan_config.neon" ]]; then
759     phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"
760     # Saving new baseline in temp if necessary.
```

```
    rm "${LFBFL_temp_phpstan_baseline}"
    phpstan --configuration="${LFBFL_subdir3}/phpstan_config.neon"\
      --generate-baseline="${LFBFL_temp_phpstan_baseline}"\
      --allow-empty-baseline
    sed --in-place --expression='s/\t/ /g' "${LFBFL_temp_phpstan_baseline}"
    cat "${LFBFL_temp_phpstan_baseline}"
  elif [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "No file phpstan_config.neon.\n"
  fi

  printf "Running phpDocumentor\n"
  # shellcheck disable=SC2312
  docker run --rm --volume="$(pwd):/data" phpdoc/phpdoc
  printf -- "---PHP end---\n"

  printf -- "---JS---\n"
  local LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_path}" npm_lint_directories\
    --result-variable-prefix=LFBFL_
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd -- "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  [[ LFBFL_i_directory_changed -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "${LFBFL_subdir3}/post_build.sh" ]]; then
    "./${LFBFL_subdir3}/post_build.sh" "${LFBFL_verbose}"
  fi

  printf "Running xmllint\n"
  # In some cases, we need to handle a cache of common DTDs by hand.
  # XHTML DTDs validation can yield errors:
  # see https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119
  # If you use Ubuntu/Debian, install w3c-sgml-lib instead.
  LFBFL_file_name="xhtml1-strict.dtd"
  LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"
  LFBFL_script_download_URL="${LFBFL_a}"
  LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
  LFBFL_correct_sha512='fe11db06c0e1741a9c460fe4f9cf167bca57f4dbe9746'
  LFBFL_correct_sha512+='d8352bf19947168c9bcf1ae212eca035824293abb1f9'
  LFBFL_correct_sha512+='7a0ac02d67a82d098c34905c8cf1b213c891cc0'
  wrapped_wget_sha512

  LFBFL_file_name="xhtml1-transitional.dtd"
```

```
822   LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"
823   LFBFL_script_download_URL="${LFBFL_a}"
824   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
825   LFBFL_correct_sha512='4bece71fbf75d4a68a900ed29b8ba4efae1a9dfa447a5'
826   LFBFL_correct_sha512+='6fc8583da2531f9eadb0fb57bc4cf4990a82ac1da3b3'
827   LFBFL_correct_sha512+='c2892e38c88d70a80f8464b21f0bdef79a04843'
828   wrapped_wget_sha512
829
830   LFBFL_file_name="xhtml1-frameset.dtd"
831   LFBFL_a="https://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"
832   LFBFL_script_download_URL="${LFBFL_a}"
833   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
834   LFBFL_correct_sha512='f969c36d5ba7f6de42261c223f1a8958c8205cb938127'
835   LFBFL_correct_sha512+='ea5b7fce6bb2e03a529259e41aa3a429c64177893614'
836   LFBFL_correct_sha512+='f05e3f22bd9c46c61b37a83eb702a6d72ce2872'
837   wrapped_wget_sha512
838
839   LFBFL_file_name="xhtml11.dtd"
840   LFBFL_a="https://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"
841   LFBFL_script_download_URL="${LFBFL_a}"
842   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
843   LFBFL_correct_sha512='aa08bb274fd8c4d33bbcaf8993b2357c4e6c03fbdc34b'
844   LFBFL_correct_sha512+='1c46bb98332e21db824a11de43da9c6388032149a148'
845   LFBFL_correct_sha512+='42992c6d92e728e275817b656d1c333ca7f5d04'
846   wrapped_wget_sha512
847
848   LFBFL_file_name="xhtml-lat1.ent"
849   LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-lat1.ent"
850   LFBFL_script_download_URL="${LFBFL_a}"
851   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
852   LFBFL_correct_sha512='3243f33c056e71ae995d51cd2418979a505875d53ec9b'
853   LFBFL_correct_sha512+='17310268232dfbda19f5a52794973db47e51058cbcb4'
854   LFBFL_correct_sha512+='4a86838fed4937e6adc29636e12741bdc93af04'
855   wrapped_wget_sha512
856
857   LFBFL_file_name="xhtml-special.ent"
858   LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-special.ent"
859   LFBFL_script_download_URL="${LFBFL_a}"
860   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
861   LFBFL_correct_sha512='ec8e78745673044993b2d57242f834c94f1615a0df653'
862   LFBFL_correct_sha512+='5882b84c1df223f652ce536cc6b17eece23ab6dae518'
863   LFBFL_correct_sha512+='60a6e562b20f7ef5440004d56f9dbe4ea58913f'
864   wrapped_wget_sha512
865
866   LFBFL_file_name="xhtml-symbol.ent"
867   LFBFL_a="https://www.w3.org/MarkUp/DTD/xhtml-symbol.ent"
868   LFBFL_script_download_URL="${LFBFL_a}"
869   LFBFL_file_path="./${LFBFL_subdir3}/temp/${LFBFL_file_name}"
870   LFBFL_correct_sha512='47ca9edba10776c5ea8bba5c0e3ea8dc1a94b2830eada'
871   LFBFL_correct_sha512+='2ebf2bec1101ca05066ce09c0e06ecef1af7785c1179'
872   LFBFL_correct_sha512+='0a61c2efdc959d89610b7b4b084bb60af89cf72'
873   wrapped_wget_sha512
874
875   declare -r LFBFL_s_files_for_xmllint=$(
876     find . -type f -iregex ".*\.\(xml\|xhtml\)"\
877     | relevant_find\
878     | sort
879   )
880   if [[ -n "${LFBFL_s_files_for_xmllint}" ]]; then
881     LFBFL_xmllint_use_downloaded_DTDs=""
882     grep_variable "${LFBFL_data_file_path}" xmllint_use_downloaded_DTDs\
```

```
    --result-variable-prefix=LFBFL_

  declare -a LFBFL_arr_files_for_xmllint
  mapfile -t LFBFL_arr_files_for_xmllint\
    <<< "${LFBFL_s_files_for_xmllint}"
  readonly LFBFL_arr_files_for_xmllint
  local LFBFL_xmllint_files_without_DTD=""
  grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
    xmllint_files_without_DTD\
    --result-variable-prefix=LFBFL_
  declare -a LFBFL_arr_xmllint_files_without_DTD
  if [[ -n "${LFBFL_xmllint_files_without_DTD}" ]]; then
    mapfile -t LFBFL_arr_xmllint_files_without_DTD\
      <<< "${LFBFL_xmllint_files_without_DTD}"
  fi
  for LFBFL_file_path in "${LFBFL_arr_xmllint_files_without_DTD[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "xmllint on %s\n" "${LFBFL_file_path}"
    # see https://gitlab.gnome.org/GNOME/libxml2/-/issues/1100
    # https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1100
    # xmllint --pedantic --noout -- "${LFBFL_file_path}"
    xmllint --pedantic --noout "${LFBFL_file_path}"
  done

  local LFBFL_file_path2
  for LFBFL_file_path in "${LFBFL_arr_files_for_xmllint[@]}"; do
    for LFBFL_file_path2 in "${LFBFL_arr_xmllint_files_without_DTD[@]}";
    do
      if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
        LFBFL_file_path=""
      fi
    done
    if [[ "${LFBFL_file_path}" == "" ]]; then
      continue
    fi
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "xmllint on %s\n" "${LFBFL_file_path}"
    if [[
      "${LFBFL_file_path}" == *.xhtml
      && "${LFBFL_xmllint_use_downloaded_DTDs}" == "true"
    ]]; then
      if grep --fixed-strings\
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd"\
        "${LFBFL_file_path}"
      then
        xmllint --pedantic --noout\
          --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-strict.dtd"\
          "${LFBFL_file_path}"
      elif grep --fixed-strings\
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd"\
        "${LFBFL_file_path}"
      then
        xmllint --pedantic --noout\
          --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-transitional.dtd"\
          "${LFBFL_file_path}"
      elif grep --fixed-strings\
        "http://www.w3.org/TR/xhtml1/DTD/xhtml1-frameset.dtd"\
        "${LFBFL_file_path}"
      then
        xmllint --pedantic --noout\
          --dtdvalid "./${LFBFL_subdir3}/temp/xhtml1-frameset.dtd"\
```

```
            "${LFBFL_file_path}"
        elif grep --fixed-strings\
          "http://www.w3.org/TR/xhtml1/DTD/xhtml11.dtd"\
          "${LFBFL_file_path}"
        then
          xmllint --pedantic --noout\
            --dtdvalid "./${LFBFL_subdir3}/temp/xhtml11.dtd"\
            "${LFBFL_file_path}"
        else
          printf "Unknown XHTML DTD.\n"
        fi
      else
        xmllint --pedantic --noout --valid "${LFBFL_file_path}"
      fi
    done
  fi
  # ---------------------------------------------------------------------

  # ---------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_path}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix=LFBFL_

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_i_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_path}" vnu_jar_path\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_i_vnu_jar_ts=$(stat --format %Y -- "${LFBFL_vnu_jar_path}")
    LFBFL_i_current_ts=$(date +%s)
    ((LFBFL_i_current_ts -= LFBFL_i_vnu_jar_ts))
    ((
      LFBFL_i_current_ts -= LFBFL_upgrade_vnu_jar_time_interval_in_seconds
    ))
    if [[ LFBFL_i_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    # vnu has false positives for xhtml.
    local LFBFL_use_vnu_for_xhtml=""
    grep_variable "${LFBFL_data_file_path}" use_vnu_for_xhtml\
      --result-variable-prefix=LFBFL_
    local LFBFL_find_regexp
    if [[ "${LFBFL_use_vnu_for_xhtml}" == "true" ]]; then
      LFBFL_find_regexp=".*\.\(css\|html\|svg\|xhtml\)"
    else
      LFBFL_find_regexp=".*\.\(css\|html\|svg\)"
    fi
    declare -r LFBFL_s_files_for_Nu=$(
      find . -type f -iregex "${LFBFL_find_regexp}"\
```

```
      | relevant_find
    )
    if [[ -n "${LFBFL_s_files_for_Nu}" ]]; then
      declare -a LFBFL_arr_files_for_Nu
      mapfile -t LFBFL_arr_files_for_Nu <<< "${LFBFL_s_files_for_Nu}"
      readonly LFBFL_arr_files_for_Nu
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose -- "${LFBFL_arr_files_for_Nu[@]}"
    fi
  fi
  # ----------------------------------------------------------------------

  if [[ LFBFL_i_upgrade_venvs -eq 1 ]]; then
    touch -- "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

3.12 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

```

```
# RETURN traps handling
# Add the following lines at the start of a function.
#   declare -a LFBFL_return_traps_stack
#   init_return_trap

init_return_trap(){
  local LFBFL_previous_return_trap
  local LFBFL_previous_return_trap2
  LFBFL_previous_return_trap=$(trap -p RETURN)
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s previous_return_trap is: %s.\n"\
      "${LFBFL_where_was_i}"\
      "${LFBFL_previous_return_trap}"
  fi
  if [[ -z "${LFBFL_previous_return_trap}" ]]; then
    LFBFL_previous_return_trap="trap - RETURN"
  fi
  readonly LFBFL_previous_return_trap
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  LFBFL_previous_return_trap2="${LFBFL_function_depth}:"
  LFBFL_previous_return_trap2+="${LFBFL_previous_return_trap}"
  readonly LFBFL_previous_return_trap2
  LFBFL_return_traps_stack+=("${LFBFL_previous_return_trap2}")
  trap 'execute_return_traps' RETURN
}

execute_return_traps(){
  if [[ "${FUNCNAME[1]}" == "init_return_trap" ]]; then
    return
  fi
  # -----------------------------------------------------------------------
  # See https://lists.gnu.org/archive/html/help-bash/2026-03/msg00052.html
  if [[ "${FUNCNAME[1]}" == "source" ]]; then
    # Not Working
    return
  fi
  if [[ ${LFBFL_i_no_return_traps:-0} -eq 1 ]]; then
    return
  fi
  # -----------------------------------------------------------------------
  local LFBFL_some_return_trap
  local LFBFL_where_was_i
  get_where_was_i 2
  declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
  while [[ ${#LFBFL_return_traps_stack[@]} -gt 0 ]]; do
    LFBFL_some_return_trap="${LFBFL_return_traps_stack[-1]}"
    if [[ "${LFBFL_some_return_trap}" != ${LFBFL_function_depth}:* ]]; then
      # We only treat the top of the stack that should correspond to the
      # current function.
      break
    fi
    LFBFL_some_return_trap=${LFBFL_some_return_trap#*:}
    # shellcheck disable=SC2154,SC2309
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ ${#LFBFL_return_traps_stack[@]} -gt 1 ]]; then
        printf "%s executing some return trap: %s.\n"\
          "${LFBFL_where_was_i}"\
          "${LFBFL_some_return_trap}"
```

```
      else
        printf "%s setting back to previous_return_trap: %s.\n"\
          "${LFBFL_where_was_i}"\
          "${LFBFL_some_return_trap}"
      fi
    fi
    eval -- "${LFBFL_some_return_trap}"
    unset 'LFBFL_return_traps_stack[-1]'
  done
}


is_top_dirstack_directory(){
  # Altough "${DIRSTACK[0]}" may seem to be already a realpath,
  # possible use of symbolic links makes this function necessary.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needed to be "realpathed" already
  # but I removed this micro-optimization.
  # A compiled language could have a "realpathed property" on
  # "path object/string" to optimize when possible.
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath -- "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  local LFBFL_real_path2
  LFBFL_real_path2=$(realpath -- "$1")
  readonly LFBFL_real_path2
  [[ "${LFBFL_real_path}" == "${LFBFL_real_path2}" ]]
}

enhanced_pushd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory (needed to be sanitized with realpath before call)
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  # $1 can be any path now, see my comment on the micro-optimization in
  # is_top_dirstack_directory() just above.
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s.\n" "$1"
  # Value 0 will always be overwritten, it is superfluous.
  declare -gi i_enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    i_enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath -- "$1")
  readonly LFBFL_real_path
  # shellcheck disable=SC2154,SC2309
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd passed validation 1.\n"
  is_top_dirstack_directory "${LFBFL_real_path}" && {
    i_enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${i_enhanced_pushd_result}
```

```
163     }
164     # shellcheck disable=SC2154,SC2309
165     [[ LFBFL_i_verbose -eq 1 ]]\
166       && printf "enhanced_pushd passed validation 2.\n"
167
168     pushd -- "${LFBFL_real_path}" || {
169       i_enhanced_pushd_result=$?
170       local LFBFL_where_was_i
171       get_where_was_i "$2"
172       printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3"\
173         "${LFBFL_real_path}"
174       # shellcheck disable=SC2248
175       return ${i_enhanced_pushd_result}
176     }
177     # shellcheck disable=SC2154,SC2309
178     [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed.\n"
179     # shellcheck disable=SC2248
180     return ${i_enhanced_pushd_result}
181   }
182
183   enhanced_popd(){
184     # Auxiliary function needed below.
185     # See usages below if you want to reuse it.
186     # $1=to_directory
187     # $2=offset_for_where_was_i
188     # shellcheck disable=SC2154,SC2309
189     [[ LFBFL_i_verbose -eq 1 ]]\
190       && printf "enhanced_popd requested: %s.\n" "$1"
191     if [[ -z "$1" ]]; then
192       return 0
193     fi
194
195     declare -i LFBFL_i_popd_result
196     popd || {
197       LFBFL_i_popd_result=$?
198       readonly LFBFL_i_popd_result
199       local LFBFL_where_was_i
200       get_where_was_i "$2"
201       printf "%s popd failed.\n" "${LFBFL_where_was_i}"
202       # shellcheck disable=SC2248
203       return ${LFBFL_i_popd_result}
204     }
205     # shellcheck disable=SC2154,SC2309
206     [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed.\n"
207     return 0
208   }
209
210   can_continue_after_enhanced_pushd(){
211     [[
212       i_enhanced_pushd_result -eq 0
213       || i_enhanced_pushd_result -eq 110
214       || i_enhanced_pushd_result -eq 111
215     ]]
216   }
217
218   # Main usage for the next functions relative to verbose mode:
219   # Add the two following lines at the start of a function.
220   #   declare -i LFBFL_i_verbose=0
221   #   get_verbose_option "$@"
222   # /!\ You really need all of the two lines. /!\
223   # Otherwise, the call to get_verbose_option may impact a function above;
```

```
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

get_some_flag(){
  # $1=variable_name
  # $2=some_flag_long_name
  # $3=read_only 0 or 1 to make the variable readonly
  # This command is to be called in another one with same arguments on top
  # of the three previous.
  # This test:
  # if [[ "$*" == *--verbose* ]]; then
  # is faster than below.
  # But it is not bug free in a calling function,
  # and even incompatible with the current function where
  # the flag_long_name is a mandatory argument.
  printf -v "$1" "%d" 0
  declare -i LFBFL_i
  local LFBFL_argument
  for ((LFBFL_i = 4; LFBFL_i <= $#; ++LFBFL_i)); do
    LFBFL_argument="${!LFBFL_i}"
    if [[ "${LFBFL_argument}" == "$2" ]]; then
      printf -v "$1" "%d" 1
      break
    fi
  done
  if [[ $3 -eq 1 ]]; then
    readonly "$1"
  fi
}

get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  get_some_flag LFBFL_i_verbose --verbose 1 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
  fi
}


get_some_option(){
```

```
285   # $1=variable_name
286   # $2=some_option_long_name
287   # $3=option_default_value_if_flag_without_option_value_or_empty_value
288   # $4=option_default_value_if_nothing
289   # $5=read_only 0 or 1 to make the variable readonly
290   # This command is to be called in another one with same arguments on top
291   # of the four previous.
292   # If you truly have a need to distinguish:
293   # option_default_value_if_flag_without_option_value --thing
294   # option_default_value_if_flag_with_empty_option_value --thing= or =''
295   # then think twice before modifying this function.
296   printf -v "$1" "%s" "$4"
297   declare -i LFBFL_i
298   local LFBFL_argument
299   for ((LFBFL_i = 6; LFBFL_i <= $#; ++LFBFL_i)); do
300     LFBFL_argument="${!LFBFL_i}"
301     if [[ "${LFBFL_argument}" == "$2" ]]; then
302       printf -v "$1" "%s" "$3"
303       break
304     fi
305     if [[ "${LFBFL_argument}" == "$2"=* ]]; then
306       printf -v "$1" "%s" "${LFBFL_argument#"$2"=}"
307       printf -v "$1" "%s" "${!1:-$3}"
308       break
309     fi
310   done
311   if [[ $5 -eq 1 ]]; then
312     readonly "$1"
313   fi
314 }
315
316 keep_options(){
317   # $1=variable_name of array of options
318   # Other arguments: the options to keep.
319   declare -a LFBFL_array_of_filtered_options
320   declare -n LFBFL_array_of_options=$1
321   local LFBFL_some_argument
322   local LFBFL_some_argument2
323   for LFBFL_some_argument in "${LFBFL_array_of_options[@]}"; do
324     for LFBFL_some_argument2 in "$@"; do
325       if [[
326         "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"
327         || "${LFBFL_some_argument}" == "${LFBFL_some_argument2}"=*
328       ]]; then
329         LFBFL_array_of_filtered_options+=("${LFBFL_some_argument}")
330       fi
331     done
332   done
333   LFBFL_array_of_options=("${LFBFL_array_of_filtered_options[@]}")
334 }
335
336
337 # Main usage for the next functions relative to work_directory:
338 # Add the five following lines at the start of a function.
339 #   local LFBFL_work_directory=""
340 #   get_work_directory_option "$@"
341 #   pushd_to_work_directory\
342 #     && trap 'popd_from_work_directory' RETURN
343 #   can_continue_after_enhanced_pushd || return 1
344 # Or:
345 #   local LFBFL_work_directory=""
```

```
#   get_work_directory_option "$@"
#   pushd_to_work_directory --trap-popd
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  get_some_option LFBFL_work_directory --work-directory "." "." 0 "$@"
  LFBFL_work_directory=$(realpath -- "${LFBFL_work_directory}")
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
  # Options:
  #   --trap-popd
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
  if [[ i_enhanced_pushd_result -eq 0 ]]; then
    declare -i LFBFL_i_trap_popd
    get_some_flag LFBFL_i_trap_popd --trap-popd 1 "$@"
    if [[ LFBFL_i_trap_popd -eq 1 ]]; then
      declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
      local LFBFL_some_return_trap
      LFBFL_some_return_trap="${LFBFL_function_depth}:"
      LFBFL_some_return_trap+="popd_from_work_directory 4"
      readonly LFBFL_some_return_trap
      LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
    fi
  fi
  # shellcheck disable=SC2248
  return ${i_enhanced_pushd_result}
}

popd_from_work_directory(){
  # $1=offset for where_was_i
  enhanced_popd "${LFBFL_work_directory}" "${1:-3}"
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


get_offset_option(){
  # This command is to be called in another one with same arguments on top
  # of the default value.
  # $1=option_default_value
  # Options:
  #   --offset=2
  get_some_option LFBFL_i_offset --offset "$1" "$1" 1 "$@"
}

get_max_line_length_option(){
  # This command is to be called in another one with same arguments.
  # Its most useful feature is that it factorizes the default maximum line
  # length of 70 characters.
  # Options:
  #   --max-line-length
  get_some_option LFBFL_i_max_line_length --max-line-length 70 70 1 "$@"
}

```

```
407
408 # Functions relative to shell and bash options
409 # (See SHELLOPTS and BASHOPTS, set and shopt):
410 # - shell options usage, example with pipefail:
411 #     enhanced_set_shell_option pipefail\
412 #       && trap 'enhanced_unset_shell_option pipefail' RETURN
413 #   or maybe:
414 #     enhanced_unset_shell_option pipefail\
415 #       && trap 'enhanced_set_shell_option pipefail' RETURN
416 #   if you need to unset it in some function.
417 #   or better if you use my RETURN traps handler:
418 #     enhanced_set_shell_option pipefail --trap-unset
419 #     enhanced_unset_shell_option pipefail --trap-set
420 # - bash options usage, example with globstar:
421 #     enhanced_set_bash_option globstar\
422 #       && trap 'enhanced_unset_bash_option globstar' RETURN
423 #   or maybe:
424 #     enhanced_unset_bash_option globstar\
425 #       && trap 'enhanced_set_bash_option globstar' RETURN
426 #   or better if you use my RETURN traps handler:
427 #     enhanced_set_bash_option globstar --trap-unset
428 #     enhanced_unset_bash_option globstar --trap-set
429
430 enhanced_set_shell_option(){
431   # $1=optname some shell option for set
432   # Options:
433   #   --offset=2
434   #   --trap-unset
435   declare -r LFBFL_result_name="i_enhanced_set_shell_option_$1_result"
436   declare -gi "${LFBFL_result_name}"=0
437   if [[ -o "$1" ]]; then
438     printf -v "${LFBFL_result_name}" "%d" 1
439     # shellcheck disable=SC2086
440     return ${!LFBFL_result_name}
441   fi
442   set -o "$1"
443   declare -i LFBFL_i_offset=2
444   get_offset_option 2 "$@"
445   # shellcheck disable=SC2154,SC2309
446   if [[ LFBFL_i_verbose -eq 1 ]]; then
447     local LFBFL_where_was_i
448     # shellcheck disable=SC2248
449     get_where_was_i ${LFBFL_i_offset}
450     printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
451   fi
452   declare -i LFBFL_i_trap_unset
453   get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
454   if [[ LFBFL_i_trap_unset -eq 1 ]]; then
455     declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
456     local LFBFL_some_return_trap
457     LFBFL_some_return_trap="${LFBFL_function_depth}:"
458     LFBFL_some_return_trap+="enhanced_unset_shell_option '$1' --offset=3"
459     readonly LFBFL_some_return_trap
460     LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
461   fi
462   # shellcheck disable=SC2086
463   return ${!LFBFL_result_name}
464 }
465
466 enhanced_unset_shell_option(){
467   # $1=optname some shell option for set
```

```
468     # Options:
469     #   --offset=2
470     #   --trap-set
471     declare -r LFBFL_result_name="i_enhanced_unset_shell_option_$1_result"
472     declare -gi "${LFBFL_result_name}"=0
473     if [[ ! -o "$1" ]]; then
474       printf -v "${LFBFL_result_name}" "%d" 1
475       # shellcheck disable=SC2086
476       return ${!LFBFL_result_name}
477     fi
478     set +o "$1"
479     declare -i LFBFL_i_offset=2
480     get_offset_option 2 "$@"
481     # shellcheck disable=SC2154,SC2309
482     if [[ LFBFL_i_verbose -eq 1 ]]; then
483       local LFBFL_where_was_i
484       # shellcheck disable=SC2248
485       get_where_was_i ${LFBFL_i_offset}
486       printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
487     fi
488     declare -i LFBFL_i_trap_set
489     get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
490     if [[ LFBFL_i_trap_set -eq 1 ]]; then
491       declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
492       local LFBFL_some_return_trap
493       LFBFL_some_return_trap="${LFBFL_function_depth}:"
494       LFBFL_some_return_trap+="enhanced_set_shell_option '$1' --offset=3"
495       readonly LFBFL_some_return_trap
496       LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
497     fi
498     # shellcheck disable=SC2086
499     return ${!LFBFL_result_name}
500   }
501
502   enhanced_set_bash_option(){
503     # $1=optname some bash option for shopt
504     # Options:
505     #   --offset=2
506     #   --trap-unset
507     declare -r LFBFL_result_name="i_enhanced_set_bash_option_$1_result"
508     declare -gi "${LFBFL_result_name}"=0
509     # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
510     # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
511     if shopt -q "$1"; then
512       printf -v "${LFBFL_result_name}" "%d" 1
513       # shellcheck disable=SC2086
514       return ${!LFBFL_result_name}
515     fi
516     shopt -s "$1"
517     declare -i LFBFL_i_offset=2
518     get_offset_option 2 "$@"
519     # shellcheck disable=SC2154,SC2309
520     if [[ LFBFL_i_verbose -eq 1 ]]; then
521       local LFBFL_where_was_i
522       # shellcheck disable=SC2248
523       get_where_was_i ${LFBFL_i_offset}
524       printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
525     fi
526     declare -i LFBFL_i_trap_unset
527     get_some_flag LFBFL_i_trap_unset --trap-unset 1 "$@"
528     if [[ LFBFL_i_trap_unset -eq 1 ]]; then
```

```
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_unset_bash_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}

enhanced_unset_bash_option(){
  # $1=optname some bash option for shopt
  # Options:
  #   --offset=2
  #   --trap-set
  declare -r LFBFL_result_name="i_enhanced_unset_bash_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if ! shopt -q "$1"; then
    printf -v "${LFBFL_result_name}" "%d" 1
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  declare -i LFBFL_i_offset=2
  get_offset_option 2 "$@"
  # shellcheck disable=SC2154,SC2309
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    # shellcheck disable=SC2248
    get_where_was_i ${LFBFL_i_offset}
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  declare -i LFBFL_i_trap_set
  get_some_flag LFBFL_i_trap_set --trap-set 1 "$@"
  if [[ LFBFL_i_trap_set -eq 1 ]]; then
    declare -ir LFBFL_function_depth=$((${#FUNCNAME[@]} - 1))
    local LFBFL_some_return_trap
    LFBFL_some_return_trap="${LFBFL_function_depth}:"
    LFBFL_some_return_trap+="enhanced_set_bash_option '$1' --offset=3"
    readonly LFBFL_some_return_trap
    LFBFL_return_traps_stack+=("${LFBFL_some_return_trap}")
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

3.13 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
```

```
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
  # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
  # J'ai rerepensé au 3-way-merge
  # et à mon article sur le principe de première différence
  # appliqué à la largeur modulaire, de clique ou de rang avec
  # l'idée d'un principe de première différence ternaire.
  declare -i LFBFL_i_first=1
  local LFBFL_current
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    if [[ LFBFL_i_first -eq 1 ]]; then
      LFBFL_current="${LFBFL_argument}"
      LFBFL_i_first=0
      continue
    fi
    if [[ "${LFBFL_current}" != "${LFBFL_argument}" ]]; then
      return 0
    fi
  done
  return 1
}

some_distinct(){
  ! equal "$@"
}

all_distinct(){
  # Returns 1 if all the arguments strings are distinct.
  declare -ir LFBFL_ij_max=$#
  declare -i LFBFL_i
  declare -i LFBFL_j
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_ij_max; ++LFBFL_i)); do
    for ((LFBFL_j = LFBFL_i + 1; LFBFL_j <= LFBFL_ij_max; ++LFBFL_j)); do
      [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
    done
  done
  return 1
}

all_distinct2(){
  # Returns 1 if all the arguments strings are distinct.
  local LFBFL_element
  local LFBFL_previous_element
  if [[ $# -le 1 ]]; then
```

```
    return 1
  fi
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i = 1; LFBFL_i < LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
    | uniq --repeated
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

```

```
some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval -- "$1 --reverse"\
    | head --lines=1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=sort_command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval -- "$1"\
    | head --lines=1
}

is_substring(){
  # $1=string
  # $2=substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed --expression='s/\\/\\&/g' --expression="s/'/\\\\&/g"
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php --run="${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=file_path
  # $2=sub_file_path
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
```

```
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php --run="${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

3.14 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
50  declare -a LFBFL_return_traps_stack
51  init_return_trap
52  pushd_to_work_directory --trap-popd
53  can_continue_after_enhanced_pushd || return 1
54
55  enhanced_set_shell_option pipefail --trap-unset
56
57  declare -a LFBFL_common_options=("$@")
58  keep_options LFBFL_common_options --verbose
59
60  local LFBFL_subdir2="build_and_checks_variables"
61  local LFBFL_data_file_path="${LFBFL_subdir2}/repository_data.txt"
62  local LFBFL_repository_name=""
63  grep_variable "${LFBFL_data_file_path}" repository_name\
64    --result-variable-prefix="LFBFL_"\
65    --replace-line-returns-by=""
66
67  local LFBFL_repository_web_url=""
68  grep_variable "${LFBFL_data_file_path}" repository_web_url\
69    --result-variable-prefix="LFBFL_"\
70    --replace-line-returns-by=""
71
72  local LFBFL_repository_git_url=""
73  grep_variable "${LFBFL_data_file_path}" repository_git_url\
74    --result-variable-prefix="LFBFL_"\
75    --replace-line-returns-by=""
76
77  local LFBFL_abstract=""
78  grep_variable "${LFBFL_data_file_path}" abstract\
79    --result-variable-prefix="LFBFL_"
80
81  local LFBFL_acknowledgments=""
82  grep_variable "${LFBFL_data_file_path}" acknowledgments\
83    --result-variable-prefix="LFBFL_"
84
85  local LFBFL_author_email=""
86  grep_variable "${LFBFL_data_file_path}" author_email\
87    --result-variable-prefix="LFBFL_"\
88    --replace-line-returns-by=""
89
90  local LFBFL_author_full_name=""
91  grep_variable "${LFBFL_data_file_path}" author_full_name\
92    --result-variable-prefix="LFBFL_"\
93    --replace-line-returns-by=" "
94
95  local LFBFL_author_website=""
96  grep_variable "${LFBFL_data_file_path}" author_website\
97    --result-variable-prefix="LFBFL_"\
98    --replace-line-returns-by=""
99
100 declare -i LFBFL_i_max_line_length
101 grep_variable "${LFBFL_data_file_path}" max_line_length\
102   --result-variable-prefix="LFBFL_i_"
103
104 declare -r LFBFL_current_date=$(date --iso-8601="date")
105
106 declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)
107
108 declare -ir LFBFL_i_number_of_commits=$(
109   git shortlog\
110   | space_starting_lines\
```

```
    | wc --lines
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "${LFBFL_common_options[@]}"
    code_lines_count_not_empty "${LFBFL_common_options[@]}"
    code_lines_count_empty "${LFBFL_common_options[@]}"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${i_code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${i_code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  declare -r LFBFL_temp_path="${LFBFL_subdir2}/temp"

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree_light.txt"

  tree -a -Fh --timefmt='%F %R' --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree.txt.temp"

  local LFBFL_s_files_paths
  # Remove line returns that are here to keep lines short.
  LFBFL_s_files_paths=$(
    sed --regexp-extended --null-data\
      --expression='s/(.)\\\n/\1/Mg'\
      -- "${LFBFL_subdir2}/files_names_listing.txt"\
    | grep --invert-match '^//'
  )
  readonly LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  readonly LFBFL_arr_files_paths

  local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
  LFBFL_temp_files_listing+="files_listing.tex.sub.temp"
```

```
172     readonly LFBFL_temp_files_listing
173     : > "${LFBFL_temp_files_listing}"
174     local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
175     LFBFL_temp_files_listing2+="files_listing.html.sub.temp"
176     readonly LFBFL_temp_files_listing2
177     : > "${LFBFL_temp_files_listing2}"
178     # HTML <li> elements, hence "lis".
179     local LFBFL_temp_files_lis
180     LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.sub.temp"
181     readonly LFBFL_temp_files_lis
182     : > "${LFBFL_temp_files_lis}"
183     # shellcheck disable=SC2248
184     get_split_score_simple 1 ${LFBFL_i_max_line_length} /
185     declare -r LFBFL_score_command="${get_split_score_result}"
186     declare -i LFBFL_i_score_command_properties
187     LFBFL_i_score_command_properties=${i_get_split_score_result2}
188     readonly LFBFL_i_score_command_properties
189     # shellcheck disable=SC2248
190     get_split_score_simple 1 ${LFBFL_i_max_line_length} ':'
191     declare -r LFBFL_score_command2="${get_split_score_result}"
192     declare -i LFBFL_i_score_command_properties2
193     LFBFL_i_score_command_properties2=${i_get_split_score_result2}
194     readonly LFBFL_i_score_command_properties2
195     declare -r LFBFL_suffix='%'
196     declare -i LFBFL_i=0
197     local LFBFL_cleaned_path1
198     local LFBFL_cleaned_path2
199     local LFBFL_new_lines
200     local LFBFL_new_lines2
201     local LFBFL_new_lines3
202     local LFBFL_file_path
203     local LFBFL_s_format
204     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
205       printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
206       # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
207       # LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
208       LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
209       LFBFL_cleaned_path2=$(
210         printf "%s" "${LFBFL_file_path}"\
211         | sed --expression='s/\//:/g;s/\.//g'
212       )
213       LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
214       if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
215         # shellcheck disable=SC2248
216         repeated_split_last_line "${LFBFL_new_lines}"\
217           ""\
218           ${LFBFL_i_max_line_length}\
219           "${LFBFL_suffix}"\
220           "${LFBFL_score_command}"\
221           ${LFBFL_i_score_command_properties}\
222           "\\"
223         LFBFL_new_lines=${repeated_split_last_line_result}
224       fi
225       LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
226       if [[ ${#LFBFL_new_lines2} -gt LFBFL_i_max_line_length ]]; then
227         # shellcheck disable=SC2248
228         repeated_split_last_line "${LFBFL_new_lines2}"\
229           ""\
230           ${LFBFL_i_max_line_length}\
231           "${LFBFL_suffix}"\
232           "${LFBFL_score_command2}"\
```

```
233         ${LFBFL_i_score_command_properties2}
234       LFBFL_new_lines2=${repeated_split_last_line_result}
235     fi
236     LFBFL_new_lines3="${LFBFL_file_path}"
237     if [[ ${#LFBFL_new_lines3} -gt LFBFL_i_max_line_length ]]; then
238       # shellcheck disable=SC2248
239       repeated_split_last_line "${LFBFL_new_lines3}"\
240         ""\
241         ${LFBFL_i_max_line_length}\
242         "${LFBFL_suffix}"\
243         "${LFBFL_score_command}"\
244         ${LFBFL_i_score_command_properties}
245       LFBFL_new_lines3=${repeated_split_last_line_result}
246     fi
247     LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
248     LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
249     LFBFL_s_format+="{\n%s\n}\n\n\n"
250     # shellcheck disable=SC2059
251     printf "${LFBFL_s_format}"\
252       "${LFBFL_new_lines}"\
253       "${LFBFL_new_lines2}"\
254       "${LFBFL_new_lines3}"\
255       >> "${LFBFL_temp_files_listing}"
256 
257     # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
258     ((++LFBFL_i))
259     LFBFL_new_lines="${LFBFL_file_path}"
260     if [[ ${#LFBFL_file_path} -gt LFBFL_i_max_line_length ]]; then
261       # shellcheck disable=SC2248
262       repeated_split_last_line "${LFBFL_new_lines}"\
263         "-->"\
264         ${LFBFL_i_max_line_length}\
265         "<!--"\
266         "${LFBFL_score_command}"\
267         ${LFBFL_i_score_command_properties}
268       LFBFL_new_lines=${repeated_split_last_line_result}
269     fi
270     {
271       LFBFL_s_format="<h3 id=\"subsection3.%s\">3.%s\n%s\n</h3>\n"
272       LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
273       # shellcheck disable=SC2059
274       printf "${LFBFL_s_format}"\
275         "${LFBFL_i}"\
276         "${LFBFL_i}"\
277         "${LFBFL_new_lines}"
278       sed --expression='s/&/\&amp;/g;s/</\&lt;/g'\
279         < "${LFBFL_file_path}"
280       printf "</pre>\n\n\n"
281     } >> "${LFBFL_temp_files_listing2}"
282     LFBFL_s_format="      <li><a href=\"#subsection3.%s\">\n%s\n"
283     LFBFL_s_format+="      </a></li>\n"
284     # shellcheck disable=SC2059
285     printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
286       >> "${LFBFL_temp_files_lis}"
287   done
288   overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.tex.sub"\
289     "${LFBFL_temp_files_listing}"
290   # Example of generated content of files_listing.tex.sub:
291   # ------------------------------------------------------------------------
292   # \subsection{
293   #   build\_and\_checks\_dependencies/licenses\_templates/%
```

```
    # build\_licenses\_templates.exec.sh
    # }
    # \label{
    #   build_and_checks_dependencies:licenses_templates:%
    # build_licenses_templatesexecsh
    # }
    #
    # \VerbatimInput[numbers=left,xleftmargin=-5mm]{
    # build_and_checks_dependencies/licenses_templates/%
    # build_licenses_templates.exec.sh
    # }
    # ------------------------------------------------------------------------
    sed --in-place --expression="s|\\n</pre>|</pre>|g"\
      -- "${LFBFL_temp_files_listing2}"
    overwrite_if_not_equal "${LFBFL_temp_path}/files_listing.html.sub"\
      "${LFBFL_temp_files_listing2}"
    # Example of generated content of files_listing.html.sub:
    # ------------------------------------------------------------------------
    # <h3 id="subsection3.15">3.15
    # build_and_checks_dependencies/licenses_templates/<!--
    # -->build_licenses_templates.exec.sh
    # </h3>
    # ...
    # <h3 id="subsection3.33">3.33
    # build_and_checks_variables/.gitignore
    # </h3>
    # <pre class="numbered_lines">
    # temp/
    # upgrade_venvs_ts
    # </pre>
    # ------------------------------------------------------------------------
    overwrite_if_not_equal "${LFBFL_temp_path}/files_lis.html.sub"\
      "${LFBFL_temp_files_lis}"
    # Example of generated content of files_lis.html.sub:
    # ------------------------------------------------------------------------
    #       <li><a href="#subsection3.15">
    # build_and_checks_dependencies/licenses_templates/<!--
    # -->build_licenses_templates.exec.sh
    #       </a></li>
    # ------------------------------------------------------------------------

    # shellcheck disable=SC2248
    get_overlength_regexp ${LFBFL_i_max_line_length}

    # We verify if some lines are beyond max_line_length characters
    # in current_tree_light.txt and current_tree.txt.
    declare -ar LFBFL_trees=(
      "current_tree_light.txt"
      "current_tree.txt.temp"
    )
    local LFBFL_tree
    local LFBFL_tree_path
    local LFBFL_line
    local LFBFL_prefix
    declare -i LFBFL_i_prefix_last_position
    local LFBFL_char
    local LFBFL_line_start
    local LFBFL_file_name
    local LFBFL_s_lines
    declare -a LFBFL_arr_lines
    declare -i LFBFL_i_keep_length
```

```
355   for LFBFL_tree in "${LFBFL_trees[@]}"; do
356     LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
357     # shellcheck disable=SC2248
358     LFBFL_s_lines=$(
359       grep --regexp="${get_overlength_regexp_result}" --\
360         "${LFBFL_tree_path}"
361     )
362     if [[ -z "${LFBFL_s_lines}" ]]; then
363       continue
364     fi
365     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
366     for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
367       # printf "LFBFL_line: %s\n" "${LFBFL_line}"
368       LFBFL_prefix=$(
369         printf "%s" "${LFBFL_line}"\
370         | sed --regexp-extended --expression='s/(.*)──[^─]+$/\1/g'
371       )
372       # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
373       LFBFL_i_prefix_last_position=$((${#LFBFL_prefix}-1))
374       LFBFL_char="${LFBFL_prefix:${LFBFL_i_prefix_last_position}:1}"
375       # printf "LFBFL_char: %s\n" "${LFBFL_char}"
376       LFBFL_prefix=$(
377         printf "%s" "${LFBFL_prefix}"\
378         | sed --regexp-extended --expression='s/[^ ]+$//g'
379       )
380       if [[ "${LFBFL_char}" == "└" ]]; then
381         LFBFL_prefix+="  "
382       else
383         LFBFL_prefix+="│ "
384       fi
385       # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
386       LFBFL_file_name=$(
387         printf "%s" "${LFBFL_line}"\
388         | sed --regexp-extended --expression='s|.* ([^ ]+)$|\1|g'
389       )
390       # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
391       LFBFL_i_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
392       LFBFL_line_start=${LFBFL_line:0:${LFBFL_i_keep_length}}
393       LFBFL_line_start=$(
394         printf "%s" "${LFBFL_line_start}"\
395         | sed --expression='s/ *$//g'
396       )
397       # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
398       LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
399       if [[ ${#LFBFL_new_lines} -gt LFBFL_i_max_line_length ]]; then
400         # shellcheck disable=SC2248
401         repeated_split_last_line "${LFBFL_new_lines}"\
402           "${LFBFL_prefix}"\
403           ${LFBFL_i_max_line_length}\
404           ""\
405           ""\
406           ""
407         LFBFL_new_lines=${repeated_split_last_line_result}
408       fi
409       printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
410         > "${LFBFL_tree_path}.some_line.temp"
411       insert_file_at_token "${LFBFL_tree_path}"\
412         "${LFBFL_line}"\
413         "${LFBFL_tree_path}.some_line.temp"\
414         ""\
415         --quiet
```

```
    done
    rm --force -- "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" --keep-temp --tree-mode

  local LFBFL_tex_path_start
  LFBFL_tex_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
  readonly LFBFL_tex_path_start
  local LFBFL_html_path_start
  LFBFL_html_path_start="${LFBFL_temp_path}/${LFBFL_repository_name}.html"
  readonly LFBFL_html_path_start

  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]]; then
    # If there is a template, we init the process from it.
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]]; then
    # Otherwise if there is a tex file, we init the process from it.
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/.\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]]; then
    cp -- "${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "${LFBFL_repository_name}.html" ]]; then
    cp -- "${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./.\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed --expression='s/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed --expression='s/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed --in-place\
      --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
      --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
      --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
      --expression="s|@author_email@|${LFBFL_author_email}|g"\
      --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
      --expression="s|@author_website@|${LFBFL_author_website}|g"\
      --expression="s|@current_date@|${LFBFL_current_date}|g"\
      --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
```

```
477       --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
478       --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
479       -- "${LFBFL_tex_path_start}.1"
480
481     insert_file_at_token "${LFBFL_tex_path_start}.1"\
482       @abstract@\
483       "${LFBFL_temp_path}/abstract_temp"\
484       "${LFBFL_tex_path_start}.2"
485
486     insert_file_at_token "${LFBFL_tex_path_start}.2"\
487       @acknowledgments@\
488       "${LFBFL_temp_path}/acknowledgments_temp"\
489       "${LFBFL_tex_path_start}.3"
490
491     insert_file_at_token "${LFBFL_tex_path_start}.3"\
492       @current_tree_light@\
493       "${LFBFL_temp_path}/current_tree_light.txt"\
494       "${LFBFL_tex_path_start}.4"
495
496     insert_file_at_token "${LFBFL_tex_path_start}.4"\
497       @current_tree@\
498       "${LFBFL_temp_path}/current_tree.txt"\
499       "${LFBFL_tex_path_start}.5"
500
501     insert_file_at_token "${LFBFL_tex_path_start}.5"\
502       @dependencies_notes@\
503       "${LFBFL_temp_path}/dependencies_notes.tex.sub"\
504       "${LFBFL_tex_path_start}.6"
505
506     insert_file_at_token "${LFBFL_tex_path_start}.6"\
507       @files_listing_VerbatimInput@\
508       "${LFBFL_temp_path}/files_listing.tex.sub"\
509       "${LFBFL_tex_path_start}.7"
510
511     overwrite_if_not_equal\
512       "${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
513       "${LFBFL_tex_path_start}.7" --keep-temp
514   fi
515
516   # HTML filling:
517   declare -i LFBFL_i_HTML_updated=0
518   if [[ -f "${LFBFL_html_path_start}.1" ]]; then
519     sed --in-place\
520       --expression="s|@repository_name@|${LFBFL_repository_name}|g"\
521       --expression="s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
522       --expression="s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
523       --expression="s|@author_email@|${LFBFL_author_email}|g"\
524       --expression="s|@author_full_name@|${LFBFL_author_full_name}|g"\
525       --expression="s|@author_website@|${LFBFL_author_website}|g"\
526       --expression="s|@current_date@|${LFBFL_current_date}|g"\
527       --expression="s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
528       --expression="s|@number_of_commits@|${LFBFL_i_number_of_commits}|g"\
529       --expression="s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
530       -- "${LFBFL_html_path_start}.1"
531
532     insert_file_at_token "${LFBFL_html_path_start}.1"\
533       @abstract@\
534       "${LFBFL_temp_path}/abstract_temp"\
535       "${LFBFL_html_path_start}.2"
536
537     insert_file_at_token "${LFBFL_html_path_start}.2"\
```

```
538       @acknowledgments@\
539       "${LFBFL_temp_path}/acknowledgments_temp"\
540       "${LFBFL_html_path_start}.3"
541
542     insert_file_at_token "${LFBFL_html_path_start}.3"\
543       @files_lis@\
544       "${LFBFL_temp_path}/files_lis.html.sub"\
545       "${LFBFL_html_path_start}.4"
546
547     insert_file_at_token "${LFBFL_html_path_start}.4"\
548       @current_tree_light@\
549       "${LFBFL_temp_path}/current_tree_light.txt"\
550       "${LFBFL_html_path_start}.5"
551
552     insert_file_at_token "${LFBFL_html_path_start}.5"\
553       @current_tree@\
554       "${LFBFL_temp_path}/current_tree.txt"\
555       "${LFBFL_html_path_start}.6"
556
557     insert_file_at_token "${LFBFL_html_path_start}.6"\
558       @dependencies_notes@\
559       "${LFBFL_temp_path}/dependencies_notes.html.sub"\
560       "${LFBFL_html_path_start}.7"
561
562     insert_file_at_token "${LFBFL_html_path_start}.7"\
563       @files_listing_HTMLPreInput@\
564       "${LFBFL_temp_path}/files_listing.html.sub"\
565       "${LFBFL_html_path_start}.8"
566
567     overwrite_if_not_equal "${LFBFL_repository_name}.html"\
568       "${LFBFL_html_path_start}.8" --keep-temp
569     LFBFL_i_HTML_updated=$?
570   fi
571
572   # The HTML contains everything including listings with source code.
573   # If it doesn't get updated,
574   # then it is useless to recompute the PDF.
575   if [[ LFBFL_i_HTML_updated -eq 0 ]]; then
576     return
577   fi
578
579   local LFBFL_latex_command=""
580   grep_variable "${LFBFL_data_file_path}" latex_command\
581     --result-variable-prefix="LFBFL_"\
582     --replace-line-returns-by=""
583
584   case ${LFBFL_latex_command} in
585     pdflatex) : ;;
586     lualatex) : ;;
587     *) printf "Unsupported latex command" ; return 1 ;;
588   esac
589
590   declare -a LFBFL_arr_commands
591   local LFBFL_command
592
593   local LFBFL_latex_targets=""
594   grep_variable "${LFBFL_data_file_path}" latex_targets\
595     --result-variable-prefix="LFBFL_"
596   declare -a LFBFL_arr_latex_targets
597   if [[ -z "${LFBFL_latex_targets}" ]]; then
598     LFBFL_command="${LFBFL_latex_command}"
```

```
    LFBFL_command+=" '${LFBFL_subdir2}/${LFBFL_repository_name}.tex'"
    if [[ LFBFL_i_verbose -eq 0 ]]; then
      LFBFL_command+=" > /dev/null"
    fi
    LFBFL_arr_commands+=("${LFBFL_command}")
  else
    mapfile -t LFBFL_arr_latex_targets <<< "${LFBFL_latex_targets}"
    local LFBFL_latex_target
    local LFBFL_jobname_suffix
    local LFBFL_buildtarget
    for LFBFL_latex_target in "${LFBFL_arr_latex_targets[@]}"; do
      LFBFL_jobname_suffix=${LFBFL_latex_target%:*}
      LFBFL_buildtarget=${LFBFL_latex_target##*:}
      LFBFL_command="${LFBFL_latex_command}"
      LFBFL_command+=" --jobname="
      LFBFL_command+="'${LFBFL_repository_name}-${LFBFL_jobname_suffix}'"
      LFBFL_command+=" '\def\buildtarget{${LFBFL_buildtarget}}"
      LFBFL_command+=" \input{"
      LFBFL_command+="${LFBFL_subdir2}/${LFBFL_repository_name}.tex}'"
      if [[ LFBFL_i_verbose -eq 0 ]]; then
        LFBFL_command+=" > /dev/null"
      fi
      LFBFL_arr_commands+=("${LFBFL_command}")
    done
  fi

  for LFBFL_command in "${LFBFL_arr_commands[@]}"; do
    for ((LFBFL_i = 0; LFBFL_i < 3; ++LFBFL_i)); do
      eval -- "${LFBFL_command}"
    done
  done

  # LFBFL_files_to_temp=(\
  #   "${LFBFL_repository_name}.aux"\
  #   "${LFBFL_repository_name}.log"\
  #   "${LFBFL_repository_name}.out"\
  #   "${LFBFL_repository_name}.toc"
  # )
  # for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
  #   mv -- "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  # done

  LFBFL_files_suffix_to_temp=(\
    ".aux"\
    ".log"\
    ".out"\
    ".toc"
  )
  for LFBFL_file_suffix in "${LFBFL_files_suffix_to_temp[@]}"; do
    mv -- "${LFBFL_repository_name}"*"${LFBFL_file_suffix}"\
      "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

3.15 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
```

```
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat -- "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_path
  # $2=target_file_path
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    printf "%s\n" "$5"\
      > "${LFBFL_temp}"
  fi
  sed --expression="s~^~$3~g" -- "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
```

```
    eval -- "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_part1_of_file
  # $4=path_of_part2_of_file
  # We cannot add --line-regexp with --fixed-strings because of the
  # possible indentation in file,
  # instead we use grep_fixed_string_with_anchor.
  declare -ir LFBFL_i_line_number=$(
    grep_fixed_string_with_anchor "$1" "$2"\
      --enforce-line-ends-with-fixed-string --line-number\
    | head --lines=1\
    | cut --delimiter=':' --fields=1
  )
  declare -ir LFBFL_i_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_i_lines_after=$((
    LFBFL_i_line_count - LFBFL_i_line_number
  ))
  head --lines="$((LFBFL_i_line_number - 1))" -- "$1"\
    > "$3"
  tail --lines="${LFBFL_i_lines_after}" -- "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=file_path
  # $2=token assume that token is the only thing on his line,
  #    except for a possible indentation.
  # $3=path_of_file_to_insert
  # $4=path_of_result_file
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_path="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_path="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_path
  if [[ -n "$4" ]]; then
    LFBFL_result_file_path="$4"
  else
    LFBFL_result_file_path="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_path
  split_file_in_two "$1" "$2" "${LFBFL_start_file_path}"\
    "${LFBFL_end_file_path}"
  cat -- "${LFBFL_start_file_path}" "$3" "${LFBFL_end_file_path}"\
    > "${LFBFL_result_file_path}"
  rm -- "${LFBFL_start_file_path}" "${LFBFL_end_file_path}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_path}" "$5"
  fi
}
```

3.16 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
```

```
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.neon'
    '**/*.php'
    '**/*.py'
    '**/*.rs'
    '**/*.rlib'
    '**/*.sh'
    '**/*.sql'
    '**/*.sub'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.xslt'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
```

```
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
    '*.neon'
    '*.php'
    '*.py'
    '*.rs'
    '*.rlib'
    '*.sh'
    '*.sql'
    '*.sub'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.xslt'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

3.17 build_and_checks_dependencies/grammar_and_spelling_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2026-03-22T15:30+01:00: This file was renamed from
# "grammar_and_spell_check.libr.sh"
# to
# "grammar_and_spelling_check.libr.sh".
# This file was renamed from "grammar_and_spell_check.sh"
# to "grammar_and_spell_check.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

grammar_and_spelling_check(){
  # $1=configuration_path build_and_checks_variables/repository_data.txt
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_grammar_and_spelling_checker_command=""
  grep_variable "$1" grammar_and_spelling_checker_command\
    --result-variable-prefix=LFBFL_\
    --replace-line-returns-by=""
  declare -r LFBFL_command="${LFBFL_grammar_and_spelling_checker_command}"
  if [[ -z "${LFBFL_command}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: No command in configuration.\n"
    fi
    return
  fi
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "grammar_and_spelling_check: Command found: %s.\n"\
      "${LFBFL_command}"
  fi
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  local LFBFL_pattern
  local LFBFL_eval_string
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
```

```
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "grammar_and_spelling_check: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    LFBFL_s_files_paths=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spelling_check: No file found.\n"
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval -- "${LFBFL_eval_string}"
    done
  done
}
```

3.18 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
```

```
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ i_enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

build_licenses_templates(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap

  enhanced_set_shell_option pipefail --trap-unset

  local LFBFL_s_format

  # Source
  local LFBFL_license_source_subdir
  LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
  readonly LFBFL_license_source_subdir
  local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
  LFBFL_source_license_prefix+="license_file_header_"
  readonly LFBFL_source_license_prefix
  # Target
  local LFBFL_license_target_subdir
  LFBFL_license_target_subdir="${LFBFL_work_directory}/"
  LFBFL_license_target_subdir="build_and_checks_variables/temp/"
  readonly LFBFL_license_target_subdir
  local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
  LFBFL_target_license_prefix+="license_file_header_"
  readonly LFBFL_target_license_prefix

  declare -ar LFBFL_licenses=(
    "AGPLv3+"
    "GPLv3+"
    "LGPLv3+"
  )

  # ----------------------------------------------------------------------
  # Various language definitions for text file formats
  declare -ar LFBFL_block_comment_languages=(
    "c"
    "py"
    "xml"
  )
  declare -ar LFBFL_block_comment_enters=(
    '/*'
    '"""'
    '<!--'
  )
  declare -ar LFBFL_block_comment_exits=(
    '*/'
    '"""'
```

```
100       '-->'
101     )
102     equal "${#LFBFL_block_comment_languages[@]}"\
103           "${#LFBFL_block_comment_enters[@]}"\
104           "${#LFBFL_block_comment_exits[@]}"
105     declare -i LFBFL_i_equal_result=$?
106     if [[ LFBFL_i_equal_result -eq 0 ]]; then
107       printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
108       return 1
109     fi
110
111     declare -ar LFBFL_line_comment_languages=(
112       "rs"
113       "sh"
114       "sql"
115       "tex"
116     )
117     declare -ar LFBFL_line_comment_prefixes=(
118       '// '
119       '# '
120       '-- '
121       '% '
122     )
123     equal "${#LFBFL_line_comment_languages[@]}"\
124           "${#LFBFL_line_comment_prefixes[@]}"
125     LFBFL_i_equal_result=$?
126     if [[ LFBFL_i_equal_result -eq 0 ]]; then
127       printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
128       return 1
129     fi
130     # ----------------------------------------------------------------------
131
132     # ----------------------------------------------------------------------
133     # We generate the license header templates adapted to various text
134     # files formats.
135     # Kamoulox do endfor, Jacques Beauregard XD
136     # (C'est toujours mieux que l'inverse :) !)
137     # for do endfor -> for do done (Fort Doux Donne).
138     local LFBFL_some_license
139     local LFBFL_license_prefix
140     local LFBFL_license_prefix2
141     local LFBFL_extension
142     local LFBFL_license_file_path
143     local LFBFL_enter_string
144     local LFBFL_exit_string
145     local LFBFL_temp2
146     local LFBFL_prefix_string
147     local LFBFL_intermediate_file_path
148     local LFBFL_file_prefix
149     declare -i LFBFL_i_generate_from_template_result
150     declare -i LFBFL_i
151     for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
152       LFBFL_license_prefix="${LFBFL_source_license_prefix}"
153       LFBFL_license_prefix+="${LFBFL_some_license}"
154       LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
155       LFBFL_license_prefix2+="${LFBFL_some_license}"
156       for ((
157         LFBFL_i = 0;
158         LFBFL_i < ${#LFBFL_block_comment_languages[@]};
159         ++LFBFL_i
160       )); do
```

```
      LFBFL_extension=${LFBFL_block_comment_languages[LFBFL_i]}
      LFBFL_license_file_path="${LFBFL_license_prefix2}"
      LFBFL_license_file_path+=".${LFBFL_extension}"
      LFBFL_enter_string=${LFBFL_block_comment_enters[LFBFL_i]}
      LFBFL_exit_string=${LFBFL_block_comment_exits[LFBFL_i]}
      generate_from_template_with_block_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_path}.tpl"\
        "${LFBFL_enter_string}" "${LFBFL_exit_string}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_path}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_path}"
        fi
      fi
    done
    LFBFL_temp2=".generate_from_template_with_line_comments.temp"
    for ((
      LFBFL_i = 0;
      LFBFL_i < ${#LFBFL_line_comment_languages[@]};
      ++LFBFL_i
    )); do
      LFBFL_extension=${LFBFL_line_comment_languages[LFBFL_i]}
      LFBFL_license_file_path="${LFBFL_license_prefix2}"
      LFBFL_license_file_path+=".${LFBFL_extension}"
      LFBFL_prefix_string=${LFBFL_line_comment_prefixes[LFBFL_i]}
      LFBFL_intermediate_file_path="${LFBFL_license_file_path}.tpl"
      LFBFL_intermediate_file_path+="${LFBFL_temp2}"
      LFBFL_file_prefix=""
      if [[ "${LFBFL_extension}" == "sh" ]]; then
        LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
      fi
      # The sed command is here to remove trailing spaces of previously
      # empty lines after comment prefix is added to the line.
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_path}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_path}'"\
        "${LFBFL_file_prefix}"
      LFBFL_i_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_i_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_path}"
        elif [[ LFBFL_i_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_path}"
        fi
      fi
    done
  done
  # -----------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in work directory, we can go there.
  pushd_to_work_directory --trap-popd
```

```
222   can_continue_after_enhanced_pushd || return 1
223
224   local LFBFL_data_file_path="build_and_checks_variables/"
225   LFBFL_data_file_path+="repository_data.txt"
226   readonly LFBFL_data_file_path
227   local LFBFL_repository_name=""
228   grep_variable "${LFBFL_data_file_path}" repository_name\
229     --result-variable-prefix="LFBFL_"\
230     --replace-line-returns-by=""
231   local LFBFL_license=""
232   grep_variable "${LFBFL_data_file_path}" license\
233     --result-variable-prefix="LFBFL_"
234   local LFBFL_license2=""
235   grep_variable "${LFBFL_data_file_path}" license2\
236     --result-variable-prefix="LFBFL_"
237   local LFBFL_author_full_name=""
238   grep_variable "${LFBFL_data_file_path}" author_full_name\
239     --result-variable-prefix="LFBFL_"\
240     --replace-line-returns-by=" "
241
242   LFBFL_license_prefix="build_and_checks_variables/temp/"
243   LFBFL_license_prefix+="license_file_header_"
244   readonly LFBFL_license_prefix
245   LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
246   readonly LFBFL_license_prefix2
247   local LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"
248   readonly LFBFL_license_prefix3
249
250   # First year according to current state of git repository.
251   declare -ir LFBFL_i_first_year=$(
252     git log\
253     | grep 'Date:'\
254     | tail --lines=1\
255     | cut --delimiter=' ' --fields=8
256   )
257   # Last year according to current state of git repository.
258   declare -ir LFBFL_i_last_year=$(
259     git log\
260     | grep 'Date:'\
261     | head --lines=1\
262     | cut --delimiter=' ' --fields=8
263   )
264   local LFBFL_copyright_string
265   LFBFL_copyright_string="${LFBFL_i_first_year}-${LFBFL_i_last_year}"
266   LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
267   readonly LFBFL_copyright_string
268
269   LFBFL_files_without_license_header=""
270   grep_variable_with_multiple_files "${LFBFL_data_file_path}"\
271     files_without_license_header\
272     --result-variable-prefix="LFBFL_"
273   declare -a LFBFL_arr_files_without_license_header
274   if [[ -n "${LFBFL_files_without_license_header}" ]]; then
275     mapfile -t LFBFL_arr_files_without_license_header\
276       <<< "${LFBFL_files_without_license_header}"
277   fi
278
279   prepare_filled_license_file_block(){
280     # $1=license_file_path
281     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
282         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
```

```
283         -- "$1.tpl"\
284       > "$1.temp"
285     overwrite_if_not_equal "$1" "$1.temp"
286     declare -ir LFBFL_i_result=$?
287     if [[ LFBFL_i_verbose -eq 1 ]]; then
288       if [[ LFBFL_i_result -eq 1 ]]; then
289         printf "License template %s updated.\n" "$1"
290       elif [[ LFBFL_i_result -eq 2 ]]; then
291         printf "License template %s created.\n" "$1"
292       fi
293     fi
294   }
295
296   prepare_filled_license_file_block_post(){
297     # $1=license_file_path
298     head --lines=-1 -- "$1"\
299       | tail --lines=+2\
300       > "$1.temp"
301     # Strange world: tail --lines=-1 was working
302     # to exclude first line.
303     # I tested it months ago, and saw with my eyes the start and end
304     # block comments removed with:
305     # head --lines=-1 "$1" | tail --lines=-1.
306     # Now, man page says to use --lines=+2 instead.
307     # Welcome in the world where AI change your environment and your
308     # documentation on the fly...
309   }
310
311   declare -Ar LFBFL_all_block_comment_languages=(
312     ["c"]="c"
313     ["css"]="c"
314     ["html"]="xml"
315     ["php"]="c"
316     ["py"]="py"
317     ["js"]="c"
318     ["ts"]="c"
319     ["xml"]="xml"
320   )
321
322   local LFBFL_key
323   local LFBFL_dest
324   local LFBFL_license_file_path2
325   declare -i LFBFL_i_not_subfile
326   declare -i LFBFL_i_not_subfile2
327   local LFBFL_file_path
328   local LFBFL_file_path2
329   local LFBFL_s_files_paths
330   declare -a LFBFL_arr_files_paths
331   declare -i LFBFL_i_is_whitelisted
332   for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
333     LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
334     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
335     prepare_filled_license_file_block "${LFBFL_license_file_path}"
336     if [[ -n "${LFBFL_license2}" ]]; then
337       LFBFL_license_file_path2="${LFBFL_license_prefix3}"
338       LFBFL_license_file_path2+=".${LFBFL_dest}"
339       prepare_filled_license_file_block "${LFBFL_license_file_path2}"
340     fi
341
342     LFBFL_s_files_paths=$(
343       find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
```

```
344         -printf '%P\n'\
345       | relevant_find
346     )
347     if [[ -z "${LFBFL_s_files_paths}" ]]; then
348       if [[ LFBFL_i_verbose -eq 1 ]]; then
349         LFBFL_s_format="build_licenses_templates: "
350         LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
351         # shellcheck disable=SC2059
352         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
353       fi
354       continue
355     fi
356     prepare_filled_license_file_block_post "${LFBFL_license_file_path}"
357     if [[ -n "${LFBFL_license2}" ]]; then
358       prepare_filled_license_file_block_post "${LFBFL_license_file_path2}"
359     fi
360     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
361     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
362       LFBFL_i_is_whitelisted=0
363       for LFBFL_file_path2 in\
364         "${LFBFL_arr_files_without_license_header[@]}";
365       do
366         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
367           LFBFL_i_is_whitelisted=1
368         fi
369       done
370       if [[ LFBFL_i_is_whitelisted -eq 1 ]]; then
371         continue
372       fi
373       is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}.temp"
374       LFBFL_i_not_subfile=$?
375       LFBFL_i_not_subfile2=1
376       if [[ -n "${LFBFL_license2}" ]]; then
377         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}.temp"
378         LFBFL_i_not_subfile2=$?
379       fi
380       if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
381       then
382         printf "File %s has no/wrong license header?\n"\
383           "${LFBFL_file_path}"
384       fi
385     done
386     rm -- "${LFBFL_license_file_path}.temp"
387     if [[ -n "${LFBFL_license2}" ]]; then
388       rm -- "${LFBFL_license_file_path2}.temp"
389     fi
390   done
391
392   prepare_filled_license_file_line(){
393     # $1=LFBFL_license_file_path
394     sed --expression="s/@repository_name@/${LFBFL_repository_name}/g"\
395         --expression="s/@copyright_string@/${LFBFL_copyright_string}/g"\
396         -- "$1.tpl"\
397       > "$1.temp"
398     overwrite_if_not_equal "$1" "$1.temp"
399     declare -ir LFBFL_i_result=$?
400     if [[ LFBFL_i_verbose -eq 1 ]]; then
401       if [[ LFBFL_i_result -eq 1 ]]; then
402         printf "License template %s updated.\n" "$1"
403       elif [[ LFBFL_i_result -eq 2 ]]; then
404         printf "License template %s created.\n" "$1"
```

```
405       fi
406     fi
407   }
408
409   declare -Ar LFBFL_all_line_comment_languages=(
410     ["rlib"]="rs"
411     ["rs"]="rs"
412     ["sh"]="sh"
413     ["sql"]="sql"
414     ["tex"]="tex"
415     ["toml"]="sh"
416     ["yaml"]="sh"
417   )
418
419   for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
420     LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
421     LFBFL_license_file_path="${LFBFL_license_prefix2}.${LFBFL_dest}"
422     prepare_filled_license_file_line "${LFBFL_license_file_path}"
423     if [[ -n "${LFBFL_license2}" ]]; then
424       LFBFL_license_file_path2="${LFBFL_license_prefix3}"
425       LFBFL_license_file_path2+=".${LFBFL_dest}"
426       prepare_filled_license_file_line "${LFBFL_license_file_path2}"
427     fi
428
429     LFBFL_s_files_paths=$(
430       find . -type f -iregex ".*\.\(${LFBFL_key}\|${LFBFL_key}\.tpl\)"\
431         -printf '%P\n'\
432       | relevant_find
433     )
434     if [[ -z "${LFBFL_s_files_paths}" ]]; then
435       if [[ LFBFL_i_verbose -eq 1 ]]; then
436         LFBFL_s_format="build_licenses_templates: "
437         LFBFL_s_format+="No file found with extension .%s or .%s.tpl.\n"
438         # shellcheck disable=SC2059
439         printf "${LFBFL_s_format}" "${LFBFL_key}" "${LFBFL_key}"
440       fi
441       continue
442     fi
443     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
444     for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
445       LFBFL_i_is_whitelisted=0
446       for LFBFL_file_path2 in\
447         "${LFBFL_arr_files_without_license_header[@]}";
448       do
449         if [[ "${LFBFL_file_path}" == "${LFBFL_file_path2}" ]]; then
450           LFBFL_i_is_whitelisted=1
451         fi
452       done
453       if [[ LFBFL_i_is_whitelisted -eq 1 ]]; then
454         continue
455       fi
456       is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path}"
457       LFBFL_i_not_subfile=$?
458       LFBFL_i_not_subfile2=1
459       if [[ -n "${LFBFL_license2}" ]]; then
460         is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_path2}"
461         LFBFL_i_not_subfile2=$?
462       fi
463       if [[ LFBFL_i_not_subfile -ge 1 && LFBFL_i_not_subfile2 -ge 1 ]];
464       then
465         printf "File %s has no/wrong license header?\n"\
```

```
            "${LFBFL_file_path}"
        fi
      done
    done
}

build_licenses_templates "$@"
```

3.19 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

```
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```

3.20 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

3.21 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

3.22 build_and_checks_dependencies/lines_counts.libr.sh

```
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

```

```
all_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "all_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | tr '\n' '\0'\
      | xargs -0 grep --invert-match --perl-regexp --with-filename\
        -- 'a(?!a)a'
  done
}

all_self_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  # below and another lines, see SC2145 false positive
  all_code_lines "${LFBFL_common_options[@]}" \
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
```

```
96    get_verbose_option "$@"
97    local LFBFL_work_directory=""
98    get_work_directory_option "$@"
99    declare -a LFBFL_return_traps_stack
100   init_return_trap
101   pushd_to_work_directory --trap-popd
102   can_continue_after_enhanced_pushd || return 1
103
104   enhanced_set_shell_option pipefail --trap-unset
105
106   declare -a LFBFL_common_options=("$@")
107   keep_options LFBFL_common_options --verbose
108
109   all_self_code_lines "${LFBFL_common_options[@]}" \
110     | empty_lines_after_file_name
111 }
112
113 # shellcheck disable=SC2120
114 all_self_not_empty_code_lines(){
115   # Options:
116   #   --verbose
117   #   --work-directory=""
118   declare -i LFBFL_i_verbose=0
119   get_verbose_option "$@"
120   local LFBFL_work_directory=""
121   get_work_directory_option "$@"
122   declare -a LFBFL_return_traps_stack
123   init_return_trap
124   pushd_to_work_directory --trap-popd
125   can_continue_after_enhanced_pushd || return 1
126
127   enhanced_set_shell_option pipefail --trap-unset
128
129   declare -a LFBFL_common_options=("$@")
130   keep_options LFBFL_common_options --verbose
131
132   all_self_code_lines "${LFBFL_common_options[@]}" \
133     | not_empty_lines_after_file_name
134 }
135
136 code_lines_count_all(){
137   # Options:
138   #   --verbose
139   #   --work-directory=""
140   declare -i LFBFL_i_verbose=0
141   get_verbose_option "$@"
142   local LFBFL_work_directory=""
143   get_work_directory_option "$@"
144   declare -a LFBFL_return_traps_stack
145   init_return_trap
146   pushd_to_work_directory --trap-popd
147   can_continue_after_enhanced_pushd || return 1
148
149   enhanced_set_shell_option pipefail --trap-unset
150
151   declare -gi i_code_lines_count_all_result
152   i_code_lines_count_all_result=$(
153     all_self_code_lines\
154     | wc --lines
155   )
156   # The printf is for command line use.
```

```
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_all_result}
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail --trap-unset

  declare -gi i_code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  i_code_lines_count_not_empty_result=$(
    all_self_not_empty_code_lines\
    | wc --lines
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${i_code_lines_count_not_empty_result}
}
```

3.23 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_i_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
```

```
      ll_wc_var_i_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_i_number_only=1
    elif [[ ll_wc_var_i_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_i_number_only}"
  if [[ ll_wc_var_i_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat -- "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed --expression='s|domain|lists.gnu.org|' \
#       --expression='s|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  # Same arguments and options as grep.
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
```

```
    > "${LFBFL_temp}"
  # declare -ir LFBFL_i_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_i_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_i_lines_before}"
  # printf "%s\n" "${LFBFL_i_lines_after}"
  # if [[ LFBFL_i_lines_before -eq LFBFL_i_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent -- "${LFBFL_temp}" "${!#}"; then
    rm -- "${LFBFL_temp}"
    return
  fi
  mv -- "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=file_path
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below,
  #   you don't need to supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_i_enforce_line_starts=0
  declare -i LFBFL_i_enforce_line_ends=0
  declare -i LFBFL_i_quiet=0
  local LFBFL_argument
  for LFBFL_argument in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-starts-with-fixed-string"
    ]]; then
      LFBFL_i_enforce_line_starts=1
      continue
    fi
    if [[
      "${LFBFL_argument}" == "--enforce-line-ends-with-fixed-string"
    ]]; then
      LFBFL_i_enforce_line_ends=1
      continue
    fi
    LFBFL_grep_options+=("${LFBFL_argument}")
    if [[
      "${LFBFL_argument}" == "-q"
      || "${LFBFL_argument}" == "--quiet"
    ]]; then
      LFBFL_i_quiet=1
      continue
    fi
    LFBFL_grep_options2+=("${LFBFL_argument}")
  done
```

```
182
183   if [[
184     LFBFL_i_enforce_line_starts -eq 0
185     && LFBFL_i_enforce_line_ends -eq 0
186   ]]; then
187     grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
188     return
189   fi
190   declare -r LFBFL_s_lines=$(
191     grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
192       "$2" "$1"
193   )
194   declare -i LFBFL_i_result=1
195   if [[ -z "${LFBFL_s_lines}" ]]; then
196     # shellcheck disable=SC2248
197     return ${LFBFL_i_result}
198   fi
199   declare -a LFBFL_arr_lines
200   mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
201   readonly LFBFL_arr_lines
202
203   declare -i LFBFL_i_fixed_string_length=${#2}
204   local LFBFL_some_line
205   local LFBFL_line_part
206   for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
207     if [[ "${LFBFL_some_line}" == "$2" ]]; then
208       LFBFL_i_result=0
209       if [[ LFBFL_i_quiet -eq 1 ]]; then
210         break
211       fi
212       printf "%s\n" "${LFBFL_some_line}"
213       continue
214     fi
215     if [[
216       LFBFL_i_enforce_line_starts -eq 1
217       && LFBFL_i_enforce_line_ends -eq 1
218     ]]; then
219       continue
220     fi
221     if [[ LFBFL_i_enforce_line_starts -eq 1 ]]; then
222       LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_i_fixed_string_length}}
223     fi
224     if [[ LFBFL_i_enforce_line_ends -eq 1 ]]; then
225       LFBFL_line_part=${LFBFL_some_line: -${LFBFL_i_fixed_string_length}}
226     fi
227     if [[ "${LFBFL_line_part}" == "$2" ]]; then
228       LFBFL_i_result=0
229       if [[ LFBFL_i_quiet -eq 1 ]]; then
230         break
231       fi
232       printf "%s\n" "${LFBFL_some_line}"
233     fi
234   done
235   # shellcheck disable=SC2248
236   return ${LFBFL_i_result}
237 }
238
239 grep_variable(){
240   # $1=file_path
241   # $2=variable_name
242   # Options :
```

```
243   #   --replace-line-returns-by=""
244   #   --result-variable-prefix="LFBFL_" for example
245   declare -r LFBFL_regexp="(?<=^$2=).*$"
246   # printf "%s\n" "${LFBFL_regexp}"
247   local LFBFL_variable_value
248   local LFBFL_s_replace
249   get_some_option LFBFL_s_replace --replace-line-returns-by '' $'\n' 1 "$@"
250   if [[ "${LFBFL_s_replace}" != $'\n' ]]; then
251     # sed -Ez -e...
252     LFBFL_variable_value=$(
253       grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"\
254       | sed --null-data --regexp-extended\
255         --expression='s/\n/'"${LFBFL_s_replace}"'/Mg'
256     )
257     # We remove the trailing replace string,
258     # since the intent is only to replace the intermediate \n.
259     if [[ -n "${LFBFL_s_replace}" ]]; then
260     LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
261     fi
262   else
263     LFBFL_variable_value=$(
264       grep --only-matching --perl-regexp -- "${LFBFL_regexp}" "$1"
265     )
266   fi
267   local LFBFL_prefix=""
268   get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
269   declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
270   # printf "%s : %s\n" "${LFBFL_variable_name}" "${LFBFL_variable_value}"
271   # The declare line may be dropped by a local or declare
272   # in outer scope.
273   # declare -g "${LFBFL_variable_name}"
274   # So the assignment must be done on another line.
275   printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
276   # printf "%s\n" "${!LFBFL_variable_name}"
277 }
278
279 empty_lines(){
280   grep '^$'
281 }
282
283 not_empty_lines(){
284   grep --invert-match '^$'
285 }
286
287 space_starting_lines(){
288   grep '^ '
289 }
290
291 not_space_starting_lines(){
292   grep --invert-match '^ '
293 }
294
295 # Space starting or empty lines = spampty
296 spampty_lines(){
297   grep --extended-regexp '^( |$)'
298 }
299
300 not_spampty_lines(){
301   grep '^[^ ]'
302   # grep --extended-regexp --invert-match '^( |$)'
303 }
```

```
304
305 all_space_lines(){
306   grep '^[ ]\+$'
307 }
308
309 not_all_space_lines(){
310   grep --invert-match '^[ ]\+$'
311 }
312
313 # All space or empty lines = aspampty
314 aspampty_lines(){
315   grep '^[ ]*$'
316 }
317
318 not_aspampty_lines(){
319   grep '[^ ]'
320   # grep --invert-match '^[ ]*$'
321 }
322
323 empty_lines_after_file_name(){
324   grep '^[^:]\+:$'
325 }
326
327 not_empty_lines_after_file_name(){
328   grep --invert-match '^[^:]\+:$'
329 }
330
331 space_starting_lines_after_file_name(){
332   grep '^[^:]\+: '
333 }
334
335 not_space_starting_lines_after_file_name(){
336   grep --invert-match '^[^:]\+: '
337 }
338
339 spampty_lines_after_file_name(){
340   grep --extended-regexp '^[^:]+:( |$)'
341 }
342
343 not_spampty_lines_after_file_name(){
344   # The two variants results differ on lines without filename before.
345   # Since the intent of the function is only on lines "blabla.sh:whatever",
346   # use the fastest one. I didn't do extensive testing and just assumed
347   # that the "positive" one is faster as usual when querying database.
348   # But there is no index here ;) XD.
349   grep '^[^:]\+:[^ ]'
350   # grep --extended-regexp --invert-match '^[^:]+:( |$)'
351 }
352
353 all_space_lines_after_file_name(){
354   grep '^[^:]\+:[ ]\+$'
355 }
356
357 not_all_space_lines_after_file_name(){
358   grep --invert-match '^[^:]\+:[ ]\+$'
359 }
360
361 aspampty_lines_after_file_name(){
362   grep '^[^:]\+:[ ]*$'
363 }
364
```

```
not_aspampty_lines_after_file_name(){
  grep --invert-match '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# -E = --show-ends :). Nice command :).

not_JS_dependencies_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)node_modules/"\
    --regexp="(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?node_modules/"\
    --regexp="^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_php_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)\.phpdoc/cache/"
}

not_php_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?\.phpdoc/cache/"
}

not_python_cache_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)__pycache__/"\
    --regexp="(^|/)\.mypy_cache/"\
    --regexp="(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?__pycache__/"\
    --regexp="^([^:]+/)?\.mypy_cache/"\
    --regexp="^([^:]+/)?\.ruff_cache/"
}

not_cache_find(){
  not_php_cache_find\
    | not_python_cache_find
}

not_cache_grep(){
  not_php_cache_grep\
```

```
    | not_python_cache_grep
}

not_git_find(){
  grep --extended-regexp --invert-match "(^|/)\.git/"
}

not_git_grep(){
  grep --extended-regexp --invert-match "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep --extended-regexp --invert-match "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep --extended-regexp --invert-match\
    --regexp="(^|/)COPYING$"\
    --regexp="(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?COPYING:"\
    --regexp="^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep --extended-regexp --invert-match\
    --regexp="(^|/)${LFBFL_repository_name}\.html$"
}
```

```
487
488 not_main_html_grep(){
489   local LFBFL_repository_name=""
490   grep_variable build_and_checks_variables/repository_data.txt\
491     repository_name\
492     --result-variable-prefix="LFBFL_"\
493     --replace-line-returns-by=""
494   grep --extended-regexp --invert-match\
495     --regexp="^([^:]+/)?${LFBFL_repository_name}\.html:"
496 }
497
498 not_temp_file_find(){
499   grep --extended-regexp --invert-match\
500     --regexp="(^|/)build_and_checks_variables/temp/"
501 }
502
503 not_temp_file_grep(){
504   grep --extended-regexp --invert-match\
505     --regexp="^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
506 }
507
508 not_unchecked_builds_find(){
509   grep --extended-regexp --invert-match\
510     --regexp="(^|/)\.phpdoc/build/"
511 }
512
513 not_unchecked_builds_grep(){
514   grep --extended-regexp --invert-match\
515     --regexp="^([^:]+/)?\.phpdoc/build/"
516 }
517
518 relevant_find(){
519   enhanced_set_shell_option pipefail\
520     && trap 'enhanced_unset_shell_option pipefail' RETURN
521   not_dependencies_find\
522     | not_cache_find\
523     | not_git_find\
524     | not_archive_find\
525     | not_temp_file_find\
526     | not_unchecked_builds_find
527 }
528
529 relevant_grep(){
530   enhanced_set_shell_option pipefail\
531     && trap 'enhanced_unset_shell_option pipefail' RETURN
532   not_dependencies_grep\
533     | not_cache_grep\
534     | not_git_grep\
535     | not_archive_grep\
536     | not_temp_file_grep\
537     | not_unchecked_builds_grep
538 }
```

3.24 build_and_checks_dependencies/lines_maps.libr.sh

```
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2 # This file is part of DevOrSysAdminScripts.
3 #
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```

```
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed\
    --expression='s/\xc2\xa0/ /g'\
    --expression='s/\xe2\x80\x80/ /g'\
    --expression='s/\xe2\x80\x81/ /g'\
    --expression='s/\xe2\x80\x82/ /g'\
    --expression='s/\xe2\x80\x83/ /g'\
    --expression='s/\xe2\x80\x84/ /g'\
    --expression='s/\xe2\x80\x85/ /g'\
    --expression='s/\xe2\x80\x86/ /g'\
    --expression='s/\xe2\x80\x87/ /g'\
    --expression='s/\xe2\x80\x88/ /g'\
    --expression='s/\xe2\x80\x89/ /g'\
    --expression='s/\xe2\x80\x8a/ /g'\
    --expression='s/\xe2\x80\x8b/ /g'\
    --expression='s/\xe2\x80\x8c/ /g'\
    --expression='s/\xe2\x80\x8d/ /g'\
    --expression='s/\xe2\x80\x8e/ /g'\
    --expression='s/\xe2\x80\x8f/ /g'\
    --expression='s/\xe2\x80\xa8/ /g'\
    --expression='s/\xe2\x80\xa9/ /g'\
    --expression='s/\xe2\x80\xaa/ /g'\
    --expression='s/\xe2\x80\xab/ /g'\
    --expression='s/\xe2\x80\xac/ /g'\
    --expression='s/\xe2\x80\xad/ /g'\
    --expression='s/\xe2\x80\xae/ /g'\
    --expression='s/\xe2\x80\xaf/ /g'\
    --expression='s/\xe2\x81\x9f/ /g'\
    --expression='s/\xe2\x81\xa0/ /g'\
    --expression='s/\xe2\x81\xa1/ /g'\
    --expression='s/\xe2\x81\xa2/ /g'\
```

```
    --expression='s/\xe2\x81\xa3/ /g'\
    --expression='s/\xe2\x81\xa4/ /g'\
    --expression='s/\xe2\x81\xa5/ /g'\
    --expression='s/\xe2\x81\xa6/ /g'\
    --expression='s/\xe2\x81\xa7/ /g'\
    --expression='s/\xe2\x81\xa8/ /g'\
    --expression='s/\xe2\x81\xa9/ /g'\
    --expression='s/\xe2\x81\xaa/ /g'\
    --expression='s/\xe2\x81\xab/ /g'\
    --expression='s/\xe2\x81\xac/ /g'\
    --expression='s/\xe2\x81\xad/ /g'\
    --expression='s/\xe2\x81\xae/ /g'\
    --expression='s/\xe2\x81\xaf/ /g'
}
```

3.25 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"

overwrite_if_not_equal(){
  # $1=target_file_path
  # $2=temp_file_path
  # Options:
  #   --keep-temp : keep the temporary file
  #   --quiet : we even don't want the message of diff saying that
  #             files are distinct
  #   --tree-mode : when the two files are the output of tree command,
  #   we ignore if all the differences are only on directories timestamps.
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  declare -i LFBFL_i_keep_temp
  get_some_flag LFBFL_i_keep_temp --keep-temp 1 "$@"
```

```
  declare -i LFBFL_i_quiet
  get_some_flag LFBFL_i_quiet --quiet 1 "$@"
  declare -i LFBFL_i_tree_mode
  get_some_flag LFBFL_i_tree_mode --tree-mode 1 "$@"

  if [[ ! -f "$1" ]]; then
    if [[ LFBFL_i_keep_temp -eq 1 ]]; then
      cp --preserve=mode,ownership,timestamps -- "$2" "$1"
    else
      mv -- "$2" "$1"
    fi
    return 2
  fi
  declare -i LFBFL_i_is_equal
  if [[ LFBFL_i_tree_mode -eq 1 ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff -- "$1" "$2"\
      | grep --extended-regexp "^(>|<)"\
      | grep --invert-match "/$"
    LFBFL_i_is_equal=${PIPESTATUS[2]}
    if [[ LFBFL_i_is_equal -eq 1 ]]; then
      grep --only-matching --regexp="[^ ]*/$" -- "$1"\
        > "overwrite_if_not_equal.file1.temp"
      grep --only-matching --regexp="[^ ]*/$" -- "$2"\
        > "overwrite_if_not_equal.file2.temp"
      if [[ LFBFL_i_quiet -eq 1 ]]; then
        diff overwrite_if_not_equal.file1.temp\
          overwrite_if_not_equal.file2.temp\
          > /dev/null
      else
        diff overwrite_if_not_equal.file1.temp\
          overwrite_if_not_equal.file2.temp
      fi
      LFBFL_i_is_equal=1-$?
      rm overwrite_if_not_equal.file1.temp\
        overwrite_if_not_equal.file2.temp
    fi
  else
    if [[ LFBFL_i_quiet -eq 1 ]]; then
      diff --brief -- "$1" "$2" > /dev/null
    else
      diff --brief -- "$1" "$2"
    fi
    LFBFL_i_is_equal=1-$?
  fi
  if [[ LFBFL_i_is_equal -eq 1 ]]; then
    if [[ LFBFL_i_keep_temp -eq 0 ]]; then
      rm -- "$2"
    fi
    return 0
  fi
  if [[ LFBFL_i_keep_temp -eq 1 ]]; then
    cp --preserve=mode,ownership,timestamps -- "$2" "$1"
  else
    mv -- "$2" "$1"
  fi
  return 1
}
```

3.26 build_and_checks_dependencies/phpmd_baseline.xslt

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
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-->
<xsl:stylesheet xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
    version="1.1">
  <xsl:output method="xml"/>
  <xsl:template match="phpmd-baseline">
    <xsl:copy>
      <xsl:text>
</xsl:text>
      <xsl:apply-templates select="node()">
        <xsl:sort select="@file"/>
        <xsl:sort select="@rule"/>
        <xsl:sort select="@method"/>
      </xsl:apply-templates>
    </xsl:copy>
  </xsl:template>
  <xsl:template match="violation">
    <xsl:text>  </xsl:text>
    <xsl:copy>
      <xsl:copy-of select="@*"/>
    </xsl:copy>
    <xsl:text>
</xsl:text>
  </xsl:template>
  <!--
  Remove all indentation and line returns in text nodes,
  because otherwise it will be output first after sorting above.
  -->
  <xsl:template match="text()"></xsl:template>
</xsl:stylesheet>
```

3.27 build_and_checks_dependencies/python_black_complement.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
```

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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep --multiline --\
    $'def ([^"]|"(?!""))*"""([^"]|"(?!""))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
```

```
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_no_empty_line_after_python_function_docstring\
    "${LFBFL_common_options[@]}"
}
```

3.28 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

check_collections_abc_place(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # This function should be useless in a few months:
  # https://github.com/PyCQA/isort/issues/2250
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
```

```
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
      "Checking that import of _collections_abc is at the right place.\n"

  enhanced_set_shell_option pipefail --trap-unset

  declare -r LFBFL_temp=".check_collections_abc_place.temp"
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.py" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_collections_abc_place: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    sed --regexp-extended --null-data\
      --expression='s/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
      -- "${LFBFL_file_path}"\
      > "${LFBFL_file_path}${LFBFL_temp}"
    overwrite_if_not_equal "${LFBFL_file_path}"\
      "${LFBFL_file_path}${LFBFL_temp}"
  done
}

python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose

  check_collections_abc_place "${LFBFL_common_options[@]}"
}
```

3.29 build_and_checks_dependencies/repository_data.libr.sh

```
#!/usr/bin/env bash
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_path
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap

  enhanced_set_bash_option extglob --trap-unset

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}

grep_variable_with_multiple_files(){
  # $1=data_file_path
  # $2=variable_name
  # Options :
```

```
  #   --result-variable-prefix="LFBFL_" for example
  local LFBFL_prefix=""
  get_some_option LFBFL_prefix --result-variable-prefix '' '' 1 "$@"
  grep_variable "$1" "$2" --result-variable-prefix="${LFBFL_prefix}"
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  declare -a LFBFL_arr
  if [[ -z "${!LFBFL_variable_name}" ]]; then
    return
  fi
  mapfile -t LFBFL_arr <<< "${!LFBFL_variable_name}"
  local LFBFL_s=""
  local LFBFL_file_path
  local LFBFL_previous_file_path
  for LFBFL_file_path in "${LFBFL_arr[@]}"; do
    if [[
      -n "${LFBFL_previous_file_path}"
      && "${LFBFL_previous_file_path}" != */
    ]]; then
      LFBFL_s+=$'\n'
    fi
    LFBFL_s+="${LFBFL_file_path}"
    LFBFL_previous_file_path="${LFBFL_file_path}"
  done
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_s}"
}
```

3.30 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
#!/usr/bin/env bash
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
```

```
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion.\n"
  # shellcheck disable=SC2059
  [[ LFBFL_i_verbose -eq 1 ]] && printf "${LFBFL_s}"

  pcre2grep --multiline -- $'\\$\\{#(|.*_)arr_(?!.*\\[@\\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
  get_max_line_length_option "$@"
  declare -i LFBFL_i_max_previous_line_length
  LFBFL_i_max_previous_line_length=$((LFBFL_i_max_line_length - 5))

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_i_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];[^\n]{0,${LFBFL_i_max_previous_line_length}})"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_i_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"
  # pcre2grep --multiline --
  # '^[^\n]{1,70}(?<=\]\];[^\n]{0,70})'
  # '(?<!^\s{0,70}\]\];)\n\s*then'
  pcre2grep --multiline -- "${LFBFL_regexp}" **/*.sh
}

check_no_negation_before_bash_test(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
```

```
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar --trap-unset

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking that shell scripts do not contain ! before [[.\n"

  pcre2grep --multiline -- $'!\\s*\\[\\[' **/*.sh
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -a LFBFL_common_options=("$@")
  keep_options LFBFL_common_options --verbose --max-line-length

  check_no_size_of_array_first_element "${LFBFL_common_options[@]}"
  check_no_misplaced_then "${LFBFL_common_options[@]}"
  check_no_negation_before_bash_test "${LFBFL_common_options[@]}"
}
```

3.31 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php

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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".

@category Executable
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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*/

declare(strict_types=1);
declare(encoding='UTF-8');

require_once './build_and_checks_dependencies/split_score.libr.php';

if(PHP_SAPI !== 'cli'){
  fwrite(
    STDERR,
    "split_score.exec.php must be run in a shell.\n",
  );
  die(1);
}

if(count($argv) !== 7){
  fwrite(
    STDERR,
    "split_score.exec.php must be given exactly 6 arguments.\n",
  );
  die(1);
}

/*
php -r\
$'require_once \'./build_and_checks_dependencies/split_score.libr.php\';'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'

php -r \
$'require_once \'./build_and_checks_dependencies/split_score.libr.php\';'\
$'echo(\SplitScore\generate_split_score(true, 70, [\'/\'])'\
$'(\'/\', 60, true));'

php -r \
$'require_once \'./build_and_checks_dependencies/split_score.libr.php\';'\
$'$o_split_score_closure ='\
$'  new \SplitScore\SplitScoreClosure(true, 70, [\'/\']);'\
$'echo($o_split_score_closure(\'/\', 60, true));'
*/

/*
```

```php
// The first cast "(int) $func..." is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1],
    (int) $argv[2],
    explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
//*/
// AHAHAH PHPCS...
/* AHAHAH PHPCS XD */

$o_split_score_closure = new \SplitScore\SplitScoreClosure(
  (bool) $argv[1],
  (int) $argv[2],
  explode(',', $argv[3]),
);
echo($o_split_score_closure($argv[4], (int) $argv[5], (bool) $argv[6]));

?>
```

3.32 build_and_checks_dependencies/split_score.libr.php

```php
<?php

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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
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@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
```

```
38  */
39
40  declare(strict_types=1);
41  declare(encoding='UTF-8');
42
43
44
45  namespace SplitScore;
46
47  /**
48  Given the 3 following arguments, this function returns a closure that can
49  be used to compute the split score given the current string
50  (hopefully matching a delimiter), the current cut/split position,
51  and if that split position is after or before the current string.
52
53  @param bool          $b_larger_after
54                       Score is larger when splitting after,
55                       previous name of this parameter was $b_after_before.
56  @param int           $i_max_length
57                       The maximum length allowed (goal length).
58  @param array<string> $arr_delimiter_strings_domain
59                       An array of "delimiters".
60
61  @return \Closure {
62    The "split_score" closure to use repeatedly afterward.
63
64    @CLparam string $s_delimiter_string The string tested to be a delimiter.
65    @CLparam int    $i_cut_position     The offset of the cut in the string.
66    @CLparam bool   $b_is_cut_after     Is the offset taken after the string?
67
68    @CLreturn int The split score.
69  }
70  */
71  function generate_split_score(
72    bool $b_larger_after,
73    int $i_max_length,
74    array $arr_delimiter_strings_domain,
75  ) : \Closure {
76    return function (
77      string $s_delimiter_string,
78      int $i_cut_position,
79      bool $b_is_cut_after,
80    ) use (
81      $b_larger_after,
82      $i_max_length,
83      $arr_delimiter_strings_domain,
84    ) : int {
85      if(
86        !in_array(
87          $s_delimiter_string,
88          $arr_delimiter_strings_domain,
89          true,
90        )
91      ){
92        // I do not handle regexps for the moment.
93        // I'm too brainfucked. I just want to finish this feature fast
94        // for HTML and Tex/PDF listings generation with HTML and Tex on
95        // at most 70 characters.
96        return 0;
97      }
98      if($b_is_cut_after === $b_larger_after){
```

```
99        // When $b_larger_after,
100       // always way better to cut after '/' than before it.
101       // When !$b_larger_after (aka larger_before),
102       // always way better to cut before '/' than after it.
103       return 1 + $i_max_length + $i_cut_position;
104     }
105     return 1 + $i_cut_position;
106   };
107 }//end generate_split_score()
108
109
110
111
112 /**
113 Given the type system of PHP, a better way to have a closure computing
114 the split score .
115
116 @category Library
117 @package DevOrSysAdminScripts
118 @class SplitScoreClosure
119 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
120 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
121 */
122 final class SplitScoreClosure {
123   /**
124   Score is larger when splitting after.
125
126   @var bool $b_larger_after
127   */
128   private bool $b_larger_after;
129
130   /**
131   The maximum length allowed (goal length).
132
133   @var int $i_max_length
134   */
135   private int $i_max_length;
136
137   /**
138   An array of "delimiters".
139
140   @var array<string> $arr_delimiter_strings_domain
141   */
142   private array $arr_delimiter_strings_domain;
143
144
145
146   /**
147   Given the 3 following arguments,
148   this constructor creates an object that will behave like a Closure
149   that can be used to compute the split score given the current string
150   (hopefully matching a delimiter), the current cut/split position,
151   and if that split position is after or before the current string.
152
153   @param bool          $b_larger_after
154                        Score is larger when splitting after,
155                        previous name of this parameter was $b_after_before.
156   @param int           $i_max_length
157                        The maximum length allowed (goal length).
158   @param array<string> $arr_delimiter_strings_domain
159                        An array of "delimiters".
```

```
160
161   @return void
162   */
163   public function __construct(
164     bool $b_larger_after,
165     int $i_max_length,
166     array $arr_delimiter_strings_domain,
167   ) {
168     $this->b_larger_after = $b_larger_after;
169     $this->i_max_length = $i_max_length;
170     $this->arr_delimiter_strings_domain = $arr_delimiter_strings_domain;
171   }//end __construct() SplitScoreClosure
172
173
174
175   /**
176   The "split_score" closure to use repeatedly afterward.
177
178   @param string $s_delimiter_string The string tested to be a delimiter.
179   @param int    $i_cut_position     The offset of the cut in the string.
180   @param bool   $b_is_cut_after     Is the offset taken after the string?
181
182   @return int The split score.
183   */
184   public function __invoke(
185     string $s_delimiter_string,
186     int $i_cut_position,
187     bool $b_is_cut_after,
188   ) : int {
189     if(
190       !in_array(
191         $s_delimiter_string,
192         $this->arr_delimiter_strings_domain,
193         true,
194       )
195     ){
196       return 0;
197     }
198     if($b_is_cut_after === $this->b_larger_after){
199       // When $b_larger_after,
200       // always way better to cut after '/' than before it.
201       // When !$b_larger_after (aka larger_before),
202       // always way better to cut before '/' than after it.
203       return 1 + $this->i_max_length + $i_cut_position;
204     }
205     return 1 + $i_cut_position;
206   }//end __invoke() SplitScoreClosure
207 }//end class SplitScoreClosure
208
209 ?>
```

3.33 build_and_checks_dependencies/strings_functions.libr.sh

```
1 #!/usr/bin/env bash
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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

split_line_at(){
  # $1=line
  # $2=position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

#-----------------------------------------------------------------------
#-----------------------------------------------------------------------
# Properties for the score for a possible split/cut in the string between
# two characters.
#-----------------------------------------------------------------------
# Some preliminary remarks:
# a) If the character before the split position is a forbidden previous
#   character, the score is -1.
#   (Usually, either we have no or one forbidden previous character that is
#   a backslash.)
# b) If the previous character is not forbidden,
#   and either the character before the position is among the delimiters,
#   or the character after the position is among the delimiters,
#   then the score is computed according to the delimiter found,
#   the position, and the fact that the position is after or before the
#   delimiter.
# c) If two consecutive delimiters are found in the string,
#   the maximum score between the one after the first delimiter
#   and before the second delimiter is taken for the score between the
#   two delimiters.
# d) Otherwise, by default,
#   the score for cutting/splitting at some position is 0.
```

```
# e) A substring is given by the goal length given for the resulting
#   strings/lines after split.
# f) When some scores are equal,
#   we split at the rightmost position with maximum score in the substring.
# g) Remark f) implies that below we do not need to consider strict
#    inequalities for the score for the result.
#    But if you look at split_line_at_most code,
#    you will see a micro-optimization where distinguishing < and <= could
#    have been interesting.
# h) The score function SF(position) which will dictate our choice for
#   split/cut position is in reality
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   )
#   but at the cost of expliciting its internal and adding verbosity,
#   we can see that the score really depends on the following arguments
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   )
#   If it is needed,
#   then matched_delimiter_before or matched_delimiter_after is NULL.
#   Somehow previous_character can be seen as position.previous_character,
#   and next_character can be seen as position.next_character.
#   We can see that the previous or next character without
#   knowing that it matches a delimiter is irrelevant.
#   The fact that there is 1 or more delimiters may be interesting.
#   The fact that there is 0 or more forbidden previous characters
#   may be interesting, but we haven't integrated it in the split_score
#   functions.
#   SF1a when the previous character is not forbidden
#   needs a score function SF2a with the following parameters
#   SF2a(position, current_character, delimiter, is_after):
#   SF1a(
#     position, previous_character, next_character, delimiters, forbiddens
#   ) = max(
#     For all delimiter in delimiters,
#     SF2a(position, previous_character, delimiter, TRUE),
#     SF2a(position, next_character, delimiter, FALSE),
#   )
#   =  max(
#     For all delimiter in delimiters,
#     For all couple C in {
#       (position.previous_character, TRUE),
#       (position.next_character, FALSE)
#     }
#     SF2a(position, C.1, delimiter, C.2),
#   )
#   Similarly,
#   SF1b(
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2b(position, matched_delimiter_before, TRUE),
#     SF2b(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2b has the following parameters
#   SF2b(cut_position, matched_delimiter, is_after).
#   We have also:
#   SF1b(
```

```
#     position, matched_delimiter_before, matched_delimiter_after,
#     previous_character_is_forbidden
#   ) = max(
#     For all delimiter in delimiters,
#     SF2c(position-1, matched_delimiter_before, TRUE),
#     SF2c(position, matched_delimiter_after, FALSE),
#   )
#   where the score function SF2c has the following parameters
#   SF2c(character_position, matched_delimiter, is_after).
#   current_character is only used to check if it is equal to the
#   delimiter, otherwise the score is zero;
#   but when expressing properties it makes sense to compare after
#   and before the same character position.
#   position, delimiter, and is_after may all three impact the score
#   once the current character matches the delimiter.
#   Position is an integer from 1 to goal length.
#   Cutting at position 0 would be stupid,
#   and moreover there would be no previous character.
#   Since we ask for a split, the current length is above goal length,
#   and there is a next character at the last considered position.
#   So, in search of properties that may have an impact on the algorithm,
#   we discard the characters to only keep matched delimiters.
#   Let us use p1 and p2 for position1 and position2,
#   mdb1 and mdb2 for matched_delimiter_before1, etc.,
#   mda1 and mda2 for matched_delimiter_after1, etc.,
#   pcif for previous_character_is_forbidden, etc.,
#   md1 and md2 for matched_delimiter1, etc.,
#   ia1 and ia2 for is_after1, etc.
#
#   What we consider as properties are of 3 forms.
#   The first form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all mdb1 in delimiters union {NULL},
#     for all mdb2 in delimiters union {NULL},
#     for all mda1 in delimiters union {NULL},
#     for all mda2 in delimiters union {NULL},
#     RelationP(p1, p2) AND RelationM(mdb1, mdb2, mda1, mda2)
#     implies RelationS(
#       SF1b(p1, mdb1, mda1, false),
#       SF1b(p2, mdb2, mda2, false),
#     )
#   There are many choices for RelationP, RelationM and RelationS.
#   We can apply the without loss of generality argument only once when
#   breaking symmetry between equivalent properties.
#   Breaking symmetry on another relation that RelationS first
#   is a challenge that we do not tackle.
#   Indeed RelationS is the only relation where all variables appear.
#   And breaking symmetry on RelationP would not allow to break symmetry
#   on RelationM freely since some variables are correlated by the
#   suffixes 1 or 2.
#   But some choices for some relation may allow to break symmetry on
#   another relation.
#   Breaking symmetry on the result, there are only 2 choices for
#   RelationS:
#   RS= : SF1b(p1, mdb1, mda1, false) = SF1b(p2, mdb2, mda2, false),
#   RS<= : SF1b(p1, mdb1, mda1, false) <= SF1b(p2, mdb2, mda2, false).
#   But then there are 5+1 choices for RelationP:
#   RP1 : p1 = p2,
#   RP2 : p1 < p2,
#   RP3 : p1 <= p2,
#   RP4 : p1 > p2,
```

```
#   RP5 : p1 >= p2,
#   plus the choice that p1 and p2 are not necessarily related
#   (empty RP or RPe).
#   We can add RP6 : p1 = p2 - 1 and RP7 : p1 = p2 + 1, since it implies
#   some equalities between matched delimiters.
#   Note that  p1 = p2 + i or p2 = p1 + i with i >= 2 brings no additional
#   logical relations compared to p1 < p2 or p2 < p1;
#   it only yields subslices of other properties and open a slightly silly
#   door on infinity.
#   If we have RS=, by symmetry we can consider only RP1, RP2, RP3 and RPe,
#   and maybe RP6.
#   Note that RSe or empty RS would have no interest at all.
#   There are many choices for RelationM:
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is NULL is not
#   interesting at all since both scores are 0 or -1.
#   mdb1 is NULL AND mdb2 is NULL AND mda1 is NULL AND mda2 is not NULL
#   implies that second score is always larger, that's not an additional
#   property, that is a theorem.
#   Hence, at least one of mdb1 or mda1, resp. mdb2 or mda2,
#   must (have the possibility to) be non-NULL.
#   If we start listing the remaining relations, we have:
#   1) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#   RMA1A2
#   2) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 = mda2
#   RMA1A2=
#   3) mdb1 is NULL AND mdb2 is NULL AND mda1 is not NULL
#     AND mda2 is not NULL
#     AND mda1 != mda2 -> This case is not interesting (alone,
#     full analysis could combine it with some other case,
#     but it seems gratis combinatorial explosion without insight):
#     If we only know they are different, we cannot say anything on
#     properties related to delimiters that are different unless the score
#     doesn't depend on the delimiter, and that case is already handled by
#     1) that contains it. The same remark applies to some other choices.
#     The only thing that could be said would be if we had an order on
#     delimiters scores that are uniform for the position, or the score
#     before some delimiter would always be =, >=, <=, etc. the score
#     before or after another one. So it would only make sense with RPe,
#     and mda1 <= mda2 for example, but that would trivially imply RS<=.
#     It would end up with max(mda1, mdb1) <= max(mda2, mdb2) where max
#     ignores NULL value. You do not deduce or specify anything new here.
#   4) mdb1 is not NULL AND mdb2 is not NULL AND mda1 is NULL
#     AND mda2 is NULL
#   RMB1B2
#   and so on. So if we use the short notation RM followed by A1, resp. A2,
#   resp. B1, resp. B2, when mda1, resp. mda2, resp. mdb1, resp. mdb2,
#   is not null, we have the following possibilities:
#   RMA1A2, RMB1B2, RMA1B2, RMA2B1, all 4 with variant with = like RMA1A2=,
#   RMA1A2B1, RMA1A2B2, RMA1B1B2, RMA2B1B2, all 4 (RMXXYYZZ) with variants
#   with suffixes _1 (XX = YY), _2 (XX = ZZ), _3 (YY = ZZ),
#   and _4 (XX = YY = ZZ)
#   for equalities on the 1st, 2nd or 3rd couple or between the three
#   variables by transitivity,
#   RMALL where none is null, with suffixes A1A2 (A1 = A2), etc.
#   RMALL_A1A2, RMALL_A1B1, RMALL_A1B2, RMALL_A2B1, RMALL_A2B2, RMALL_B1B2,
#   for one edge;
#   RMALL_A1A2B1, RMALL_A1A2B2, RMALL_A1B1B2, RMALL_A2B1B2, one triangle;
#   RMALL_A1A2_B1B2, RMALL_A1B1_A2B2, RMALL_A1B2_A2B1, 2 disjoint edges;
```

```
#   RMALL_ALL square with diagonals.
#   So we see that we listed at least 39 possibilities only for RM.
#   We did not list yet the possibilities where the variables not present
#   in some equality can be NULL or not NULL, etc.
#   We see 9 of them : RMe (40), and the following 8:
#   RMA1A2+ = RMA1A2= OR RMA1A2B1_1 OR RMA1A2B2_1 OR RMALL_A1A2
#   RMB1B2+ = RMB1B2= OR RMA1B1B2_3 OR RMA2B1B2_3 OR RMALL_B1B2
#   RMA1B2+ = RMA1B2= OR RMA1A2B2_2 OR RMA1B1B2_2 OR RMALL_A1B2
#   RMA2B1+ = RMA2B1= OR RMA1A2B1_3 OR RMA2B1B2_1 OR RMALL_A2B1
#   RMA1A2B1+ = RMA1A2B1_4 OR RMALL_A1A2B1
#   RMA1A2B2+ = RMA1A2B2_4 OR RMALL_A1A2B2
#   RMA1B1B2+ = RMA1B1B2_4 OR RMALL_A1B1B2
#   RMA2B1B2+ = RMA2B1B2_4 OR RMALL_A2B1B2
#   We don't do complete analysis at the moment.
#   If we wanted to filter out some possibilities,
#   we would keep only the RM possibilities, among the 38,
#   that are symmetric between 1 and 2:
#   RMA1A2, RMB1B2, RMA1A2=, RMB1B2=,
#   RMALL, RMALL_A1A2, RMALL_A2B2, RMALL_A1A2_B1B2, RMALL_A1B1_A2B2,
#   RMALL_ALL. Still 10 possibilities.
#   We have 1 (RS=) times 5 times 10 + 1 (RS<=) times 8 times 10 = 130
#   combinations (life is too short ;) :) );
#   if we keep 48 RM possibilities, there is 624 combinations,
#   but we did not analyze totally which RM relations are useless.
#
#   The second form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2b(p1, md1, ia1)
#       SF2b(p2, md2, ia2)
#     )
#   The third form is like:
#     For all p1 in [1, goal_length], for all p2 in [1, goal_length],
#     for all md1 in delimiters union {NULL},
#     for all md2 in delimiters union {NULL},
#     for all ia1 in {TRUE,FALSE}, for all ia2 in {TRUE,FALSE},
#     RelationPP(p1, p2) AND RelationMM(md1, md2) AND RelationIA(ia1, ia2)
#     implies RelationS(
#       SF2c(p1, md1, ia1)
#       SF2c(p2, md2, ia2)
#     )
#   This time the combinatorics is much more pleasant :).
#   As above, we have RS= and RS<=, RPe, RP1 to RP7;
#   For RelationMM, assume that the possible delimiters correspond to
#   integer numbers 1 to i; hence md1, resp. md2, can be seen as an integer
#   in [0,i], either 0 for NULL or the delimiter index.
#   Clearly, if md1 = k in [1,i] and md2 = l in [1,i], then the only
#   information we can use is k = l or k != l.
#   If we allow that md1 != 0 and md2 = 0,
#   then both RS= and RS<= are false.
#   Thus RMMe is not possible.
#   If we allow that md1 = 0 and md2 != 0, then RS<= is always true,
#   whilst RS= is false.
#   Thus properties with RS<= can trivially be extended with
#   RMM-"allow-md1-0", not useful to explicit this for the moment.
#   If both md1 = 0 and md2 = 0, then both scores are 0 and we also learn
#   nothing more.
```

```
#   Thus we can focus on: md1 and md2 are not NULL,
#   and are equal RMM= or not RMM!=. It seems that a property with RMM!=
#   could be extended to a property RMM=or!=/RMMNotNULL.
#   and finally RIA0 to RIA8 where TRUE = 1, FALSE = 0 and the digit is
#   3 * ia1 + ia2, and both variables can be 2 if they can be either TRUE
#   or FALSE. Wrong ! Not pleasant ! XD
#   So for RIA, we keep either: empty RIAe (RIA8),
#   ia1 = ia2 RIA= (RIA0 or RIA4),
#   ia1 = ia2 = TRUE RIA=T (RIA4), ia1 = ia2 = FALSE RIA=F (RIA0),
#   ia1 != ia2 RIA!= (RIA1 or RIA3), ia1 != ia2 = FALSE RIA!=T1 (RIA3),
#   ia1 != ia2 = TRUE RIA!=T2 (RIA1),
#   ia1 = TRUE (RIA5), ia1=FALSE (RIA2),
#   ia2 = TRUE (RIA7), ia2=FALSE (RIA6),
#   even less pleasant XD : 11 possibilities.
#   In total, we have now 1 (RS=) times 5 times 2 times 11
#   + 1 (RS<=) times 8 times 2 times 11 = 286
#   combinations (life is even shorter ;) :) ).
#   We only listed below properties of the third form.
# i) By the previous remark, there is a good chance that we missed some
#   possible property.
#   We hope we did found the properties that had a simple meaning,
#   in particular related to the algorithm where either you
#   loop on the string forward and keep last position with maximum score,
#   or you loop backward and abort as soon as possible as soon you know
#   that you found some position at the end that must have maximum score.
# j) These properties are "boolean" flags (integers that are powers of 2).
#   We first put the exact integers,
#   then wondered about switching everything to exponentiation 2**i,
#   then we took the solution that minimizes the character length between
#   [1-9][0-9]* and 2**i, then reverted to initial before committing.
#   Most efficient interpreted code is what we did first,
#   if there is no opcode cache for constants.
#   Most safe code is when the language hides the details and handles
#   families of flags,
#   but beware if some flag goes into the database.
#   Most readable code is a mix: 1 is easier to read than 2**0 ;) :P,
#   or $((2<<0))...
#   What I would love to see in many languages is a flag type, as a subtype
#   of integers, flag32, flag64 for C for example.
#   I think there is no such thing because sooner or later you play with
#   pseudo-flags that are combination of flags, hence integers.
#   But just for the declaration to let the type checker check that the
#   integer is a power of 2: flag32 MY_FLAG = 4;
#   Or better flag32 MY_FLAG = 2; as an equivalent of int32 MY_FLAG = 1<<2;
#   Another fact is that practice makes perfect, so working with the powers
#   of 2 written out in base 10 gets you accustomed to them.
#   Safe code should be favored since (pre-)compilation optimizes it,
#   but for this project, I prefer to keep the integers in decimal and
#   check that it doubles when reading my listing, because I'm familiar
#   with that and the powers of 2 doesn't go beyond what I can check fast
#   enough in my brain.
# k) Some of the properties below are union of other properties,
#   and are here to simplify not only code but understanding.
#   split_score_properties_logical_closure() function below checks that
#   all flags that are consequences of already activated flags are set.
# l) We use SSP_ prefix for SPLIT_SCORE_PROPERTY_.
# Some of these remarks are repeated below in the corresponding functions,
# and some remarks are added just after the constants.
#------------------------------------------------------------------------
# Any score after a character that matches a delimiter
# is larger than or equal to
```

```
# any score before a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T2 RS<=
declare -gir SSP_ALWAYS_LARGER_AFTER=1

# Any score before a character that matches a delimiter
# is larger than or equal to
# any score after a character that matches a delimiter.
# RPPe RMMNotNull RIA!=T1 RS<=
declare -gir SSP_ALWAYS_LARGER_BEFORE=2

# The score doesn't depend on the delimiter that is found.
# RPP1 RMMNotNull RIA= RS=
declare -gir SSP_DELIMITER_UNIFORM=4

# Whenever a character that matches a delimiter is found,
# the score after is equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is equal to
# the score before the character.
# RPP1 RMM= RIA!= RS=
declare -gir SSP_EQUAL_AFTER_OR_BEFORE=8
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_EQUAL_AFTER_OR_BEFORE=

# Whenever a character that matches a delimiter is found,
# the score after is larger than or equal to the score before.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score after the character
# is larger than or equal to
# the score before the character.
# RPP1 RMM= RIA!=T2 RS<=
declare -gir SSP_LARGER_AFTER=16
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_AFTER=

# Whenever a character that matches a delimiter is found,
# the score before is larger than or equal to the score after.
# Or equivalently:
# For any delimiter, and for any character position,
# if the character at this position matches this delimiter,
# the score before the character
# is larger than or equal to
# the score after the character.
# RPP1 RMM= RIA!=T1 RS<=
declare -gir SSP_LARGER_BEFORE=32
# declare -gir SSP_SAME_DELIMITER_AND_POSITION_LARGER_BEFORE=

# The score after, resp. before, a character that matches a delimiter
# doesn't decrease when the position in the string increase.
# (RPP2 or RPP3) RMM= RIA= RS<=
declare -gir SSP_POSITION_NOT_DECREASING=64

# The score after a character that matches a delimiter doesn't decrease
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=T RS<=
declare -gir SSP_POSITION_NOT_DECREASING_AFTER=128

# The score before a character that matches a delimiter doesn't decrease
```

```
# when the position in the string increases.
# (RPP2 or RPP3) RMM= RIA=F RS<=
declare -gir SSP_POSITION_NOT_DECREASING_BEFORE=256

# The score after, resp. before, a character that matches a delimiter
# doesn't depend on the position in the string.
# (RPP2 or RPP3) RMM= RIA= RS=
declare -gir SSP_POSITION_UNIFORM=512

# The score after a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=T RS=
declare -gir SSP_POSITION_UNIFORM_AFTER=1024

# The score before a character that matches a delimiter doesn't depend
# on the position in the string.
# (RPP2 or RPP3) RMM= RIA=F RS=
declare -gir SSP_POSITION_UNIFORM_BEFORE=2048

# For any delimiter,
# any score after a character that matches this delimiter
# is equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!= RS=
# <=> RPPe RMM= RIAe RS=
declare -gir SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE=4096

# For any delimiter,
# any score after a character that matches this delimiter
# is larger than or equal to
# any score before a character that matches this delimiter.
# RPPe RMM= RIA!=T2 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_AFTER=8192

# For any delimiter,
# any score before a character that matches this delimiter
# is larger than or equal to
# any score after a character that matches this delimiter.
# RPPe RMM= RIA!=T1 RS<=
declare -gir SSP_SAME_DELIMITER_LARGER_BEFORE=16384

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!= RS=
declare -gir SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE=32768

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score after a character at this position that matches a delimiter
# is larger than or equal to
# the score before a character at this position that matches a delimiter.
# RPP1 RMMNotNull RIA!=T2 RS<=
declare -gir SSP_SAME_POSITION_LARGER_AFTER=65536

# For any character position,
# assuming that a character at this position matches some delimiter,
# the score before a character at this position that matches a delimiter
# is larger than or equal to
```

```
497 # the score after a character at this position that matches a delimiter.
498 # RPP1 RMMNotNull RIA!=T1 RS<=
499 declare -gir SSP_SAME_POSITION_LARGER_BEFORE=131072
500
501 # For all possible split/cut positions, the score is equal as long as
502 # at least one of the previous and next characters is a delimiter.
503 # Or equivalently:
504 # Any score after a character that matches a delimiter
505 # is equal to
506 # any score before a character that matches a delimiter.
507 # is_after doesn't change anything.
508 # So we cut on the last position adjacent to a delimiter without
509 # forbidden previous character (edge case, on last position).
510 # RPPe RMMNotNull RIAe RS=
511 declare -gir SSP_UNIFORM=262144
512 # declare -gir SSP_ALWAYS_EQUAL_AFTER_OR_BEFORE=1
513
514 #-------------------------------------------------------------------------
515 # m) Note that a property telling that the score is always the same
516 #   even if the current character isn't among delimiters is not needed,
517 #   since, in that case, a null split score command will split at the goal
518 #   length.
519 #-------------------------------------------------------------------------
520 #-------------------------------------------------------------------------
521
522 split_score_properties_logical_closure(){
523   # $1=split_score_properties
524   # Since this code is not performance critical,
525   # we followed lexicographic order on constants,
526   # instead of an order to minimize the operations.
527   # It is also safer to avoid forgetting some relation.
528   declare -i LFBFL_i_result=$(($1))
529   declare -i LFBFL_i_result_old=0
530   declare -i LFBFL_i_all_flags=$((
531     SSP_ALWAYS_LARGER_AFTER
532     | SSP_ALWAYS_LARGER_BEFORE
533     | SSP_DELIMITER_UNIFORM
534     | SSP_EQUAL_AFTER_OR_BEFORE
535     | SSP_LARGER_AFTER
536     | SSP_LARGER_BEFORE
537     | SSP_POSITION_NOT_DECREASING
538     | SSP_POSITION_NOT_DECREASING_AFTER
539     | SSP_POSITION_NOT_DECREASING_BEFORE
540     | SSP_POSITION_UNIFORM
541     | SSP_POSITION_UNIFORM_AFTER
542     | SSP_POSITION_UNIFORM_BEFORE
543     | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
544     | SSP_SAME_DELIMITER_LARGER_AFTER
545     | SSP_SAME_DELIMITER_LARGER_BEFORE
546     | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
547     | SSP_SAME_POSITION_LARGER_AFTER
548     | SSP_SAME_POSITION_LARGER_BEFORE
549     | SSP_UNIFORM
550   ))
551   until [[ LFBFL_i_result_old -eq LFBFL_i_result ]]; do
552     LFBFL_i_result_old=$((LFBFL_i_result))
553     if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER)); then
554       LFBFL_i_result=$((
555         LFBFL_i_result
556         | SSP_LARGER_AFTER
557         | SSP_SAME_DELIMITER_LARGER_AFTER
```

```
558           | SSP_SAME_POSITION_LARGER_AFTER
559         ))
560       fi
561       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE)); then
562         LFBFL_i_result=$((
563           LFBFL_i_result
564           | SSP_LARGER_BEFORE
565           | SSP_SAME_DELIMITER_LARGER_BEFORE
566           | SSP_SAME_POSITION_LARGER_BEFORE
567         ))
568       fi
569       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING)); then
570         LFBFL_i_result=$((
571           LFBFL_i_result
572           | SSP_POSITION_NOT_DECREASING_AFTER
573           | SSP_POSITION_NOT_DECREASING_BEFORE
574         ))
575       fi
576       if ((LFBFL_i_result & SSP_POSITION_UNIFORM)); then
577         LFBFL_i_result=$((
578           LFBFL_i_result
579           | SSP_POSITION_UNIFORM_AFTER
580           | SSP_POSITION_UNIFORM_BEFORE
581           | SSP_POSITION_NOT_DECREASING
582           | SSP_POSITION_NOT_DECREASING_AFTER
583           | SSP_POSITION_NOT_DECREASING_BEFORE
584         ))
585       fi
586       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER)); then
587         LFBFL_i_result=$((
588           LFBFL_i_result
589           | SSP_POSITION_NOT_DECREASING_AFTER
590         ))
591       fi
592       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
593         LFBFL_i_result=$((
594           LFBFL_i_result
595           | SSP_POSITION_NOT_DECREASING_BEFORE
596         ))
597       fi
598       if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
599         LFBFL_i_result=$((
600           LFBFL_i_result
601           | SSP_EQUAL_AFTER_OR_BEFORE
602           | SSP_LARGER_AFTER
603           | SSP_LARGER_BEFORE
604           | SSP_SAME_DELIMITER_LARGER_AFTER
605           | SSP_SAME_DELIMITER_LARGER_BEFORE
606         ))
607       fi
608       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
609         LFBFL_i_result=$((
610           LFBFL_i_result
611           | SSP_LARGER_AFTER
612         ))
613       fi
614       if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
615         LFBFL_i_result=$((
616           LFBFL_i_result
617           | SSP_LARGER_BEFORE
618         ))
```

```
619       fi
620       if ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
621         LFBFL_i_result=$((
622           LFBFL_i_result
623           | SSP_DELIMITER_UNIFORM
624           | SSP_EQUAL_AFTER_OR_BEFORE
625           | SSP_LARGER_AFTER
626           | SSP_LARGER_BEFORE
627           | SSP_SAME_POSITION_LARGER_AFTER
628           | SSP_SAME_POSITION_LARGER_BEFORE
629         ))
630       fi
631       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
632         LFBFL_i_result=$((
633           LFBFL_i_result
634           | SSP_LARGER_AFTER
635         ))
636       fi
637       if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
638         LFBFL_i_result=$((
639           LFBFL_i_result
640           | SSP_LARGER_BEFORE
641         ))
642       fi
643       if ((LFBFL_i_result & SSP_UNIFORM)); then
644         LFBFL_i_result=$((LFBFL_i_all_flags))
645         break
646       fi
647       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_AFTER))\
648       && ((
649         LFBFL_i_result & (
650           SSP_ALWAYS_LARGER_BEFORE
651           | SSP_EQUAL_AFTER_OR_BEFORE
652           | SSP_LARGER_BEFORE
653           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
654           | SSP_SAME_DELIMITER_LARGER_BEFORE
655           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
656           | SSP_SAME_POSITION_LARGER_BEFORE
657         )
658       )); then
659         LFBFL_i_result=$((LFBFL_i_all_flags))
660         break
661       fi
662       if ((LFBFL_i_result & SSP_ALWAYS_LARGER_BEFORE))\
663       && ((
664         LFBFL_i_result & (
665           SSP_EQUAL_AFTER_OR_BEFORE
666           | SSP_LARGER_AFTER
667           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
668           | SSP_SAME_DELIMITER_LARGER_AFTER
669           | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
670           | SSP_SAME_POSITION_LARGER_AFTER
671         )
672       )); then
673         LFBFL_i_result=$((LFBFL_i_all_flags))
674         break
675       fi
676       if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
677       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
678         LFBFL_i_result=$((
679           LFBFL_i_result
```

```
680         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
681       ))
682     fi
683     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
684     && ((LFBFL_i_result & SSP_LARGER_AFTER)); then
685       LFBFL_i_result=$((
686         LFBFL_i_result
687         | SSP_SAME_POSITION_LARGER_AFTER
688       ))
689     fi
690     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
691     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
692       LFBFL_i_result=$((
693         LFBFL_i_result
694         | SSP_SAME_POSITION_LARGER_BEFORE
695       ))
696     fi
697     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
698     && ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE)); then
699       LFBFL_i_result=$((LFBFL_i_all_flags))
700       break
701     fi
702     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
703     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER)); then
704       LFBFL_i_result=$((
705         LFBFL_i_result
706         | SSP_ALWAYS_LARGER_AFTER
707       ))
708     fi
709     if ((LFBFL_i_result & SSP_DELIMITER_UNIFORM))\
710     && ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE)); then
711       LFBFL_i_result=$((
712         LFBFL_i_result
713         | SSP_ALWAYS_LARGER_BEFORE
714       ))
715     fi
716     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
717     && ((LFBFL_i_result & SSP_LARGER_BEFORE)); then
718       LFBFL_i_result=$((
719         LFBFL_i_result
720         | SSP_EQUAL_AFTER_OR_BEFORE
721       ))
722     fi
723     if ((LFBFL_i_result & SSP_LARGER_AFTER))\
724     && ((
725       LFBFL_i_result & (
726         SSP_POSITION_UNIFORM
727         | SSP_POSITION_UNIFORM_AFTER
728       )
729     )); then
730       LFBFL_i_result=$((
731         LFBFL_i_result
732         | SSP_SAME_DELIMITER_LARGER_AFTER
733       ))
734     fi
735     if ((LFBFL_i_result & SSP_LARGER_BEFORE))\
736     && ((
737       LFBFL_i_result & (
738         SSP_POSITION_UNIFORM
739         | SSP_POSITION_UNIFORM_BEFORE
740       )
```

```
741       )); then
742         LFBFL_i_result=$((
743           LFBFL_i_result
744           | SSP_SAME_DELIMITER_LARGER_BEFORE
745         ))
746       fi
747       if ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_AFTER))\
748       && ((LFBFL_i_result & SSP_POSITION_NOT_DECREASING_BEFORE)); then
749         LFBFL_i_result=$((
750           LFBFL_i_result
751           | SSP_POSITION_NOT_DECREASING
752         ))
753       fi
754       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
755       && ((LFBFL_i_result & SSP_EQUAL_AFTER_OR_BEFORE)); then
756         LFBFL_i_result=$((
757           LFBFL_i_result
758           | SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE
759         ))
760       fi
761       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
762       && ((LFBFL_i_result & SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE)); then
763         LFBFL_i_result=$((LFBFL_i_all_flags))
764         break
765       fi
766       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
767       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
768         LFBFL_i_result=$((
769           LFBFL_i_result
770           | SSP_ALWAYS_LARGER_AFTER
771         ))
772       fi
773       if ((LFBFL_i_result & SSP_POSITION_UNIFORM))\
774       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
775         LFBFL_i_result=$((
776           LFBFL_i_result
777           | SSP_ALWAYS_LARGER_BEFORE
778         ))
779       fi
780       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
781       && ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE)); then
782         LFBFL_i_result=$((
783           LFBFL_i_result
784           | SSP_POSITION_UNIFORM
785         ))
786       fi
787       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_AFTER))\
788       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
789         LFBFL_i_result=$((
790           LFBFL_i_result
791           | SSP_ALWAYS_LARGER_AFTER
792         ))
793       fi
794       if ((LFBFL_i_result & SSP_POSITION_UNIFORM_BEFORE))\
795       && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
796         LFBFL_i_result=$((
797           LFBFL_i_result
798           | SSP_ALWAYS_LARGER_BEFORE
799         ))
800       fi
801       if ((LFBFL_i_result & SSP_SAME_DELIMITER_EQUAL_AFTER_OR_BEFORE))\
```

```
802     && ((
803       LFBFL_i_result & (
804         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
805         | SSP_SAME_POSITION_LARGER_AFTER
806         | SSP_SAME_POSITION_LARGER_BEFORE
807       )
808     )); then
809       LFBFL_i_result=$((LFBFL_i_all_flags))
810       break
811     fi
812     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
813     && ((
814       LFBFL_i_result & (
815         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
816         | SSP_SAME_POSITION_LARGER_BEFORE
817       )
818     )); then
819       LFBFL_i_result=$((LFBFL_i_all_flags))
820       break
821     fi
822     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_AFTER))\
823     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER)); then
824       LFBFL_i_result=$((
825         LFBFL_i_result
826         | SSP_ALWAYS_LARGER_AFTER
827       ))
828     fi
829     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
830     && ((
831       LFBFL_i_result & (
832         SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
833         |  SSP_SAME_POSITION_LARGER_AFTER
834       )
835     )); then
836       LFBFL_i_result=$((LFBFL_i_all_flags))
837       break
838     fi
839     if ((LFBFL_i_result & SSP_SAME_DELIMITER_LARGER_BEFORE))\
840     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
841       LFBFL_i_result=$((
842         LFBFL_i_result
843         | SSP_ALWAYS_LARGER_BEFORE
844       ))
845     fi
846     if ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_AFTER))\
847     && ((LFBFL_i_result & SSP_SAME_POSITION_LARGER_BEFORE)); then
848       LFBFL_i_result=$((
849         LFBFL_i_result
850         | SSP_SAME_POSITION_EQUAL_AFTER_OR_BEFORE
851       ))
852     fi
853   done
854   declare -gi i_split_score_properties_logical_closure_result=$((
855     LFBFL_i_result
856   ))
857 }
858
859 split_score_simple(){
860   # $1=larger_after (The name was "after_before" previously,
861   #                  the meaning was not modified.
862   #                  Remark not repeated below.)
```

```
  # $2=delimiters_strings_domain concatenated characters/delimiters
  # $3=delimiter_string a single character
  # $4=is_cut_after
  declare -gi i_split_score_result
  if [[ "$2" != *$3* ]]; then
    i_split_score_result=0
    return
  fi
  if [[ "$4" == "$1" ]]; then
    i_split_score_result=2
    return
  fi
  i_split_score_result=1
}

get_split_score_simple(){
  # $1=larger_after
  # $2=max_length
  # $3=delimiters_strings_domain
  declare -g get_split_score_result="split_score_simple $1 '$3'"
  declare -gi i_get_split_score_result2
  # i_get_split_score_result2=$((11-$1*4))
  # if [[ $1 -eq 1 ]]; then
  #   i_get_split_score_result2=7
  # else
  #   i_get_split_score_result2=11
  # fi
  i_get_split_score_result2=$((
    SSP_DELIMITER_UNIFORM
    + SSP_POSITION_UNIFORM
    + SSP_LARGER_BEFORE
    + (
      $1 * (
        SSP_LARGER_AFTER
        - SSP_LARGER_BEFORE
      )
    )
  ))
  # shellcheck disable=SC2248
  split_score_properties_logical_closure ${i_get_split_score_result2}
  i_get_split_score_result2=$((
    i_split_score_properties_logical_closure_result
  ))
}

get_split_score_exec(){
  declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
  SPLIT_SCORE_EXEC+="/split_score.exec.php"
}

split_score(){
  # $1=larger_after
  # $2=max_length
  # $3=delimiters_strings_domain concatenated characters/delimiters
  # $4=delimiter_string a single character
  # $5=cut_position
  # $6=is_cut_after
  get_split_score_exec
  declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
  declare -gi i_split_score_result
  i_split_score_result=$(eval -- "${LFBFL_command}")
```

```
924 }
925
926 # This function is currently not called,
927 # hence we need this shellcheck disable.
928 # shellcheck disable=SC2034
929 get_split_score(){
930   # $1=larger_after
931   # $1=max_length
932   # $2=delimiters_strings_domain
933   declare -g get_split_score_result="split_score $1 $2 '$3'"
934   declare -gi i_get_split_score_result2
935   # i_get_split_score_result2=$((9-$1*4))
936   i_get_split_score_result2=$((
937     SSP_DELIMITER_UNIFORM
938     + SSP_LARGER_BEFORE
939     + (
940       $1 * (
941         SSP_LARGER_AFTER
942         - SSP_LARGER_BEFORE
943       )
944     )
945   ))
946   # shellcheck disable=SC2248
947   split_score_properties_logical_closure ${i_get_split_score_result2}
948   i_get_split_score_result2=$((
949     i_split_score_properties_logical_closure_result
950   ))
951 }
952
953 split_line_at_most(){
954   # $1=line
955   # $2=max_length of beginning result string
956   # $3=split_score_command
957   #   negative score means do not consider as a possible split
958   # $4=split_score_command_properties :
959   #   These properties are flags.
960   #   If someday some property is not a flag,
961   #   then create split_score_command_properties2.
962   #   Old values to get a simple idea without reading everything at the
963   #   beginning of file:
964   #   - null or 0 no property;
965   #   - 1 split score doesn't depend on which delimiter matches;
966   #   - 2 split score doesn't depend on the current position;
967   #   - 4 any split score after is always larger
968   #     than any split score before;
969   #   - 8 any split score before is always larger
970   #     than any split score after;
971   # $5=forbidden_previous_character
972   # $6=split_score_command_delimiter_strings_domain
973   #   optimisation if small domain
974   #   - this argument can be "null", then we will use $3 only instead,
975   #     in that case, only mono-character strings are considered,
976   #   - this argument can be an array of characters,
977   #     optimize accordingly,
978   #   - this argument can be an array of strings,
979   #     optimize accordingly,
980   #   - this argument can be an array of regexps,
981   #     optimize accordingly.
982   #     Not sure an array of regexps makes more sense than a single
983   #     one. Maybe they could be both cases with corresponding
984   #     optimizations. Same for a single character or substring.
```

```
985     #
986     # Note that we're always looking for the last position to split
987     # among the positions with maximum score;
988     # if some position gets a score from 2 delimiters,
989     # its score is always the maximum of the two corresponding scores.
990     declare -g split_line_at_most_result_start
991     declare -g split_line_at_most_result_end
992     declare -A LFBFL_positions
993     # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
994     # For my use case in Bash scripts, I will need only an array of
995     # characters. See get_split_score().
996     if [[ -n "$6" ]]; then
997       printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
998     fi
999     if [[ $2 -ge ${#1} ]]; then
1000      split_line_at_most_result_start=$1
1001      split_line_at_most_result_end=""
1002      return
1003    fi
1004    # We know that ${#1} > $2, hence ${#1} - 1 >= $2
1005    # Edge cases:
1006    # There is always at least one character that is overlength,
1007    # but we need to look only at the score before that character that is
1008    # overlength, the score after that character is invalid,
1009    # whatever ${#1} - 1 = $2, or ${#1} - 1 > $2.
1010    # That first character that is overlength has index $2.
1011    # The cut position after that character has index $2 + 1.
1012    # Looping from character position going from 0 to $2, we need to drop
1013    # score before character 0 and drop score after character $2.
1014    # We do this with forbidden score -1.
1015    declare -ir LFBFL_i_max=$(($2))
1016    declare -i LFBFL_i
1017    declare -i LFBFL_j
1018    local LFBFL_current_char
1019
1020    if [[ $2 -lt 2 ]]; then
1021      printf "split_line_at_most: max_length should be at least 2...\n"
1022      return
1023    fi
1024    if [[ -n "$5" && "$1" =~ ^"$5"*$ ]]; then
1025      for ((LFBFL_i = 2; LFBFL_i <= $2; LFBFL_i <<= 1)); do
1026        :
1027      done
1028      ((LFBFL_i >>= 1))
1029      split_line_at_most_result_start="${1:0:LFBFL_i}"
1030      split_line_at_most_result_end="${1:LFBFL_i}"
1031      return
1032    fi
1033
1034    # At the beginning, each position has score 0.
1035    LFBFL_positions["0"]="-1"
1036    for ((LFBFL_i = 1; LFBFL_i <= LFBFL_i_max; ++LFBFL_i)) do
1037      LFBFL_positions["${LFBFL_i}"]="0"
1038    done
1039    LFBFL_positions["${LFBFL_i}"]="-1"
1040
1041    # If we have a forbidden previous character,
1042    # we mark the forbidden positions with -1.
1043    if [[ -n "$5" ]]; then
1044      for ((LFBFL_i = 0; LFBFL_i < LFBFL_i_max;)) do
1045        LFBFL_current_char="${1:${LFBFL_i}:1}"
```

```
      ((++LFBFL_i))
      if [[ "${LFBFL_current_char}" == "$5" ]]; then
        LFBFL_positions["${LFBFL_i}"]="-1"
      fi
    done
  fi

  declare -r LFBFL_sort_command="sort --numeric-sort"

  update_position_score(){
    # $1=position
    # Returns 0 if updated, 1 otherwise
    # printf "%s|%s\n" "${i_split_score_result}" "$1"
    if [[ i_split_score_result -ge 1 ]]; then
      # printf "valid result"
      if [[ ${LFBFL_positions["$1"]} != "-1" ]]; then
        LFBFL_positions["$1"]=$(
          max "${LFBFL_sort_command}"\
            "${LFBFL_positions["$1"]}"\
            "${i_split_score_result}"
        )
        return 0
      fi
    fi
    return 1
  }

  local LFBFL_split_score_command=$3
  declare -i LFBFL_split_score_command_properties=$4

  normal_or_position_uniform_call(){
    # $1=cut_position
    # $2=is_cut_after
    if ((LFBFL_split_score_command_properties & SSP_POSITION_UNIFORM));
    then
      eval --\
        "${LFBFL_split_score_command} '${LFBFL_current_char}' $2"
    else
      eval --\
        "${LFBFL_split_score_command} '${LFBFL_current_char}' $1 $2"
    fi
  }

  if (($4 & SSP_DELIMITER_UNIFORM))\
  && (($4 & SSP_LARGER_AFTER))\
  && (($4 & SSP_POSITION_NOT_DECREASING_AFTER)); then
    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      LFBFL_j=$((LFBFL_i + 1))
      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_j} 1
      # shellcheck disable=SC2248
      if update_position_score ${LFBFL_j}
      then
        break
      fi

      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_i} 0
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
```

```
    done
  elif (($4 & SSP_DELIMITER_UNIFORM))\
  && (($4 & SSP_LARGER_BEFORE))\
  && (($4 & SSP_POSITION_NOT_DECREASING_BEFORE)); then
    for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      # We still need to break in second if, after computing score after,
      # because SSP_LARGER_BEFORE implies large inequality instead of
      # strict inequality.
      # And since the algorithm is to take the largest position with
      # maximum score, there is a possibility that we'll break on a found
      # score before but for splitting on a score after.
      LFBFL_j=$((LFBFL_i + 1))
      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_j} 1
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_j}

      # shellcheck disable=SC2248
      normal_or_position_uniform_call ${LFBFL_i} 0
      # shellcheck disable=SC2248
      if update_position_score ${LFBFL_i}
      then
        break
      fi
    done
  else
    for ((LFBFL_i = 0; LFBFL_i <= LFBFL_i_max;)) do
      LFBFL_current_char="${1:${LFBFL_i}:1}"
      eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
      ((++LFBFL_i))
      eval -- "$3 '${LFBFL_current_char}' ${LFBFL_i} 1"
      # shellcheck disable=SC2248
      update_position_score ${LFBFL_i}
    done
  fi
  declare -ir LFBFL_i_max_score=$(
    max 'sort --numeric-sort' "${LFBFL_positions[@]}"
  )
  local LFBFL_value
  declare -i LFBFL_i_best_position
  for ((LFBFL_i = LFBFL_i_max; LFBFL_i >= 0; --LFBFL_i)) do
    if [[ LFBFL_i -eq 0 ]]; then
      printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
    fi
    local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
    # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
    if [[ "${LFBFL_value}" == "${LFBFL_i_max_score}" ]]; then
      LFBFL_i_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_i_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}
```

```
1168
1169 split_last_line(){
1170   # $1=new_lines
1171   # $2=prefix
1172   # $3=max_length
1173   # $4=suffix
1174   # $5=split_score_command
1175   # $6=split_score_command_properties
1176   # $7=forbidden_previous_character ('\' usually)
1177   # Options:
1178   #   --verbose
1179   # In general, a prefix/suffix can be a required line prefix/suffix
1180   # for all final lines like a comment prefix,
1181   # or it can be a continuation line prefix/suffix applied only
1182   # when splitting a line in 2 lines,
1183   # or it can be a prefix/suffix for the string made of all the lines.
1184   # Clearly, there is no gain to couple handling a prefix/suffix for
1185   # the string made of all the lines in this function.
1186   # Since we add a prefix/suffix when splitting,
1187   # it is easy to see that we do not need to distinguish between
1188   # required line prefix/suffix and continuation line prefix/suffix.
1189   # The $2 and $4 arguments are given as concatenation of the
1190   # required and continuation line prefixes/suffixes as needed by the
1191   # user.
1192   # Thus lines arguments should be given already prefixed
1193   # and suffixed.
1194   # This is logic, since it would be cumbersome to handle prefixing
1195   # or not already splitted lines in this function
1196   # (see repeated_split_last_line).
1197   # The given max_length is the goal.
1198   # This function adapts the effective max position for the split by
1199   # taking into account the length of the suffix.
1200   declare -i LFBFL_i_verbose=0
1201   get_verbose_option "$@"
1202   declare -a LFBFL_return_traps_stack
1203   init_return_trap
1204
1205   enhanced_set_shell_option pipefail --trap-unset
1206
1207   declare -g split_last_line_result="$1"
1208   declare -ir LFBFL_i_length2=$(($3 - ${#4}))
1209   get_overlength_regexp "$3"
1210
1211   # Testing if some line is too long.
1212   printf "%s" "$1"\
1213     | grep --quiet "${get_overlength_regexp_result}"
1214   if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
1215     return
1216   fi
1217
1218   declare -r LFBFL_start=$(
1219     printf "%s" "$1"\
1220     | head --lines=-1
1221   )
1222   # printf "start: %s\n" "${LFBFL_start}"
1223   declare -r LFBFL_last_line=$(
1224     printf "%s" "$1"\
1225     | tail --lines=1
1226   )
1227   # printf "last_line: %s\n" "${LFBFL_last_line}"
1228   split_last_line_result=""
```

```
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    # shellcheck disable=SC2248
    split_line_at_most "${LFBFL_last_line}" ${LFBFL_i_length2}\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_i_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_i_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}

repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_i_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_i_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

3.34 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
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```

```
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# This file was renamed from "too_long_code_lines.sh"
# to "too_long_code_lines.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

get_overlength_regexp(){
  # $1=length
  # Option:
  #   --length-is-overlength
  declare -i LFBFL_i_overlength=$1
  declare -i LFBFL_i_length_is_overlength
  get_some_flag LFBFL_i_length_is_overlength --length-is-overlength 1 "$@"
  if [[ LFBFL_i_length_is_overlength -eq 0 ]]; then
    ((++LFBFL_i_overlength))
  fi
  readonly LFBFL_i_overlength
  declare -g get_overlength_regexp_result=".\{${LFBFL_i_overlength}\}"
  # declare -g get_overlength_regexp_result="^.\{${LFBFL_i_overlength}\}"
  # declare -g get_overlength_regexp_result=".\{${LFBFL_i_overlength}\}\$"
  # First one seems to be faster by a few percents when testing like this:
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '^.{70}' COPYING; done
  # time for ((i = 0; i < 1000; ++i)); do grep '.{70}$' COPYING; done
}

too_long_code_lines(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_max_line_length
```

```
  get_max_line_length_option "$@"

  # shellcheck disable=SC2248
  get_overlength_regexp ${LFBFL_i_max_line_length}

  enhanced_set_shell_option pipefail --trap-unset

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
  local LFBFL_pattern
  local LFBFL_long_line
  local LFBFL_file_path
  local LFBFL_line
  local LFBFL_extension
  local LFBFL_base_name
  local LFBFL_s_long_lines
  declare -a LFBFL_arr_long_lines
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "too_long_code_lines: Iterating on pattern: %s.\n"\
        "${LFBFL_pattern}"
    fi
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | tr '\n' '\0'\
      | xargs -0 grep --line-number --with-filename\
        --regexp="${get_overlength_regexp_result}" --
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "too_long_code_lines: No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_path=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      # Remove file_path and line_number
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      LFBFL_line=${LFBFL_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_path##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_path%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.sub"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if [[ ! -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      elif [[ "${LFBFL_extension}" == "neon" ]]; then
        if [[ ! -f "${LFBFL_base_name}.neon.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
```

```
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
}
```

3.35 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

cd .. || exit
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./build_and_checks_dependencies/overwrite_if_not_equal.libr.sh"
cd build_and_checks_dependencies || exit

update_common_build_and_checks(){
  # Option:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  cp -- "./${LFBFL_common_file_name}.tpl"\
    "./${LFBFL_common_file_name}.temp"

```

```
52  declare -ar LFBFL_files_names=(
53    "build_dependencies_notes.exec.php"
54    "build_dependencies_notes.libr.php"
55    "build_file_from_printable_file__standard.exec.sh"
56    "build_md_from_printable_md.exec.sh"
57    "check_shell_scripts_beginnings.libr.sh"
58    "check_shell_scripts_indentation.libr.sh"
59    "check_URLs.libr.sh"
60    "common_options.libr.sh"
61    "comparisons.libr.sh"
62    "create_PDF.exec.sh"
63    "generate_from_template.libr.sh"
64    "get_common_text_glob_patterns.libr.sh"
65    "grammar_and_spelling_check.libr.sh"
66    "licenses_templates/build_licenses_templates.exec.sh"
67    "licenses_templates/license_file_header_AGPLv3+.tpl"
68    "licenses_templates/license_file_header_GPLv3+.tpl"
69    "licenses_templates/license_file_header_LGPLv3+.tpl"
70    "lines_counts.libr.sh"
71    "lines_filters.libr.sh"
72    "lines_maps.libr.sh"
73    "overwrite_if_not_equal.libr.sh"
74    "phpmd_baseline.xslt"
75    "python_black_complement.libr.sh"
76    "python_isort_complement.libr.sh"
77    "repository_data.libr.sh"
78    "shell_checks_complement.libr.sh"
79    "split_score.exec.php"
80    "split_score.libr.php"
81    "strings_functions.libr.sh"
82    "too_long_code_lines.libr.sh"
83    "update_or_check_files_names_listing.exec.sh"
84  )
85
86  local LFBFL_file_name
87  local LFBFL_file_sha512
88  local LFBFL_file_sha512_all
89  local LFBFL_base_file_name
90  for LFBFL_file_name in "${LFBFL_files_names[@]}"; do
91    LFBFL_file_sha512=$(
92      sha512sum "./${LFBFL_file_name}"\
93      | cut --delimiter=' ' --fields=1
94    )
95    LFBFL_file_sha512_all="LFBFL_correct_sha512="
96    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
97    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
98    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
99    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
100   LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
101   LFBFL_base_file_name=$(basename -- "${LFBFL_file_name}")
102   sed --in-place\
103     "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
104     "./${LFBFL_common_file_name}.temp"
105 done
106
107 overwrite_if_not_equal "./${LFBFL_common_file_name}"\
108   "./${LFBFL_common_file_name}.temp"
109
110 declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
111 cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"
112
```

```
  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut --delimiter=' ' --fields=1
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed --in-place\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

3.36 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
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# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
```

```
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  declare -a LFBFL_return_traps_stack
  init_return_trap
  pushd_to_work_directory --trap-popd
  can_continue_after_enhanced_pushd || return 1

  declare -i LFBFL_i_write
  get_some_flag LFBFL_i_write --write 1 "$@"
  declare -i LFBFL_i_append
  get_some_flag LFBFL_i_append --append 1 "$@"

  enhanced_set_shell_option pipefail --trap-unset

  LFBFL_subdir2="build_and_checks_variables"
  declare -r LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
  declare -r LFBFL_data_file_path="./${LFBFL_subdir2}/repository_data.txt"

  declare -i LFBFL_i_max_line_length
  grep_variable "${LFBFL_data_file_path}" max_line_length\
    --result-variable-prefix="LFBFL_i_"

  if [[ LFBFL_i_write -eq 1 ]]; then
    : > "${LFBFL_listing}"
  fi
  # Remove line returns that are here to keep lines short.
  sed --regexp-extended --null-data --expression='s/(.)\\\n/\1/Mg'\
    -- "${LFBFL_listing}"\
    > "${LFBFL_listing}.temp"
  # shopt -s dotglob was needed at some point but I don't see why now.
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_line_length} /
  declare -r LFBFL_split_score_command="${get_split_score_result}"
  declare -i LFBFL_i_split_score_command_properties
  LFBFL_i_split_score_command_properties=${i_get_split_score_result2}
  readonly LFBFL_i_split_score_command_properties
  local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
  readonly LFBFL_suffix
  # There is always a $'\n' final suffix from >>.
  # But when origin string ends with the suffix we must add something.
  declare -r LFBFL_final_suffix=$'\n\\'

  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find . -type f -printf '%P\n'\
    | relevant_find\
    | sort
  )
  # printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
      printf\
        "update_or_check_files_names_listing: Some files to process.\n"
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
  # printf "%s\n" "${LFBFL_arr_files_paths[@]}"

  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    # if [[ LFBFL_i_verbose -eq 1 ]]; then
    #   printf\
    #     "update_or_check_files_names_listing: Maybe ignored file: %s\n"\
    #     "${LFBFL_file_path}"
    # fi
    git check-ignore --quiet -- "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename -- "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "update_or_check_files_names_listing: Non-ignored file: %s\n"\
        "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_i_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      ${LFBFL_i_split_score_command_properties}\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_i_write -eq 1 ]]; then
      # shellcheck disable=SC2001
      printf "%s\n" "${repeated_split_last_line_result}"\
        >> "${LFBFL_listing}"
      continue
    fi
    if ! grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      printf "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
      if [[ LFBFL_i_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_files_paths=$(cat -- "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  fi
```

```
161
162   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
163     [[ "${LFBFL_file_path}" != '//'* ]] || continue
164     if [[ ! -f "${LFBFL_file_path}" ]]; then
165       printf "The non-file %s is listed in %s.\n"\
166         "${LFBFL_file_path}"\
167         "${LFBFL_listing}"
168     fi
169   done
170
171   rm -- "${LFBFL_listing}.temp"
172
173   if [[ LFBFL_i_verbose -eq 1 ]]; then
174     cat -- "${LFBFL_listing}"
175   fi
176 }
177
178 update_or_check_files_names_listing "$@"
```

3.37 build_and_checks_variables/.gitignore

```
1 temp/
2 upgrade_venvs_ts
```

3.38 build_and_checks_variables/bandit.ini

```
1 # FILE: .bandit
2 [bandit]
```

3.39 build_and_checks_variables/bandit_baseline.json

```
1 {"results":[]}
```

3.40 build_and_checks_variables/black.toml

```
1 [tool.black]
2 line-length = 75
```

3.41 build_and_checks_variables/dependencies_data.json

```
1  [
2    { "dependency_name": "Bash", "URLs": [
3        "https://www.gnu.org/software/bash/",
4        "https://tiswww.case.edu/php/chet/bash/bashtop.html",
5        "https://savannah.gnu.org/projects/bash/"
6    ] },
7    { "dependency_name": "GNU coreutils", "URLs": [
8        "https://www.gnu.org/software/coreutils/",
9        "https://savannah.gnu.org/projects/coreutils/"
10   ] },
11   { "dependency_name": "uutils coreutils", "URLs": [
12       "https://uutils.github.io/coreutils/",
13       "https://github.com/uutils/coreutils"
14   ] },
15   { "dependency_name": "diff (GNU diffutils)", "URLs": [
16       "https://www.gnu.org/software/diffutils/",
17       "https://savannah.gnu.org/projects/diffutils/"
18   ] },
19   { "dependency_name": "find, xargs (GNU findutils)", "URLs": [
20       "https://www.gnu.org/software/findutils/",
21       "https://savannah.gnu.org/projects/findutils/"
```

```
    ] },
    { "dependency_name": "grep (GNU grep)", "URLs": [
        "https://www.gnu.org/software/grep/",
        "https://savannah.gnu.org/projects/grep/"
    ] },
    { "dependency_name": "sed (GNU sed)", "URLs": [
        "https://www.gnu.org/software/sed/",
        "https://savannah.gnu.org/projects/sed/"
    ] },
    { "dependency_name": "ShellCheck", "URLs": [
        "https://www.shellcheck.net/",
        "https://github.com/koalaman/shellcheck"
    ] },
    { "dependency_name": "GNU Wget", "URLs": [
        "https://www.gnu.org/software/wget/",
        "https://savannah.gnu.org/projects/wget/"
    ] },
    { "dependency_name": "more of util-linux", "URLs": [
        "https://www.kernel.org/pub/linux/utils/util-linux/",
        "https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/",
        {
          "URL": "https://github.com/util-linux/util-linux",
          "comment": "official?"
        }
    ] },
    { "dependency_name": "pandoc", "URLs": [
        "https://pandoc.org/",
        "https://github.com/jgm/pandoc"
    ] },
    { "dependency_name": "pdftex/pdflatex", "URLs": [
        "https://www.tug.org/applications/pdftex/",
        "https://tug.org/texlive/svn/"
    ] },
    { "dependency_name": "tree", "URLs": [
        "https://oldmanprogrammer.net/source.php?dir=projects/tree",
        "https://gitlab.com/OldManProgrammer/unix-tree",
        "https://github.com/Old-Man-Programmer/tree"
    ] },
    { "dependency_name": "git", "URLs": [
        "https://git-scm.com/about",
        "https://github.com/git/git"
    ] },
    { "dependency_name": "pylint", "URLs": [
        "https://www.pylint.org/",
        "https://pylint.readthedocs.io/en/stable/",
        "https://github.com/pylint-dev/pylint",
        "https://pypi.org/project/pylint/"
    ] },
    { "dependency_name": "notify-send", "URLs": [
        "https://gitlab.gnome.org/GNOME/libnotify/",
        "https://gnome.pages.gitlab.gnome.org/libnotify/"
    ] },
    { "dependency_name": "black", "URLs": [
        "https://github.com/psf/black",
        "https://black.readthedocs.io/en/stable/",
        "https://pypi.org/project/black/"
    ] },
    { "dependency_name": "isort", "URLs": [
        "https://pycqa.github.io/isort/",
        "https://github.com/pycqa/isort/",
        "https://pypi.org/project/isort/"
```

```
  ] },
  { "dependency_name": "mypy", "URLs": [
      "https://mypy-lang.org/",
      "https://mypy.readthedocs.io/en/stable/",
      "https://github.com/python/mypy",
      "https://pypi.org/project/mypy/"
  ] },
  { "dependency_name": "phpdoc (phpDocumentor)", "URLs": [
      "https://phpdoc.org/",
      "https://docs.phpdoc.org/",
      "https://github.com/phpdocumentor/phpdocumentor"
  ] },
  { "dependency_name": "pip", "URLs": [
      "https://pip.pypa.io/en/stable/",
      "https://github.com/pypa/pip",
      {
        "URL": "https://pypi.org/project/pip/",
        "comment": "(egg and chicken ;) )"
      }
  ] },
  { "dependency_name": "ruff", "URLs": [
      "https://docs.astral.sh/ruff/",
      "https://github.com/astral-sh/ruff",
      "https://pypi.org/project/ruff/"
  ] },
  { "dependency_name": "xmllint (libxml2)", "URLs": [
      "https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home",
      "https://gitlab.gnome.org/GNOME/libxml2"
  ] },
  { "dependency_name": "bandit", "URLs": [
      "https://bandit.readthedocs.io/en/latest/",
      "https://github.com/PyCQA/bandit",
      "https://pypi.org/project/bandit/"
  ] },
  { "dependency_name": "phpcs (PHP_CodeSniffer)", "URLs": [
      "https://github.com/PHPCSStandards/PHP_CodeSniffer",
      "https://packagist.org/packages/squizlabs/php_codesniffer"
  ] },
  { "dependency_name": "phpmd", "URLs": [
      "https://phpmd.org/",
      "https://github.com/phpmd/phpmd",
      "https://packagist.org/packages/phpmd/phpmd"
  ] },
  { "dependency_name": "php", "URLs": [
      "https://www.php.net/",
      "https://github.com/php/php-src"
  ] },
  { "dependency_name": "tex", "URLs": [
      "https://ctan.org/tex-archive/systems/knuth/dist/tex"
  ] },
  { "dependency_name": "pcregrep, pcre2grep", "URLs": [
      "https://www.pcre.org/",
      "https://pcre2project.github.io/pcre2/",
      "https://github.com/PCRE2Project/pcre1",
      "https://github.com/PCRE2Project/pcre2",
      "https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
  ] },
  { "dependency_name": "python3", "URLs": [
      "https://www.python.org/",
      "https://docs.python.org/3/"
  ] }
```

```
]
```

3.42 build_and_checks_variables/dependencies_notes.txt

```
Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
```

```
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

phpdoc (phpDocumentor) can be found at the following URLs:
https://phpdoc.org/
https://docs.phpdoc.org/
https://github.com/phpdocumentor/phpdocumentor

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

xmllint (libxml2) can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libxml2/-/wikis/home
https://gitlab.gnome.org/GNOME/libxml2

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpcs (PHP_CodeSniffer) can be found at the following URLs:
https://github.com/PHPCSStandards/PHP_CodeSniffer
https://packagist.org/packages/squizlabs/php_codesniffer

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

```

```
php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

3.43 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts.

DevOrSysAdminScripts is free software:
you can redistribute it and/or modify it
under the terms of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

DevOrSysAdminScripts is distributed
in the hope that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
```

```
39    font-size: 1.17em;
40    margin: 0ex 2ex;
41  }
42  table{
43    border-collapse: collapse;
44  }
45  td{
46    padding: 0;
47  }
48  pre{
49    text-align: left;
50    margin-left: 0ex;
51    margin-right: auto;
52  }
53  pre.numbered_lines:before{
54    background: snow;
55    border-right: 1px black solid;
56    color: #010101;
57    content: attr(line-numbers);
58    float: left;
59    padding: 0 1ch;
60    text-align: right;
61  }
62  blockquote{
63    margin-left: 4ex;
64    margin-right: 4ex;
65    text-align: left;
66  }
67  ol.level1, ol.level2{
68    list-style-type: none;
69    /*
70    list-style-type: decimal;
71    Previous rule is shorter and more efficient if you need only level1.
72    But you don't have an accessible counter if you use decimal.
73    */
74  }
75  ol.level1{
76    counter-reset: level1;
77  }
78  ol.level2{
79    counter-reset: level2;
80  }
81  ol.level1 li:before{
82    content: counter(level1) ". ";
83    counter-increment: level1;
84  }
85  ol.level2 li:before{
86    content: counter(level1) "." counter(level2) " ";
87    counter-increment: level2;
88  }
89  </style>
90  <title>@repository_name@</title>
91  </head>
92  <body >
93  <table class="title">
94    <tr>
95      <td style="padding:1ex">
96        <h1 class="titlemain">@repository_name@</h1>
97        <h2 class="titlerest">
98          @author_full_name@
99          <br>
```

```
100         <a href="@author_website@">
101           <span style="font-family:monospace">
102           @author_website@
103           </span>
104         </a>
105         <br>
106         <a href="mailto:@author_email@">
107           <span style="font-family:monospace">
108           @author_email@
109           </span>
110         </a>
111       </h2>
112     </td>
113   </tr>
114 </table>
115 
116 <blockquote class="abstract">
117   <span style="font-weight:bold">Abstract:</span>
118   @abstract@
119 </blockquote>
120 <p>Current version: @current_date@</p>
121 <p>Current number of commits: @number_of_commits@</p>
122 <p>Current git SHA1: @current_git_SHA1@</p>
123 <p>
124 Code lines: @number_of_lines@
125 </p>
126 
127 
128 <h2>Contents</h2>
129 
130 <ol class="level1">
131   <li><a href="#section1">Files trees</a></li>
132   <li><a href="#section2">Notes on licenses and dependencies</a></li>
133   <li><a href="#section3">Listing of files</a><br>
134     <ol class="level2">
135       @files_lis@
136     </ol>
137   </li>
138 </ol>
139 
140 
141 <h2 id="section1">1 Files trees</h2>
142 <pre class="verbatim">
143 @current_tree_light@
144 </pre>
145 <pre class="verbatim">
146 @current_tree@
147 </pre>
148 
149 
150 <h2 id="section2">2 Notes on licenses and dependencies</h2>
151 
152 <p>
153 This repository is dual licensed.
154 All the Bash scripts source code is covered by GPLv3+.
155 All the PHP scripts source code is covered by LGPLv3+.
156 All the Python scripts source code is covered by LGPLv3+.
157 </p>
158 
159 <p>
160 The text of the licenses is available at:
```

```
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, phpdoc, pip, ruff, xmllint;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpcs, phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

@dependencies_notes@


<h2 id="section3">3 Listing of files</h2>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
```

```
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

3.44 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it
% under the terms of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed
% in the hope that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
% "./build_and_checks_variables".

\if\buildtarget0% A4 paper
\documentclass{article}
\usepackage[a4paper,vmargin=20mm,left=27mm,right=27mm,nohead]{geometry}
\else\if\buildtarget1% US Letter paper
\documentclass{article}
\usepackage[letterpaper,vmargin=20mm,left=27mm,right=27mm,nohead]{geometry}
\else\if\buildtarget2% Book with US Letter paper
\documentclass[twoside]{article}
\usepackage[letterpaper,vmargin=20mm,outer=24mm,inner=24mm,nohead,
bindingoffset=8mm]{geometry}
\fi
\fi
\fi
```

```
42
43  \usepackage[utf8]{inputenc}
44  \usepackage[T1]{fontenc}
45  \usepackage{subfigure}
46  \usepackage{amsmath}
47  \usepackage{amssymb}
48  \usepackage{amsthm}
49  \usepackage[pdftex]{hyperref}
50  \usepackage{tikz}
51  \usepackage{caption}
52  \usepackage[round]{natbib}
53  \usepackage{fancyhdr}
54  \usepackage{amsfonts}
55  \usepackage{times}
56  \usepackage{ifpdf}
57  \usepackage{latexsym}
58  \usepackage{graphicx}
59  \usepackage{enumerate}
60  \usepackage{pmboxdraw}
61  \usepackage{fancyvrb}
62  \usepackage[french,english]{babel}
63
64  % *** les environnements ***
65  %\theoremstyle{break}
66  \newtheorem{definition}{Definition}[section]
67  \newtheorem{proposition}[definition]{Proposition}
68  \newtheorem{theorem}[definition]{Theorem}
69  \newtheorem{lemma}[definition]{Lemma}
70  \newtheorem{corollary}[definition]{Corollary}
71  \newtheorem{remark}[definition]{Remark}
72  \newtheorem{openproblem}[definition]{Open problem}
73
74  \begin{document}
75
76  \author{
77    @author_full_name@\\
78    \url{@author_website@}\\
79    \href{mailto:@author_email@}{@author_email@}
80  }
81  \title{@repository_name@}
82
83  \maketitle
84  \begin{abstract}
85  @abstract@
86  \end{abstract}
87
88  Current version: @current_date@
89
90  Current number of commits: @number_of_commits@
91
92  Current git SHA1: @current_git_SHA1@
93
94  Code lines: @number_of_lines@
95
96  Buildtarget: \buildtarget
97  \if\buildtarget0
98  A4
99  \else\if\buildtarget1% US Letter paper
100 USLetter
101 \else\if\buildtarget2% Book with US Letter paper
102 Book
```

```
\fi
\fi
\fi

\tableofcontents
\newpage

\section{Files trees}
\label{section:trees}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Notes on licenses and dependencies}
\label{section:notes_on}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, phpdoc, pip, ruff, xmllint;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpcs, phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
```

```
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

@dependencies_notes@

\section{Listing of files}
\label{section:listing}

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

3.45 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
archive_libre_office_documents.exec.php
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_dependencies_notes.exec.php
build_and_checks_dependencies/build_dependencies_notes.libr.php
build_and_checks_dependencies/\
build_file_from_printable_file__standard.exec.sh
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
```

```
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spelling_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/phpmd_baseline.xslt
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/dependencies_data.json
build_and_checks_variables/dependencies_notes.txt
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpcs_ruleset.xml
build_and_checks_variables/phpmd_baseline.xml
// build_and_checks_variables/phpstan_baseline.neon
build_and_checks_variables/phpstan_baseline.neon.tpl
// build_and_checks_variables/phpstan_config.neon
build_and_checks_variables/phpstan_config.neon.tpl
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/repository_data.txt.tpl
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
files.libr.php
git_helpers/git_checks.libr.sh
git_helpers/post-commit
```

```
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.exec.py
LiensUtiles.txt
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/grep_benchmark.sh
miscellaneous/spampty_data.txt
miscellaneous/string_escaping.html
miscellaneous/string_escaping.xhtml
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
string_escaping.libr.php
unicode.libr.php
unicode_exceptions.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

3.46 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
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All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, phpdoc, pip, ruff, xmllint;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpcs, phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
```

3.47 build_and_checks_variables/phpcs_ruleset.xml

```
<?xml version="1.0"?>
<ruleset name="LFBFLCustomStandard" namespace="LFBFL\CS\CustomStandard">
  <description>A custom coding standard</description>
  <config name="tab_width" value="2"/>

  <file>../</file>

  <arg name="extensions" value="php"/>
  <arg name="report" value="summary"/>
  <arg name="colors"/>
  <arg value="sp"/>

  <!--
  I added all the standards, then excluded all the rules that gave me
  false positives or that seemed unappropriate when looking at the
  documentation.
  Some are bugs, some are conflicting with my own coding style.
  You may delete an exclude and see what pops up to maybe understand
  my choices.
  -->

  <rule ref="Generic">
    <!-- You have to allow long or short or both syntaxes. -->
    <exclude name="Generic.Arrays.DisallowShortArraySyntax"/>
    <exclude name="Generic.Commenting.DocComment.ParamGroup"/>
    <exclude name="Generic.Commenting.DocComment.TagsNotGrouped"/>
    <exclude name="Generic.Commenting.DocComment.TagValueIndent"/>
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <!-- <exclude name="Generic.PHP.ClosingPHPTag.NotFound"/> -->
    <!--
    Not standard in Unix/Linux world to keep files not ending with \n.
    This is always an option of the code editors.
    Since PHP doesn't output the final \n after ?>, it is ok.
    -->
    <exclude name="Generic.Files.EndFileNoNewline.Found"/>
    <exclude name="Generic.Files.ExecutableFile.Executable"/>
    <exclude name="Generic.Files.OneClassPerFile.MultipleFound"/>
    <exclude name="Generic.Files.OneObjectStructurePerFile.MultipleFound"/>
    <exclude name="Generic.Formatting.MultipleStatementAlignment"/>
    <exclude
     name="Generic.Functions.OpeningFunctionBraceBsdAllman.BraceOnSameLine"
    />
    <exclude
      name="Generic.NamingConventions.CamelCapsFunctionName.NotCamelCaps"
    />
    <exclude
   name="Generic.NamingConventions.CamelCapsFunctionName.ScopeNotCamelCaps"
    />
    <!-- Applies only to true, false and null. -->
    <exclude name="Generic.PHP.LowerCaseConstant.Found"/>
    <exclude name="Generic.PHP.UpperCaseConstant.Found"/>
    <!-- const_bidule_HTML_truc would be better. Everything is prefix. -->
    <exclude name="Generic.WhiteSpace.DisallowSpaceIndent.SpacesUsed"/>
  </rule>
  <rule ref="PEAR">
    <exclude name="PEAR.Classes.ClassDeclaration.OpenBraceNewLine"/>
    <exclude name="PEAR.Commenting.ClassComment.MissingLinkTag"/>
    <exclude name="PEAR.Commenting.FileComment.MissingVersion"/>
    <exclude name="PEAR.Commenting.FileComment.MissingLinkTag"/>
```

```
60      <!--
61      Look at the comment before generate_split_score.
62      PEAR and Squiz require to align the parameter names parts in the same
63      comment, and also the parameter comments parts in the same comment,
64      but with a long parameter name like $a_delimiter_strings_domain
65      and a line length limit at 75 characters, parameter comments parts
66      have only 24 characters left per line, which is ridiculous.
67      Our choice to put the parameter comments parts starting on another line
68      with same "indent" as parameter names parts is way better in those
69      cases. Alignment has always been a trap with short lines.
70      Since phpDocumentor handles the parameter comments parts correctly,
71      it is enough for me.
72      Worse is the handling of "nested doc block comment" for the Closure
73      returned by generate_split_score().
74      Our solution makes at least the source code readable.
75      Maybe one day the dev tools will handle it correctly.
76      -->
77      <exclude name="PEAR.Commenting.FunctionComment.DuplicateReturn"/>
78      <exclude name="PEAR.Commenting.FunctionComment.ExtraParamComment"/>
79      <exclude name="PEAR.Commenting.FunctionComment.MissingParamComment"/>
80      <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamName"/>
81      <exclude name="PEAR.Commenting.FunctionComment.SpacingAfterParamType"/>
82      <exclude name="PEAR.ControlStructures.ControlSignature.Found"/>
83      <exclude name="PEAR.ControlStructures.MultiLineCondition.Alignment"/>
84      <exclude
85        name="PEAR.ControlStructures.MultiLineCondition.SpaceBeforeOpenBrace"
86      />
87      <exclude
88       name="PEAR.ControlStructures.MultiLineCondition.SpacingAfterOpenBrace"
89      />
90      <exclude
91        name="PEAR.ControlStructures.MultiLineCondition.StartWithBoolean"
92      />
93      <!--
94      Don't laugh when looking at the false positives of the two exclude
95      below ;) XD.
96      They don't check for multiline function declarations apparently.
97      -->
98      <exclude
99        name="PEAR.Functions.FunctionDeclaration.NewlineBeforeOpenBrace"
100     />
101     <exclude
102       name="PEAR.Functions.FunctionDeclaration.BraceOnSameLine"
103     />
104     <exclude
105       name="PEAR.Functions.FunctionDeclaration.SpaceBeforeOpenBrace"
106     />
107     <exclude
108       name="PEAR.NamingConventions.ValidFunctionName.FunctionNoCapital"
109     />
110     <exclude
111       name="PEAR.NamingConventions.ValidFunctionName.FunctionNameInvalid"
112     />
113     <exclude
114       name="PEAR.NamingConventions.ValidFunctionName.ScopeNotCamelCaps"
115     />
116     <!--
117     You have to choose between the following exclude or another one
118     for PSR2.
119     -->
120     <exclude
```

```
      name="PEAR.NamingConventions.ValidVariableName.PrivateNoUnderscore"
    />
  </rule>
  <rule ref="PSR1">
    <exclude name="PSR1.Classes.ClassDeclaration.MultipleClasses"/>
    <exclude
      name="PSR1.Methods.CamelCapsMethodName.NotCamelCaps"
    />
  </rule>
  <rule ref="PSR2">
    <exclude name="PSR2.Classes.ClassDeclaration.OpenBraceNewLine"/>
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <exclude name="PSR2.Files.ClosingTag.NotAllowed"/>
    <exclude
name="PSR2.ControlStructures.ControlStructureSpacing.SpacingAfterOpenBrace"
    />
  </rule>
  <rule ref="PSR12">
    <exclude
 name="PSR12.ControlStructures.ControlStructureSpacing.FirstExpressionLine"
    />
    <exclude name="PSR12.Files.FileHeader.SpacingAfterDeclareBlock"/>
    <!--
    3 false positives, see:
    https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1402
    -->
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundAfterDirectiveValue"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundBeforeDirectiveValue"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.SpaceFoundBeforeSemicolon"
    />
    <!-- and even a fourth and fifth false positives -->
    <exclude
      name="PSR12.Files.DeclareStatement.CloseBracketNotAligned"
    />
    <exclude
      name="PSR12.Files.DeclareStatement.ExtraSpaceFoundAfterBracket"
    />
    <exclude name="PSR12.Classes.ClosingBrace.StatementAfter"/>
    <exclude
      name="PSR12.Functions.ReturnTypeDeclaration.SpaceBeforeColon"
    />
    <exclude name="PSR2.Namespaces.NamespaceDeclaration.BlankLineAfter"/>
    <exclude name="PSR2.Namespaces.UseDeclaration.SpaceAfterLastUse"/>
    <!-- I agree except for string concatenation. -->
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceBefore"/>
    <exclude name="PSR12.Operators.OperatorSpacing.NoSpaceAfter"/>
  </rule>
  <rule ref="Squiz">
    <exclude name="Squiz.Arrays.ArrayBracketSpacing.SpaceAfterBracket"/>
    <exclude name="Squiz.Arrays.ArrayBracketSpacing.SpaceBeforeBracket"/>
    <exclude name="Squiz.Arrays.ArrayDeclaration.DoubleArrowNotAligned"/>
    <exclude name="Squiz.Arrays.ArrayDeclaration.SingleLineNotAllowed"/>
    <exclude name="Squiz.Classes.ClassDeclaration.MultipleClasses"/>
    <exclude
      name="Squiz.Classes.ClassDeclaration.NewlinesAfterCloseBrace"
    />
```

```
182     <exclude name="Squiz.Classes.ClassDeclaration.OpenBraceNewLine"/>
183     <exclude name="Squiz.Classes.ClassDeclaration.SpaceBeforeCloseBrace"/>
184     <exclude name="Squiz.Classes.ClassFileName.NoMatch"/>
185     <exclude name="Squiz.Classes.ValidClassName.NotPascalCase"/>
186     <!--
187     It doesn't know the //*/ trick...
188     https://github.com/PHPCSStandards/PHP_CodeSniffer/issues/1403
189     -->
190     <exclude name="Squiz.Commenting.BlockComment.CloserSameLine"/>
191     <exclude name="Squiz.Commenting.BlockComment.FirstLineIndent"/>
192     <exclude name="Squiz.Commenting.BlockComment.LineIndent"/>
193     <exclude name="Squiz.Commenting.BlockComment.NoCapital"/>
194     <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineAfter"/>
195     <exclude name="Squiz.Commenting.BlockComment.NoEmptyLineBefore"/>
196     <exclude name="Squiz.Commenting.BlockComment.SingleLine"/>
197     <!-- Problem for doc block comment before namespace -->
198     <exclude name="Squiz.Commenting.BlockComment.WrongStart"/>
199     <exclude name="Squiz.Commenting.ClassComment.TagNotAllowed"/>
200     <exclude name="Squiz.Commenting.ClosingDeclarationComment.Incorrect"/>
201     <exclude name="Squiz.Commenting.DocCommentAlignment.SpaceBeforeStar"/>
202     <exclude name="Squiz.Commenting.FileComment.CopyrightTagOrder"/>
203     <exclude name="Squiz.Commenting.FileComment.IncorrectAuthor"/>
204     <exclude name="Squiz.Commenting.FileComment.IncorrectCopyright"/>
205     <exclude name="Squiz.Commenting.FileComment.MissingSubpackageTag"/>
206     <exclude name="Squiz.Commenting.FileComment.PackageTagOrder"/>
207     <exclude name="Squiz.Commenting.FileComment.SpacingAfterOpen"/>
208     <exclude name="Squiz.Commenting.FileComment.SubpackageTagOrder"/>
209     <exclude name="Squiz.Commenting.FunctionComment.DuplicateReturn"/>
210     <exclude name="Squiz.Commenting.FunctionComment.EmptyThrows"/>
211     <exclude name="Squiz.Commenting.FunctionComment.ExtraParamComment"/>
212     <!-- asks for boolean instead of bool which is incorrect -->
213     <exclude
214       name="Squiz.Commenting.FunctionComment.IncorrectParamVarName"
215     />
216     <exclude name="Squiz.Commenting.FunctionComment.IncorrectTypeHint"/>
217     <!-- asks for boolean instead of bool which is incorrect -->
218     <exclude name="Squiz.Commenting.FunctionComment.InvalidReturn"/>
219     <exclude name="Squiz.Commenting.FunctionComment.MissingParamComment"/>
220     <exclude name="Squiz.Commenting.FunctionComment.ParamCommentFullStop"/>
221     <exclude
222       name="Squiz.Commenting.FunctionComment.SpacingAfterParamName"
223     />
224     <exclude
225       name="Squiz.Commenting.FunctionComment.SpacingAfterParamType"
226     />
227     <exclude name="Squiz.Commenting.FunctionCommentThrowTag.WrongNumber"/>
228     <!-- Problem for doc block comment before namespace -->
229     <exclude name="Squiz.Commenting.InlineComment.DocBlock"/>
230     <exclude name="Squiz.Commenting.InlineComment.NotCapital"/>
231     <!-- It doesn't know the //*/ trick... -->
232     <exclude name="Squiz.Commenting.InlineComment.NoSpaceBefore"/>
233     <exclude name="Squiz.Commenting.InlineComment.InvalidEndChar"/>
234     <exclude name="Squiz.Commenting.LongConditionClosingComment.Invalid"/>
235     <exclude name="Squiz.Commenting.PostStatementComment.Found"/>
236     <!-- asks for boolean instead of bool which is incorrect -->
237     <exclude name="Squiz.Commenting.VariableComment.IncorrectVarType"/>
238     <exclude
239       name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseBrace"
240     />
241     <exclude
242   name="Squiz.ControlStructures.ControlSignature.SpaceAfterCloseParenthesis"
```

```
    />
    <exclude
      name="Squiz.ControlStructures.ControlSignature.SpaceAfterKeyword"
    />
    <exclude
      name="Squiz.ControlStructures.ForEachLoopDeclaration.SpacingBeforeAs"
    />
    <exclude
      name="Squiz.ControlStructures.InlineIfDeclaration.NotSingleLine"
    />
    <exclude name="Squiz.Files.FileExtension.ClassFound"/>
    <exclude name="Squiz.Formatting.OperatorBracket.MissingBrackets"/>
    <exclude name="Squiz.Functions.GlobalFunction.Found"/>
    <exclude
      name="Squiz.Functions.MultiLineFunctionDeclaration.BraceOnSameLine"
    />
    <exclude
  name="Squiz.Functions.MultiLineFunctionDeclaration.NewlineBeforeOpenBrace"
    />
    <exclude
   name="Squiz.Functions.MultiLineFunctionDeclaration.SpaceBeforeOpenBrace"
    />
    <exclude
      name="Squiz.NamingConventions.ValidFunctionName.NotCamelCaps"
    />
    <exclude
      name="Squiz.NamingConventions.ValidFunctionName.ScopeNotCamelCaps"
    />
    <exclude
      name="Squiz.NamingConventions.ValidVariableName.MemberNotCamelCaps"
    />
    <exclude
      name="Squiz.NamingConventions.ValidVariableName.NotCamelCaps"
    />
    <exclude name="Squiz.Operators.ComparisonOperatorUsage.NotAllowed"/>
    <exclude name="Squiz.Operators.ComparisonOperatorUsage.ImplicitTrue"/>
    <exclude name="Squiz.PHP.CommentedOutCode"/>
    <exclude name="Squiz.PHP.DisallowBooleanStatement.Found"/>
    <exclude
      name="Squiz.PHP.DisallowComparisonAssignment.AssignedComparison"
    />
    <exclude name="Squiz.PHP.DisallowInlineIf.Found"/>
    <exclude name="Squiz.Strings.EchoedStrings.HasBracket"/>
    <exclude
      name="Squiz.WhiteSpace.FunctionClosingBraceSpace.SpacingBeforeClose"
    />
    <exclude name="Squiz.WhiteSpace.FunctionSpacing"/>
    <exclude name="Squiz.WhiteSpace.MemberVarSpacing.FirstIncorrect"/>
    <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceBefore"/>
    <exclude name="Squiz.WhiteSpace.OperatorSpacing.NoSpaceAfter"/>
    <exclude name="Squiz.WhiteSpace.OperatorSpacing.SpacingBefore"/>

  </rule>
  <rule ref="Zend">
    <!-- Back to my habits at Teliae with always a closing tag. -->
    <exclude name="Zend.Files.ClosingTag.NotAllowed"/>
    <exclude
      name="Zend.NamingConventions.ValidVariableName.MemberVarNotCamelCaps"
    />
    <exclude
      name="Zend.NamingConventions.ValidVariableName.NotCamelCaps"
```

```
304       />
305       <exclude
306         name="Zend.NamingConventions.ValidVariableName.PrivateNoUnderscore"
307       />
308     </rule>
309
310     <rule ref="Generic.Arrays.ArrayIndent">
311       <properties>
312         <property name="indent" value="2" />
313       </properties>
314     </rule>
315     <rule ref="Generic.Files.LineLength">
316       <properties>
317         <property name="lineLimit" value="75"/>
318         <property name="absoluteLineLimit" value="75"/>
319       </properties>
320     </rule>
321     <rule ref="Generic.Formatting.SpaceAfterNot">
322       <properties>
323         <property name="spacing" value="0" />
324       </properties>
325     </rule>
326     <rule ref="Generic.Functions.OpeningFunctionBraceKernighanRitchie">
327       <properties>
328         <property name="checkFunctions" value="true"/>
329         <property name="checkClosures" value="true"/>
330       </properties>
331     </rule>
332     <rule ref="Generic.Metrics.CyclomaticComplexity">
333       <properties>
334         <property name="complexity" value="40" />
335         <property name="absoluteComplexity" value="80" />
336       </properties>
337     </rule>
338     <rule ref="Generic.Metrics.NestingLevel">
339       <properties>
340         <property name="nestingLevel" value="7" />
341         <property name="absoluteNestingLevel" value="10" />
342       </properties>
343     </rule>
344     <rule ref="Generic.Strings.UnnecessaryStringConcat">
345       <properties>
346         <property name="allowMultiline" value="true" />
347       </properties>
348     </rule>
349     <rule ref="Generic.WhiteSpace.ArbitraryParenthesesSpacing">
350       <properties>
351         <property name="ignoreNewlines" value="true" />
352       </properties>
353     </rule>
354     <rule ref="Generic.WhiteSpace.ScopeIndent">
355       <properties>
356         <property name="ignoreIndentationTokens" type="array">
357           <element value="T_COMMENT"/>
358           <element value="T_DOC_COMMENT_OPEN_TAG"/>
359         </property>
360         <property name="indent" value="2"/>
361       </properties>
362     </rule>
363
364     <rule ref="PEAR.Functions.FunctionCallSignature">
```

```
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
  <rule ref="PEAR.Functions.FunctionDeclaration">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
  <rule ref="PEAR.WhiteSpace.ScopeIndent">
    <properties>
      <property name="ignoreIndentationTokens" type="array">
        <element value="T_COMMENT"/>
        <element value="T_DOC_COMMENT_OPEN_TAG"/>
      </property>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="PSR2.Methods.FunctionCallSignature">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="PSR12.ControlStructures.ControlStructureSpacing">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>

  <rule ref="Squiz.Classes.ClassDeclaration">
      <properties>
          <property name="indent" value="2" />
      </properties>
  </rule>
  <rule ref="Squiz.Functions.MultiLineFunctionDeclaration">
    <properties>
      <property name="indent" value="2"/>
    </properties>
  </rule>
  <rule ref="Squiz.Strings.ConcatenationSpacing">
    <properties>
      <property name="ignoreNewlines" value="true" />
    </properties>
  </rule>
  <rule ref="Squiz.WhiteSpace.FunctionSpacing">
    <properties>
      <property name="spacing" value="3" />
      <property name="spacingBeforeFirst" value="0" />
      <property name="spacingAfterLast" value="0" />
    </properties>
  </rule>
</ruleset>
```

3.48 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation
    rule="PHPMD\Rule\CleanCode\ElseExpression"
```

```
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\CyclomaticComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Design\LongMethod"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Design\NpathComplexity"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/build_dependencies_notes.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
    method="__construct"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"
    method="__invoke"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation
    rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"
  />
  <violation
    rule="PHPMD\Rule\CleanCode\BooleanArgumentFlag"
    file="files.libr.php"
  />
  <violation
    rule="PHPMD\Rule\CleanCode\ElseExpression"
    file="files.libr.php"
  />
  <violation
```

```
      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
      file="files.libr.php"
      method="add_file_data"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
      file="files.libr.php"
      method="add_file_data_for_file_path"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
      file="files.libr.php"
      method="get_s_content"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="files.libr.php"
      method="__construct"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="files.libr.php"
      method="add_file_data"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="files.libr.php"
      method="add_file_data_for_file_path"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CyclomaticComplexity"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\LongMethod"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\NpathComplexity"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Naming\LongVariable"
      file="files.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="string_escaping.libr.php"
    />
```

```
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="string_escaping.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CleanCode\BooleanArgumentFlag"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CleanCode\ElseExpression"
      file="unicode.libr.php"
    />
    <!-- https://github.com/phpmd/phpmd/issues/910 -->
    <!-- <violation -->
    <!--   rule="PHPMD\Rule\CleanCode\MissingImport" -->
    <!--   file="unicode.libr.php" -->
    <!-- /> -->
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\CyclomaticComplexity"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\LongMethod"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\LongParameterList"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Design\NpathComplexity"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Naming\LongVariable"
      file="unicode.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseMethodName"
      file="unicode_exceptions.libr.php"
      method="get_data"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
      file="unicode_exceptions.libr.php"
      method="__construct"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCasePropertyName"
      file="unicode_exceptions.libr.php"
    />
    <violation
      rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
```

```
    file="unicode_exceptions.libr.php"
  />
</phpmd-baseline>
```

3.49 build_and_checks_variables/phpstan_baseline.neon.tpl

```
parameters:
 ignoreErrors:
  # https://github.com/phpstan/phpstan/issues/14585
  -
   rawMessage: Variable $argv might not be defined.
   identifier: variable.undefined
   count: 1
   path: ../archive_libre_office_documents.exec.php

  -
   rawMessage: Variable $argv might not be defined.
   identifier: variable.undefined
   count: 1
   path: ../build_and_checks_dependencies/build_dependencies_notes.exec.php

  -
   rawMessage: Variable $argv might not be defined.
   identifier: variable.undefined
   count: 1
   path: ../build_and_checks_dependencies/split_score.exec.php

  -
   rawMessage: 'Comparison operation "<" between int<0, 1114111> \
               and 0 is always false.'
   identifier: smaller.alwaysFalse
   count: 1
   path: ../unicode.libr.php

  -
   rawMessage: 'Comparison operation ">" between int<-283305, 283305> \
               and 1114111 is always false.'
   identifier: greater.alwaysFalse
   count: 1
   path: ../unicode.libr.php

  -
   rawMessage: 'Parameter #1 $i_unicode_code_point of function \
               DOSAS_unicode\unicode_code_point_to_UTF8 expects \
               int<0, 1114111>, int<-1114095, 1114111> given.'
   identifier: argument.type
   count: 1
   path: ../unicode.libr.php

  -
   rawMessage: 'Parameter #1 $i_unicode_code_point of function \
               DOSAS_unicode\unicode_code_point_to_UTF8 expects \
               int<0, 1114111>, int<-283305, 283305> given.'
   identifier: argument.type
   count: 1
   path: ../unicode.libr.php

  -
   rawMessage: 'Parameter #2 $i_current_octet of function \
               DOSAS_unicode\get_message_and_data_array expects int, \
               int<0, 191>|int<194, 244>|int<246, 254>|null given.'
```

```
    identifier: argument.type
    count: 1
    path: ../unicode.libr.php
```

3.50 build_and_checks_variables/phpstan_config.neon.tpl

```
includes:
    - phpstan_baseline.neon
    - /home/laurent/.config/composer/vendor/\
      phpstan/phpstan/conf/bleedingEdge.neon

parameters:
    level: max

    paths:
        - ..

    checkTooWideReturnTypesInProtectedAndPublicMethods: true
```

3.51 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

3.52 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+
```

```
15
16 author_email=Laurent.Lyaudet@gmail.com
17 author_full_name=Laurent Frédéric Bernard François Lyaudet
18 author_last_name=Lyaudet
19 author_name=Laurent Lyaudet
20 author_website=https://lyaudet.eu/laurent/
21
22 acknowledgments=Merci Dieu !
23 acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !
24
25 Great day! Now I allow myself lines of 75 characters instead of 70 :).
26 I checked how many characters fit in the PDF file generated by tex with
27 default paper size,
28 when using verbatim environment with line numbers for source code listing.
29 With a screen capture in GIMP,
30 I mesured 197px between two-digits line numbers and the left border of
31 the page.
32 (It is less than 197px for three-digits or more line numbers.)
33 The same distance from the right border yields 75 characters allowed.
34 Since when printing the HTML file, the number of allowed characters in
35 <pre> tags is way larger, 75 is the right limit with the tools I use.
36
37 max_line_length=75
38
39 latex_command=pdflatex
40
41 You can build multiple targets of the source listing according to various
42 formats and sizes.
43 Specify a jobname suffix (that will become a filename suffix)
44 and an integer that will set a buildtarget macro, separated with ':'.
45 latex_targets=A4:0
46 latex_targets=USLetter:1
47 latex_targets=Book:2
48
49 grammar_and_spelling_checker_command=
50
51
52 Why one venv for each Python dependency ?
53 Because no language of my knowledge did it right to handle all
54 possible dependency conflicts.
55 So in case of such a conflict, one specialized venv can get you out
56 of any problem.
57 The best solution would have been to allow install of different
58 versions of the same package, and have a mechanism:
59 - default behavior, use the latest available version of a package among
60 those that satisfy the declared dependency constraints of the package
61 that has this dependency;
62 it requires that all dependency constraints are declared;
63 - custom behavior, check a dependency overload configuration file
64 with tree structure: if calling python package is A version 1.1.1
65 with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
66 you could have along that another rule saying that when called
67 directly B 2.2.2 should use C 4.4.4;
68 and even more complicated:
69 if calling python package is A version 1.1.1 with dependency B and D
70 both themselves with dependency C, use B 2.2.2, D 0.1.0,
71 and C 3.3.3 for B, and C 4.4.4 for D;
72 once you've seen that it can never be more complicated than that once
73 the same version number guarantees that the code is the same,
74 you create a config language "à la CSS" for cascading dependencies ;)
75 XD fit into JSON the dumbest possible way and you stop bothering
```

```
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "3.3.3" : {}
          }
        }
      },
      "D": {
        "0.1.0": {
          "C": {
            // Dumb, you did not found it by yourself? Failed exam.
            "4.4.4" : {}
          }
        }
      }
    }
  },
  "B": {
    "2.2.2": {
      "C": {
        // Dumb, you did not found it by yourself? Failed exam.
        "4.4.4" : {}
      }
    }
  }
}
Wonderful someone rediscovered nested structures XD
Seriously so many engineers or computer scientists are dumb to cry.

bandit_venv=../venv_bandit
black_venv=../venv_black
isort_venv=../venv_isort
mypy_venv=../venv_mypy
pylint_venv=../venv_pylint
ruff_venv=../venv_ruff


upgrade venvs can be auto or ask, any other value is interpreted
as ask.

upgrade_venvs=ask


time interval can be 0 to always upgrade or ask to upgrade,
or a number at most bash max int
https://superuser.com/questions/1030122/
what-is-the-maximum-value-of-a-numeric-bash-shell-variable
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat
```

```

For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800

use_vnu_for_xhtml=false

For xmllint_files_without_DTD and files_without_license_header,
each line is a file path, except if the line ends with a /,
then it is concatenated with the following line(s) to obtain a path.

xmllint_files_without_DTD=./build_and_checks_variables/phpcs_ruleset.xml
xmllint_files_without_DTD=./build_and_checks_variables/phpmd_baseline.xml

If you use Ubuntu/Debian, install w3c-sgml-lib instead of =true.
See https://gitlab.gnome.org/GNOME/libxml2/-/work_items/1119

xmllint_use_downloaded_DTDs=false

files_without_license_header=build_and_checks_variables/phpcs_ruleset.xml
files_without_license_header=build_and_checks_variables/
files_without_license_header=phpmd_baseline.xml
files_without_license_header=README.html
files_without_license_header=miscellaneous/string_escaping.html
files_without_license_header=build_and_checks_variables/ruff.toml
files_without_license_header=build_and_checks_variables/black.toml
files_without_license_header=build_and_checks_variables/pylintrc.toml
```

3.53 build_and_checks_variables/repository_data.txt.tpl

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.
The goal of this template is to list all possible variables.

repository_name=
repository_web_url=
repository_git_url=

abstract=
abstract=

license=
license2=

author_email=
author_full_name=
author_last_name=
author_name=
author_website=

acknowledgments=
acknowledgments=
```

```
max_line_length=

latex_command=

latex_targets=
latex_targets=

grammar_and_spelling_checker_command=

bandit_venv=
black_venv=
isort_venv=
mypy_venv=
pylint_venv=
ruff_venv=

upgrade_venvs=

upgrade_venvs_time_interval_in_seconds=

vnu_jar_path=

upgrade_vnu_jar_time_interval_in_seconds=

use_vnu_for_xhtml=

npm_lint_directories=
npm_lint_directories=

xmllint_files_without_DTD=
xmllint_files_without_DTD=

xmllint_use_downloaded_DTDs=

files_without_license_header=
files_without_license_header=
```

3.54 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
```

```
  # debugged at some point I commented a part that was needed in
  # another case."...
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  "PLR0917",  # too-many-positional-arguments AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.exec.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

3.55 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"
```

3.56 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
           False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
           False positive when c contains \n.
- SC2016 : Expressions don't expand in single quotes,
           use double quotes for that.
- SC2034 : Unused variable
           False positive for globals that are declared in functions
           that are not currently called.
           Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
           Use printf '..%s..' "$foo".
           False positive when using a variable to build the format string.
           Usually because the format string is very long.
```

```
- SC2086 : Double quote to prevent globbing and word splitting.
           False positive when returning integer variable
           via an indirection like return ${!foo}.
           See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
           False positive when passing arguments would yield
           a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
           False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
           False positive when assigning a variable declared with -i.
           https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
           False positive when using "$@" line return and pipes.
           https://github.com/koalaman/shellcheck/issues/3192
           Example:
             # shellcheck disable=SC2145
             all_self_code_lines "$@"\
               | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
           Very useful but false positive when I play with
           variable variable names.
- SC2162X: read without -r will mangle backslashes.
           False positive when read is used without variable:
           https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
           False positive on returning integer values.
           https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
             Use = to compare as string (or expand explicitly with $var).
           Very useful but false positive when I play with
           variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
           value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
           https://github.com/koalaman/shellcheck/issues/3419

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2016$" -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$"\
  -e "=SC2120$" -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$"\
  -e "=SC2309$" -e "=SC2312$" -e "=SC2154,SC2309$"
```

3.57 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it
# under the terms of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed
# in the hope that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

```
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  # $1=work_directory
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --verbose
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir --parents "${LFBFL_subdir}/licenses_templates"
  local LFBFL_variables_directory
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  readonly LFBFL_variables_directory
  mkdir --parents -- "${LFBFL_variables_directory}/temp"
  if [[ ! -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\nupgrade_venvs_ts\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
```

```
  local LFBFL_script_download_URL
  LFBFL_script_download_URL="${LFBFL_dependencies_URL}/"
  LFBFL_script_download_URL+="${LFBFL_common_file_name}"
  readonly LFBFL_script_download_URL
  declare -r LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='c090463990b393a10177e9d0db26c59d45711bfdb4e08'
  LFBFL_correct_sha512+='1c7b9da75a9d14275fe5edffe1a1cddd460b6d0c92a8'
  LFBFL_correct_sha512+='d6eb76450550797861badac33847d8b24d26c67'
  wget_sha512 "${LFBFL_file_path}"\
    "${LFBFL_script_download_URL}"\
    "${LFBFL_correct_sha512}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
  chmod +x "${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}"\
    "${LFBFL_verbose}"\
    "${LFBFL_check_download}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch -- "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch -- "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

3.58 COPYING

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How to Apply These Terms to Your New Programs

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to attach them to the start of each source file to most effectively
state the exclusion of warranty; and each file should have at least
the "copyright" line and a pointer to where the full notice is found.

```
    <one line to give the program's name and a brief idea of what it does.>
    Copyright (C) <year>  <name of author>

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```

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```
    <program>  Copyright (C) <year>  <name of author>
    This program comes with ABSOLUTELY NO WARRANTY; for details type 'show w'.
    This is free software, and you are welcome to redistribute it
    under certain conditions; type 'show c' for details.
```

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3.59 COPYING.LESSER

```
                   GNU LESSER GENERAL PUBLIC LICENSE
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  An "Application" is any work that makes use of an interface provided
by the Library, but which is not otherwise based on the Library.
Defining a subclass of a class defined by the Library is deemed a mode
of using an interface provided by the Library.

  A "Combined Work" is a work produced by combining or linking an
Application with the Library.  The particular version of the Library
with which the Combined Work was made is also called the "Linked
Version".

  The "Minimal Corresponding Source" for a Combined Work means the
Corresponding Source for the Combined Work, excluding any source code
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object code and/or source code for the Application, including any data
and utility programs needed for reproducing the Combined Work from the
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that uses the facility (other than as an argument passed when the
facility is invoked), then you may convey a copy of the modified
version:

   a) under this License, provided that you make a good faith effort to
   ensure that, in the event an Application does not supply the
   function or data, the facility still operates, and performs
   whatever part of its purpose remains meaningful, or
```

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1) Use a suitable shared library mechanism for linking with the Library. A suitable mechanism is one that (a) uses at run time a copy of the Library already present on the user's computer system, and (b) will operate properly with a modified version of the Library that is interface-compatible with the Linked Version.

e) Provide Installation Information, but only if you would otherwise be required to provide such information under section 6 of the GNU GPL, and only to the extent that such information is necessary to install and execute a modified version of the Combined Work produced by recombining or relinking the

```
    Application with a modified version of the Linked Version. (If
    you use option 4d0, the Installation Information must accompany
    the Minimal Corresponding Source and Corresponding Application
    Code. If you use option 4d1, you must provide the Installation
    Information in the manner specified by section 6 of the GNU GPL
    for conveying Corresponding Source.)

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Library side by side in a single library together with other library
facilities that are not Applications and are not covered by this
License, and convey such a combined library under terms of your
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    on the Library, uncombined with any other library facilities,
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    is a work based on the Library, and explaining where to find the
    accompanying uncombined form of the same work.

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whether future versions of the GNU Lesser General Public License shall
apply, that proxy's public statement of acceptance of any version is
permanent authorization for you to choose that version for the
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```

3.60 files.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

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under the terms of the GNU Lesser General Public License
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either version 3 of the License,
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DevOrSysAdminScripts is distributed
```

```
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22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26
27 @category Library
28 @package DevOrSysAdminScripts
29 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
30 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
31 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
32 */
33
34 declare(strict_types=1);
35 declare(encoding='UTF-8');
36
37
38
39 /**
40 The namespace DOSASfiles contain additional classes and functions
41 to deal with files.
42
43 @package DevOrSysAdminScripts
44 @subpackage DOSASfiles
45 */
46 namespace DOSASfiles {
47
48   use Exception;
49
50
51
52
53   /**
54   Kind of a "dataclass" to keep the informations related with a file,
55   slightly enhanced with:
56   - a constructor that extract the distinct parts of the filepath;
57   - a method/getter handling the file content loading.
58
59   @category Library
60   @package DevOrSysAdminScripts
61   @subpackage DOSASfiles
62   @class FileData
63   @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
64   @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
65   */
66   final class FileData {
67     /**
68     The hash function to use for file contents.
69
70     @var string $s_hash_function
71     */
72     public static string $s_hash_function = 'md5';
73
74     /**
```

```
75      The filepath of the file.
76      https://www.youtube.com/watch?v=2NRlZ1ZME5U
77      Why?
78      https://github.com/squizlabs/PHP_CodeSniffer/issues/258
79
80      @var string $s_filepath
81      */
82      public string $s_filepath;
83
84      /**
85      The dirname of the filepath.
86
87      @var string $s_dirname
88      */
89      public string $s_dirname;
90
91      /**
92      The basename of the filepath.
93
94      @var string $s_basename
95      */
96      public string $s_basename;
97
98      /**
99      The filename of the basename.
100
101     @var string $s_filename
102     */
103     public string $s_filename;
104
105     /**
106     The extension of the basename.
107
108     @var string $s_extension
109     */
110     public string $s_extension;
111
112     /**
113     The filesubpath starting from a reference directory.
114
115     @var string $s_path_from_reference_directory
116     */
117     public string $s_path_from_reference_directory;
118
119     /**
120     The size of the content of the file.
121
122     @var int $i_size
123     */
124     public int $i_size;
125
126     /**
127     The hash of the content of the file.
128
129     @var string $s_hash
130     */
131     public string $s_hash;
132
133     /**
134     The content of the file as a string.
135
```

```
136         @var string $s_content
137         */
138         public $s_content;
139 
140         /**
141         Statuses relative to another directory : 1.
142 
143         Does the file exists with same relative path and same content
144         At The Same Time (ATST)?
145 
146         @var bool $b_exists_with_same__relative_path_and_content
147         */
148         public bool $b_exists_with_same__relative_path_and_content; // ATST
149 
150         /**
151         Statuses relative to another directory : 2.
152 
153         Does the file exists with same relative path?
154 
155         @var bool $b_exists_with_same_relative_path
156         */
157         public bool $b_exists_with_same_relative_path;
158 
159         /**
160         Statuses relative to another directory : 3.
161 
162         Does the file exists with same content?
163 
164         @var bool $b_exists_with_same_content
165         */
166         public bool $b_exists_with_same_content;
167 
168         /**
169         Statuses relative to another directory : 4.
170 
171         Does the file exists with suffixed relative path?
172         Only computed if some regular expression for valid suffixes
173         is supplied.
174         An empty suffix is always considered valid,
175         but it is handled if your regexp doesn't match the empty string.
176 
177         @var bool $b_exists_with_suffixed_relative_path
178         */
179         public bool $b_exists_with_suffixed_relative_path;
180 
181         /**
182         Statuses relative to another directory : 5.
183 
184         Does the file exists with suffixed relative path and same content
185         At The Same Time (ATST)?
186         Only computed if some regular expression for valid suffixes
187         is supplied.
188         An empty suffix is always considered valid,
189         but it is handled if your regexp doesn't match the empty string.
190 
191         @var bool $b_exists_with_suffixed_relative_path_and
192         */
193         public bool $b_exists_with_suffixed_relative_path_and; // ATST
194 
195 
196 
```

```
197       /**
198       Constructor for FileData.
199
200       @param string $s_filepath The filepath of the file.
201       @param string $s_reference_directory_path
202                     The directorypath of the reference directory.
203
204       @return void
205       */
206       public function __construct(
207         string $s_filepath,
208         string $s_reference_directory_path,
209       ) {
210         $this->s_filepath = $s_filepath;
211         $path_parts = pathinfo($this->s_filepath);
212         $this->s_dirname = (
213           isset($path_parts['dirname']) ? $path_parts['dirname'] : ''
214         );
215         $this->s_basename = $path_parts['basename'];
216         $this->s_filename = $path_parts['filename'];
217         if(!isset($path_parts['extension'])){
218           $this->s_extension = '';
219         }
220         else{
221           $this->s_extension = $path_parts['extension'];
222           if($this->s_filename === ''){
223             // Hidden file
224             $this->s_filename = $this->s_basename;
225             $this->s_extension = '';
226           }
227         }
228         $this->s_path_from_reference_directory = substr(
229           $this->s_filepath,
230           strlen($s_reference_directory_path),
231         );
232         // To enforce a load attempt.
233         $this->i_size = 1;
234         $this->s_hash = '';
235         $this->s_content = '';
236         $this->get_s_content();
237         $this->b_exists_with_same__relative_path_and_content = false;
238         $this->b_exists_with_same_relative_path = false;
239         $this->b_exists_with_same_content = false;
240         $this->b_exists_with_suffixed_relative_path = false;
241         $this->b_exists_with_suffixed_relative_path_and = false;
242       }//end __construct() FileData
243
244
245
246       /**
247       Returns the content of the file as a string after fetching it from
248       the filesystem.
249
250       @throws \Exception When the content of the file cannot be retrieved
251                         from the filesystem.
252
253       @return string
254       */
255       public function get_s_content() : string {
256         if($this->i_size !== strlen($this->s_content)){
257           $s_content = file_get_contents($this->s_filepath);
```

```
258         if($s_content === false){
259           throw new Exception(
260             'Unexpected failure when fetching content of '
261             .$this->s_filepath
262           );
263         }
264         $this->s_content = $s_content;
265         if($this->s_hash === ''){
266           $this->i_size = strlen($this->s_content);
267           $this->s_hash = md5($this->s_content);
268         }
269       }
270       return $this->s_content;
271     }//end get_s_content()
272   }//end class FileData
273
274
275
276
277   /**
278   Kind of a "dataclass" to keep the informations related with the files
279   under a directory.
280
281   @category Library
282   @package DevOrSysAdminScripts
283   @subpackage DOSASfiles
284   @class DataOfFilesUnderDirectory
285   @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
286   @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
287   */
288   final class DataOfFilesUnderDirectory {
289     /**
290     The path of the reference directory.
291
292     @var string $s_reference_directory_path
293     */
294     public string $s_reference_directory_path;
295
296     /**
297     The data of the files under the reference directory
298     put in order by absolute path.
299
300     @var array<string, FileData> $arr_files_by_path
301     */
302     public array $arr_files_by_path;
303
304     /**
305     The data of the files under the reference directory
306     put in order by path from reference directory.
307
308     @var array<string, FileData> $arr_files_by_path_from
309     */
310     public array $arr_files_by_path_from;
311
312     /**
313     The data of the files under the reference directory
314     put in order by size, hash, and order of addition to this object.
315
316     @var array<int, array<string, array<int, FileData>>> $arr_files_by_size
317     */
318     public array $arr_files_by_size;
```

```
319
320
321
322     /**
323     Constructor for DataOfFilesUnderDirectory.
324
325     @param string $s_reference_directory_path
326                   The directorypath of the reference directory.
327
328     @return void
329     */
330     public function __construct(string $s_reference_directory_path) {
331       $this->s_reference_directory_path = $s_reference_directory_path;
332       $this->arr_files_by_path = [];
333       $this->arr_files_by_path_from = [];
334       $this->arr_files_by_size = [];
335     }//end __construct() DataOfFilesUnderDirectory
336
337
338
339     /**
340     Adds a FileData object in the arrays used for fast access by
341     different keys.
342
343     @param FileData $o_file_data The FileData object to add.
344
345     @return void
346     */
347     public function add_file_data(FileData $o_file_data) : void {
348       $this->arr_files_by_path[$o_file_data->s_filepath] = $o_file_data;
349       $this->arr_files_by_path_from[
350         $o_file_data->s_path_from_reference_directory
351       ] = $o_file_data;
352       if(!isset($this->arr_files_by_size[$o_file_data->i_size])){
353         $this->arr_files_by_size[$o_file_data->i_size] = [];
354       }
355       if(
356         !isset(
357           $this->arr_files_by_size[$o_file_data->i_size][
358             $o_file_data->s_hash
359           ]
360         )
361       ){
362         $this->arr_files_by_size[$o_file_data->i_size][
363           $o_file_data->s_hash
364         ] = [];
365       }
366       $this->arr_files_by_size[$o_file_data->i_size][
367         $o_file_data->s_hash
368       ] []= $o_file_data;
369     }//end add_file_data()
370
371
372
373     /**
374     Create a FileData object for the given FilePath.
375     And adds this FileData object in the arrays used for fast access by
376     different keys.
377
378     @param string $s_filepath The filepath required.
379
```

```
380       @return void
381       */
382       public function add_file_data_for_file_path(
383         string $s_filepath
384       ) : void {
385         $o_file_data = new FileData(
386           $s_filepath,
387           $this->s_reference_directory_path,
388         );
389         $o_file_data->s_content = '';
390         $this->add_file_data($o_file_data);
391       }//end add_file_data_for_file_path()
392     }//end class DataOfFilesUnderDirectory
393
394
395
396
397     /**
398     Loads the data of files under a given directory,
399     but doesn't keep the contents loaded.
400     Recusivity is not implemented yet.
401
402     @param string $s_dirpath The directory to search in.
403
404     @return \DOSASfiles\DataOfFilesUnderDirectory
405     */
406     function load_files_data_under_directory(
407       string $s_dirpath,
408     ) : DataOfFilesUnderDirectory {
409       $o_data_of_files_under_directory = new DataOfFilesUnderDirectory(
410         $s_dirpath
411       );
412       $arr_s_paths = glob($s_dirpath.'*');
413       if($arr_s_paths === false){
414         return $o_data_of_files_under_directory;
415       }
416       foreach($arr_s_paths as $s_filepath){
417         $o_data_of_files_under_directory->add_file_data_for_file_path(
418           $s_filepath
419         );
420       }//end foreach($arr_s_paths as $s_filepath)
421       return $o_data_of_files_under_directory;
422     }//end load_files_data_under_directory()
423
424
425
426     /**
427     Loads the data of files under two given directories,
428     but doesn't keep the contents loaded.
429     And then compares the files under both directories.
430     Recusivity is not implemented yet.
431
432     @param string $s_source_dirpath The first directory to search in.
433     @param string $s_dest_dirpath The second directory to search in.
434     @param bool   $b_compute_also_reverse_status
435                   If you want that the status of files under dest_dirpath
436                   must also be computed.
437     @param string $s_valid_suffixes_regexp A regexp for valid suffixes.
438
439     @return array{
440       'source_files_data': \DOSASfiles\DataOfFilesUnderDirectory,
```

```
441       'dest_files_data': \DOSASfiles\DataOfFilesUnderDirectory
442     }
443     */
444     function compare_files_under_directories(
445       string $s_source_dirpath,
446       string $s_dest_dirpath,
447       bool $b_compute_also_reverse_status = false,
448       string $s_valid_suffixes_regexp = '',
449     ) : array {
450       $o_files_data_source = load_files_data_under_directory(
451         $s_source_dirpath
452       );
453       $o_files_data_dest = load_files_data_under_directory(
454         $s_dest_dirpath
455       );
456       foreach($o_files_data_source->arr_files_by_path_from as $o_file_data){
457         if(
458           isset(
459             $o_files_data_dest->arr_files_by_path_from[
460               $o_file_data->s_path_from_reference_directory
461             ]
462           )
463         ){
464           $o_file_data->b_exists_with_same_relative_path = true;
465           $o_file_data2 = (
466             $o_files_data_dest->arr_files_by_path_from[
467               $o_file_data->s_path_from_reference_directory
468             ]
469           );
470           $o_file_data2->b_exists_with_same_relative_path = true;
471           $o_file_data->b_exists_with_suffixed_relative_path = true;
472           $o_file_data2->b_exists_with_suffixed_relative_path = true;
473           if(
474             $o_file_data2->i_size === $o_file_data->i_size
475             && $o_file_data2->s_hash === $o_file_data->s_hash
476             && $o_file_data2->get_s_content()
477             === $o_file_data->get_s_content()
478           ){
479             $o_file_data->b_exists_with_same__relative_path_and_content = (
480               true
481             );
482             $o_file_data->b_exists_with_same_content = true;
483             $o_file_data->b_exists_with_suffixed_relative_path_and = (
484               true
485             );
486             $o_file_data2->b_exists_with_same__relative_path_and_content = (
487               true
488             );
489             $o_file_data2->b_exists_with_same_content = true;
490             $o_file_data2->b_exists_with_suffixed_relative_path_and = (
491               true
492             );
493             $o_file_data->s_content = '';
494             $o_file_data2->s_content = '';
495             continue;
496           }
497         }//end if(isset($o_files_data_dest->arr_files_by_path_from[...])
498         if(
499           isset($o_files_data_dest->arr_files_by_size[$o_file_data->i_size])
500           && isset(
501             $o_files_data_dest->arr_files_by_size[$o_file_data->i_size][
```

```
502             $o_file_data->s_hash
503           ]
504         )
505       ){
506         foreach(
507            $o_files_data_dest->arr_files_by_size[$o_file_data->i_size][
508             $o_file_data->s_hash
509           ]
510           as $o_file_data2
511         ){
512           if(
513             $o_file_data2->get_s_content()
514             === $o_file_data->get_s_content()
515           ){
516             $o_file_data->b_exists_with_same_content = true;
517             $o_file_data2->b_exists_with_same_content = true;
518             $o_file_data->s_content = '';
519             $o_file_data2->s_content = '';
520             if($s_valid_suffixes_regexp !== ''){
521               if(
522                 preg_match(
523                   '/'
524                   .preg_quote(
525                     $o_file_data->s_path_from_reference_directory,
526                     '/',
527                   )
528                   .$s_valid_suffixes_regexp
529                   .'/',
530                   $o_file_data2->s_path_from_reference_directory,
531                 )
532               ){
533                 $o_file_data->b_exists_with_suffixed_relative_path = true;
534                 $o_file_data->b_exists_with_suffixed_relative_path_and = (
535                   true
536                 );
537               }
538               if(
539                 $b_compute_also_reverse_status
540                 && preg_match(
541                   '/'
542                   .preg_quote(
543                     $o_file_data2->s_path_from_reference_directory,
544                     '/',
545                   )
546                   .$s_valid_suffixes_regexp
547                   .'/',
548                   $o_file_data->s_path_from_reference_directory,
549                 )
550               ){
551                 $o_file_data2->b_exists_with_suffixed_relative_path = true;
552                 $o_file_data2->b_exists_with_suffixed_relative_path_and = (
553                   true
554                 );
555               }
556             }//end if($s_valid_suffixes_regexp !== '')
557           }//end if(file_data2->content == file_data->content)
558         }//end foreach(file with same size and same hash)
559       }//end if(there is some file with same size and same hash)
560     }//end foreach($arr_files_data_source[...] as $o_file_data)
561
562     if($b_compute_also_reverse_status){
```

```
563         foreach($o_files_data_dest->arr_files_by_path_from as $o_file_data){
564           if($o_file_data->b_exists_with_suffixed_relative_path){
565             continue;
566           }
567           $arr_s_paths = glob(
568             $s_source_dirpath.$o_file_data->s_basename.'*'
569           );
570           if($arr_s_paths === false){
571             continue;
572           }
573           foreach($arr_s_paths as $s_filepath){
574             if(!isset($o_files_data_source->arr_files_by_path[$s_filepath])){
575               // A newly created file that wasn't previously loaded
576               // is ignored.
577               continue;
578             }
579             $o_file_data2 = (
580               $o_files_data_source->arr_files_by_path[$s_filepath]
581             );
582             if(
583               preg_match(
584                 '//'
585                 .preg_quote(
586                   $o_file_data->s_path_from_reference_directory,
587                   '/',
588                 )
589                 .$s_valid_suffixes_regexp
590                 .'//',
591                 $o_file_data2->s_path_from_reference_directory,
592               )
593             ){
594               $o_file_data->b_exists_with_suffixed_relative_path = true;
595             }
596           }//end foreach($arr_s_paths as $s_filepath)
597         }//end foreach($arr_files_data_dest[...] as $o_file_data)
598       }//end if($b_compute_also_reverse_status)
599
600       return [
601         'source_files_data' => $o_files_data_source,
602         'dest_files_data' => $o_files_data_dest,
603       ];
604     }//end compare_files_under_directories()
605
606
607
608     /**
609     Archive some directory files into another directory.
610
611     @param string $s_source_dirpath The source directory to search in.
612     @param string $s_dest_dirpath The archive directory to move files in.
613     @param bool   $b_archive_if_missing_relative_filepath
614                   Archive if no destination file is found with a relative
615                   path that is identical.
616     @param bool   $b_archive_if_missing_content
617                   Archive if no destination file is found with same content.
618     @param bool   $b_archive_if_missing_same_relative_path_content
619                   Archive if no destination file is found with a relative
620                   path that is identical and with same content.
621     @param bool   $b_archive_if_missing_suffixed_relative_path
622                   Archive if no destination file is found with a relative
623                   path that has a valid suffix
```

```
624                 after the source relative path.
625     @param bool   $b_archive_if_missing_suffixed_relative_path_content
626                 Archive if no destination file is found with a relative
627                 path that has a valid suffix
628                 after the source relative path
629                 and with same content.
630     @param string $s_valid_suffixes_regexp A regexp for valid suffixes.
631
632     @throws \Exception When $s_valid_suffixes_regexp is required
633                       and not supplied.
634
635     @return void
636     */
637     function archive_directory_into_another(
638       string $s_source_dirpath,
639       string $s_dest_dirpath,
640       bool $b_archive_if_missing_relative_filepath,
641       bool $b_archive_if_missing_content,
642       bool $b_archive_if_missing_same_relative_path_content,
643       bool $b_archive_if_missing_suffixed_relative_path,
644       bool $b_archive_if_missing_suffixed_relative_path_content,
645       string $s_valid_suffixes_regexp = '',
646     ) : void {
647       if(
648         (
649           $b_archive_if_missing_suffixed_relative_path
650           || $b_archive_if_missing_suffixed_relative_path_content
651         )
652         && $s_valid_suffixes_regexp === ''
653       ){
654         throw new Exception(
655           'archive_directory_into_another is missing an'
656           .' $s_valid_suffixes_regexp argument.'
657         );
658       }
659       $arr_files_data = compare_files_under_directories(
660         $s_source_dirpath,
661         $s_dest_dirpath,
662         false,
663         $s_valid_suffixes_regexp,
664       );
665       foreach(
666         $arr_files_data['source_files_data']->arr_files_by_path_from
667         as $o_file_data
668       ){
669         if(
670           (
671             $b_archive_if_missing_relative_filepath
672             && !$o_file_data->b_exists_with_same_relative_path
673           )
674           ||
675           (
676             $b_archive_if_missing_content
677             && !$o_file_data->b_exists_with_same_content
678           )
679           ||
680           (
681             $b_archive_if_missing_same_relative_path_content
682             && !$o_file_data->b_exists_with_same__relative_path_and_content
683           )
684           ||
```

```
          (
            $b_archive_if_missing_suffixed_relative_path
            && !$o_file_data->b_exists_with_suffixed_relative_path
          )
          ||
          (
            $b_archive_if_missing_suffixed_relative_path_content
            && !$o_file_data->b_exists_with_suffixed_relative_path_and
          )
        ){
          shell_exec(
            'cp --backup=numbered'
            ." '".preg_replace("/'/", "'\"'\"'", $o_file_data->s_filepath)
            ."'"
            ." '".preg_replace("/'/", "'\"'\"'", $s_dest_dirpath)."'"
          );
        }
      }//end foreach($arr_files_data['source_files_data'][] as $o_file_data)
    }//end archive_directory_into_another()
}//end namespace DOSASfiles



/*
$ php
<?php
require_once 'files.libr.php';
var_dump(
  DOSASfiles\compare_files_under_directories(
    '/home/laurent/.config/libreoffice/4/user/backup/',
    '/home/laurent/Documents/LibreOfficeArchives/',
  )
);
DOSASfiles\archive_directory_into_another(
  '/home/laurent/.config/libreoffice/4/user/backup/',
  '/home/laurent/Documents/LibreOfficeArchives/',
  false,
  true,
  false,
  false,
  false,
  '',
);
*/
?>
```

3.61 git_helpers/git_checks.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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```

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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
  declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations."
  local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
  LFBFL_send_body_2+=" dès la création du modèle."
  readonly LFBFL_send_body_2
  declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
  declare -r LFBFL_send_body_3=\
"Les nouveaux preloadings sont interdits : -> #Prefetch()."
  declare -i LFBFL_i_error=0
  git diff --cached --name-only\
    > /tmp/DOSAS_django_git_check1.temp
  if grep models /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
    LFBFL_i_error=1
  fi
  if grep serializer /tmp/DOSAS_django_git_check1.temp;
  then
    notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check1.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_no_abusive_trailing_comma(){
```

```
  # Useful for Python code, although it regularly yields
  # false positives in Python code
  # (but it is easy to distinguish by hand).
  # Use `git commit --no-verify` after check.
  declare -r LFBFL_send_summary_1="ATTENTION"
  local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
  LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
  readonly LFBFL_send_body_1
  declare -i LFBFL_i_error=0
  git diff --cached\
    > /tmp/DOSAS_django_git_check2.temp
  if grep --perl-regexp ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
  then
    notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
    LFBFL_i_error=1
  fi
  rm /tmp/DOSAS_django_git_check2.temp
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_s_files_paths=$(
    git diff --cached --name-only\
    | grep '\.py$'
  )
  printf "%s\n" "${LFBFL_s_files_paths}"
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  declare -i LFBFL_i_error=0
  local LFBFL_file_path
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Black will check formatting on file %s.\n" "${LFBFL_file_path}"
    if ! black --check --diff -- "${LFBFL_file_path}";
    then
      LFBFL_i_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

3.62 git_helpers/post-commit

```
#!/usr/bin/env bash
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#
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#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "post-commit pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=post_commit_renaming_comment.libr.sh
source "./git_helpers/post_commit_renaming_comment.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./build_and_checks_dependencies/lines_filters.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "post-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

3.63 git_helpers/post_commit_renaming_comment.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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```

```
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath -- "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname -- "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_i_pushd_result2
pushd -- "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result2=$?
  printf\
    "post_commit_renaming_comment.libr.sh pushd %s no such directory\n"\
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result2}
}
# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./build_and_checks_dependencies/strings_functions.libr.sh"
declare -i LFBFL_i_popd_result2
popd || {
  LFBFL_i_popd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result2}
}

# Never forget that the worse security is the illusion of security
# (Kevin Mitnick). So, do not expect to have an accurate trackability
# of code history just because you use this post-commit hook or
# something else more evolved. This is useful but not a guarantee.
# This function will only handle the files that were only renamed in the
# preceding commit.
# It assumes that `diff` is the current default difference utility
# parametered with `git`. This point could be enhanced.
```

```
commit_a_file_renamed_comment(){
  # Options:
  #   --verbose
  #   --log-directory-change
  #   --max-comment-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  declare -i LFBFL_i_log_directory_change
  get_some_flag LFBFL_i_log_directory_change --log-directory-change 1 "$@"
  declare -i LFBFL_i_max_comment_line_length
  get_some_option LFBFL_i_max_comment_line_length\
    --max-comment-line-length 70 70 1 "$@"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    LFBFL_s_format="LFBFL_i_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_i_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059,SC2248
    printf "${LFBFL_s_format}"\
      "${LFBFL_i_log_directory_change}"\
      ${LFBFL_i_max_comment_line_length}
  fi
  if [[ LFBFL_i_max_comment_line_length -lt 10 ]]; then
    declare -r LFBFL_send_summary="Max comment line length below 10."
    notify-send "${LFBFL_send_summary}" "${LFBFL_send_summary}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "Max comment line length below 10, aborting.\n"
    fi
    return
  fi
  local LFBFL_s_format
  #----------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log --max-count=1 --patch\
    | grep --after-context=1 '^diff --git'
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  local LFBFL_split_score_command1
  declare -i LFBFL_i_split_score_command_properties1
  # shellcheck disable=SC2248
  get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} _
  LFBFL_split_score_command1="${get_split_score_result}"
  LFBFL_i_split_score_command_properties1=${i_get_split_score_result2}
  readonly LFBFL_split_score_command1
  readonly LFBFL_i_split_score_command_properties1
  local LFBFL_split_score_command2
  declare -i LFBFL_i_split_score_command_properties2
  # shellcheck disable=SC2248
```

```
129   get_split_score_simple 1 ${LFBFL_i_max_comment_line_length} /
130   LFBFL_split_score_command2="${get_split_score_result}"
131   LFBFL_i_split_score_command_properties2=${i_get_split_score_result2}
132   readonly LFBFL_split_score_command2
133   readonly LFBFL_i_split_score_command_properties2
134   declare -i LFBFL_i_renaming_happened=0
135   declare -i LFBFL_i
136   declare -i LFBFL_j
137   declare -r LFBFL_special_file_name="files_names_listing.txt"
138   local LFBFL_diff_line
139   local LFBFL_similarity_line
140   local LFBFL_old_file_path
141   local LFBFL_old_file_name
142   local LFBFL_old_file_directory
143   local LFBFL_new_file_path
144   local LFBFL_new_file_name
145   local LFBFL_new_file_directory
146   local LFBFL_useful_file_name
147   local LFBFL_extension
148   local LFBFL_comment_prefix
149   local LFBFL_comment_prefix2
150   local LFBFL_old_file_name2
151   local LFBFL_old_file_directory2
152   local LFBFL_new_file_name2
153   local LFBFL_new_file_directory2
154   local LFBFL_new_comment
155   declare -i LFBFL_i_copyright_line_number
156   declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
157   declare -i LFBFL_i_line_count
158   declare -i LFBFL_i_lines_after
159   local LFBFL_temp_file_path
160   for ((LFBFL_i = 0; LFBFL_i < ${#LFBFL_renaming_lines_array[@]};)); do
161     LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
162     LFBFL_i=$((LFBFL_i + 1))
163     LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
164     LFBFL_i=$((LFBFL_i + 1))
165     # We need to increment 3 times since grep adds "--" lines between.
166     LFBFL_i=$((LFBFL_i + 1))
167     if [[ LFBFL_i_verbose -eq 1 ]]; then
168       printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
169         "${LFBFL_diff_line}"\
170         "${LFBFL_similarity_line}"
171     fi
172     if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
173       continue
174     fi
175
176     # Extract old file name from diff line. -------------------------
177     LFBFL_old_file_path=$(
178       printf "%s" "${LFBFL_diff_line}"\
179       | cut --delimiter=' ' --fields=3
180     )
181     # It starts by "a/".
182     LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
183     LFBFL_old_file_name=$(basename -- "${LFBFL_old_file_path}")
184     LFBFL_old_file_directory=$(dirname -- "${LFBFL_old_file_path}")
185     # Extract new file name from diff line. -------------------------
186     LFBFL_new_file_path=$(
187       printf "%s" "${LFBFL_diff_line}"\
188       | cut --delimiter=' ' --fields=4
189     )
```

```
    # It starts by "b/".
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename -- "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname -- "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. -----------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j = 0; LFBFL_j < 3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[
      "${LFBFL_extension}" == "sh"
      || "${LFBFL_extension}" == "toml"
      || "${LFBFL_extension}" == "yaml"
    ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[
      "${LFBFL_useful_file_name}" == "${LFBFL_special_file_name}"
      || "${LFBFL_extension}" == "rs"
      || "${LFBFL_extension}" == "rlib"
    ]];
    then
```

```
251       LFBFL_comment_prefix="// "
252     fi
253     if [[ LFBFL_i_verbose -eq 1 ]]; then
254       printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
255     fi
256     #-----------------------------------------------------------------
257     # Split if necessary old_file_name, new_file_name, etc. to have
258     # comment lines below max_comment_line_length.
259     LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
260     # -----
261     LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
262     LFBFL_old_file_name2+="${LFBFL_old_file_name}"
263     # shellcheck disable=SC2248
264     repeated_split_last_line "${LFBFL_old_file_name2}"\
265       "${LFBFL_comment_prefix2}"\
266       ${LFBFL_i_max_comment_line_length}\
267       '"'\
268       "${LFBFL_split_score_command1}"\
269       ${LFBFL_i_split_score_command_properties1}
270     LFBFL_old_file_name2="${repeated_split_last_line_result}"
271     # -----
272     LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
273     LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
274     # shellcheck disable=SC2248
275     repeated_split_last_line "${LFBFL_old_file_directory2}"\
276       "${LFBFL_comment_prefix2}"\
277       ${LFBFL_i_max_comment_line_length}\
278       '"'\
279       "${LFBFL_split_score_command2}"\
280       ${LFBFL_i_split_score_command_properties2}
281     LFBFL_old_file_directory2="${repeated_split_last_line_result}"
282     # -----
283     LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
284     LFBFL_new_file_name2+="${LFBFL_new_file_name}"
285     # shellcheck disable=SC2248
286     repeated_split_last_line "${LFBFL_new_file_name2}"\
287       "${LFBFL_comment_prefix2}"\
288       ${LFBFL_i_max_comment_line_length}\
289       '"'\
290       "${LFBFL_split_score_command1}"\
291       ${LFBFL_i_split_score_command_properties1}
292     LFBFL_new_file_name2="${repeated_split_last_line_result}"
293     # -----
294     LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
295     LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
296     # shellcheck disable=SC2248
297     repeated_split_last_line "${LFBFL_new_file_directory2}"\
298       "${LFBFL_comment_prefix2}"\
299       ${LFBFL_i_max_comment_line_length}\
300       '"'\
301       "${LFBFL_split_score_command2}"\
302       ${LFBFL_i_split_score_command_properties2}
303     LFBFL_new_file_directory2="${repeated_split_last_line_result}"
304     # Create new comments strings. ---------------------------------
305     LFBFL_new_comment=""
306     if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
307       LFBFL_new_comment="${LFBFL_comment_prefix}"
308       LFBFL_new_comment+="${LFBFL_timestamp}"
309       LFBFL_new_comment+=": This file was renamed from"
310       LFBFL_new_comment+=$'\n'
311       LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
```

```
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[
      LFBFL_i_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
    ]];
    then
      if [[ "${LFBFL_new_file_name}" != "${LFBFL_old_file_name}" ]]; then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. ------------------------------------
    LFBFL_i_copyright_line_number=$(
      grep --line-number -- '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut --delimiter=':' --fields=1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_copyright_line_number: %s\n"\
        "${LFBFL_i_copyright_line_number}"
    fi
    if [[ LFBFL_i_copyright_line_number -eq 0 ]]; then
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. ------------------------------------
    LFBFL_i_line_count=$(
      wc --lines\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_line_count: %s\n" "${LFBFL_i_line_count}"
    fi
    # Compute number of lines after. --------------------------------
    LFBFL_i_lines_after=$((
      LFBFL_i_line_count - LFBFL_i_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_i_lines_after: %s\n" "${LFBFL_i_lines_after}"
    fi
    # Update file. --------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
```

```
    {
      head --lines="${LFBFL_i_copyright_line_number}"\
        -- "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_i_lines_after}" -- "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv -- "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. -------------------------------------------
    git add -- "${LFBFL_new_file_path}"
    LFBFL_i_renaming_happened=1
  done
  readonly LFBFL_i_renaming_happened
  if [[ LFBFL_i_renaming_happened -eq 1 ]]; then
    # Commit -------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    printf "Commit message:\n%s\n" "${LFBFL_message}"
    git commit --no-verify --message="${LFBFL_message}"
  fi
}
```

3.64 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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# of the GNU General Public License
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# either version 3 of the License,
# or (at your option) any later version.
#
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# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_some_directory="../DevOrSysAdminScripts"
declare -i LFBFL_i_pushd_result3
pushd "${LFBFL_some_directory}" || {
  LFBFL_i_pushd_result3=$?
  printf "pre-commit pushd %s no such directory\n"\
```

```
    "${LFBFL_some_directory}"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_pushd_result3}
}
# shellcheck source=git_checks.libr.sh
source "./git_helpers/git_checks.libr.sh"
declare -i LFBFL_i_popd_result3
popd || {
  LFBFL_i_popd_result3=$?
  printf "pre-commit no popd\n"
  # shellcheck disable=SC2248
  exit ${LFBFL_i_popd_result3}
}

declare -gi LFBFL_i_git_hook_error=0
if ! check_files; then
  LFBFL_i_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_i_git_hook_error=1
fi
if ! check_black_code_formatting; then
  LFBFL_i_git_hook_error=1
fi
readonly LFBFL_i_git_hook_error
# shellcheck disable=SC2248
exit ${LFBFL_i_git_hook_error}
```

3.65 IMAP_transfert.exec.py

```
"""
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©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
2026-05-12T12:06+02:00: This file was renamed from
"IMAP_transfert.py"
to
"IMAP_transfert.exec.py".

Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""
```

```
32
33  import argparse
34  import email
35  import imaplib
36
37
38  class IMAPTransfertBadArgumentError(Exception):
39      """
40      A class to avoid a linter warning,
41      that needs an Error suffix to avoid another linter warning,
42      that needs a docstring to avoid another linter warning.
43      If you don't see its intent, try badminton.
44      """
45
46
47  class SurpriseError(Exception):
48      """
49      A class to avoid a linter warning,
50      that needs an Error suffix to avoid another linter warning,
51      that needs a docstring to avoid another linter warning.
52      If you don't see its intent after seeing its use, try volleyball.
53      But "Cheer up!", now you don't need "pass" XD.
54      """
55
56
57  def IMAP_transfert(
58      from_login: str,
59      from_password: str,
60      from_host: str,
61      from_port: int,
62      to_login: str,
63      to_password: str,
64      to_host: str,
65      to_port: int,
66      *,
67      to_new_folder: str = "",
68      verbose: bool = False,
69  ) -> None:
70      """
71      Transfert all emails from some IMAP account
72      to another IMAP account.
73      """
74      empty = ""
75      if from_login == "":
76          raise IMAPTransfertBadArgumentError("From login is empty.")
77      # if from_password == "":
78      if from_password == empty:
79          # https://github.com/PyCQA/bandit/issues/714
80          # Dumb
81          raise IMAPTransfertBadArgumentError("From password is empty.")
82      if from_host == "":
83          raise IMAPTransfertBadArgumentError("From host is empty.")
84      if from_port <= 0:
85          raise IMAPTransfertBadArgumentError(
86              "From port is not positive.",
87          )
88      if to_login == "":
89          raise IMAPTransfertBadArgumentError("To login is empty.")
90      if to_password == empty:
91          # And Dumber The infinite collection XD
92          raise IMAPTransfertBadArgumentError("To password is empty.")
```

```
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print('Connecting to "from account".')
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print('Creating new folder in "to account".')
        to_account.create(to_new_folder)

    if verbose:
        print('Fetching all emails ids in "from account".')
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print('Fetching and transfering each email in "from account".')
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
```

```
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
        ),
    )
    parser.add_argument(
        "from_login",
        type=str,
        help="Email account login of user one.",
    )
    parser.add_argument(
        "from_password",
        type=str,
        help="Email account password of user one.",
    )
    parser.add_argument(
        "to_login",
        type=str,
        help="Email account login of user two.",
    )
    parser.add_argument(
        "to_password",
        type=str,
        help="Email account password of user two.",
    )
    parser.add_argument(
        "--to_new_folder",
        type=str,
        default="",
        help="Transfert emails into a new folder of this name.",
    )
    parser.add_argument(
        "--from_host",
        type=str,
        default="imap.gmail.com",
        help="Host of first email account",
    )
    parser.add_argument(
        "--from_port",
        type=int,
```

```
215         default=993,
216         help="Port on host of first email account",
217     )
218     parser.add_argument(
219         "--to_host",
220         type=str,
221         default="imap.gmail.com",
222         help="Host of second email account",
223     )
224     parser.add_argument(
225         "--to_port",
226         type=int,
227         default=993,
228         help="Port on host of second email account",
229     )
230     parser.add_argument(
231         "-v",
232         "--verbose",
233         action="store_true",
234         help="Print details of operations",
235     )
236
237     args = parser.parse_args()
238
239     if args.verbose:
240         print(
241             "-----------------------------------------------------\n"
242             "IMAP transfert:\n"
243             "-----------------------------------------------------\n"
244             "From account:\n"
245             f" - Login: {args.from_login}\n"
246             f" - Password: {args.from_password}\n"
247             f" - Host: {args.from_host}\n"
248             f" - Port: {args.from_port}\n"
249             "To account:\n"
250             f" - Login: {args.to_login}\n"
251             f" - Password: {args.to_password}\n"
252             f" - Host: {args.to_host}\n"
253             f" - Port: {args.to_port}\n"
254             "-----------------------------------------------------\n",
255         )
256         if args.to_new_folder is not None:
257             print(
258                 f"Emails will go in folder {args.to_new_folder}"
259                 " after it is created.",
260             )
261     try:
262         IMAP_transfert(
263             args.from_login,
264             args.from_password,
265             args.from_host,
266             args.from_port,
267             args.to_login,
268             args.to_password,
269             args.to_host,
270             args.to_port,
271             to_new_folder=args.to_new_folder,
272             verbose=args.verbose,
273         )
274     except (IMAPTransfertBadArgumentError, SurpriseError) as e:
275         print(e)
```

```
276
277
278 if __name__ == "__main__":
279     main()
```

3.66 LiensUtiles.txt

```
1  https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
2  -configurez-le-service-de-migration-de-donn%C3%A9es
3  ------------------------------------------------------------------------
4  https://stackoverflow.com/a/78413334/5796086
5  https://stackoverflow.com/questions/21612980/\
6  why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
7  https://web.archive.org/web/20240501105801/https://stackoverflow.com\
8  /questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
9  78413334#78413334
10 ------------------------------------------------------------------------
11 https://news.ycombinator.com/item?id=22214690
12 https://github.com/koalaman/shellcheck/issues/2005
13 https://opensource.stackexchange.com/questions/6271/\
14 calling-gpld-programs-from-a-shell-script
15 ------------------------------------------------------------------------
16 https://pubs.opengroup.org/onlinepubs/9699919799/
17 When looking `man sed` GNU sed says to use POSIX -E instead of
18 -r or --regexp-extended for portability.
19 But the link above says that still now only -e, -f, -n are POSIX options
20 for sed.
21 For this reason we replaced all short option names by long option names
22 when possible, even for sed.
```

3.67 miscellaneous/funny.sh

```
1  #!/usr/bin/env bash
2  # This file is part of DevOrSysAdminScripts.
3  #
4  # DevOrSysAdminScripts is free software:
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9  # or (at your option) any later version.
10 #
11 # DevOrSysAdminScripts is distributed
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14 # without even the implied warranty of
15 # MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
16 # See the GNU General Public License for more details.
17 #
18 # You should have received a copy of
19 # the GNU General Public License
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24
25 unprotected_funny(){
26   # Works in a script but not in shell if job control is active.
27   shopt -s lastpipe
28   declare -i LFBFL_i_result=0
29   local LFBFL_line
```

```
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

3.68 miscellaneous/grep_benchmark.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
#
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# the GNU General Public License
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# If not, see <https://www.gnu.org/licenses/>.
#
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some_diff(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='<!-- '\
        --regexp='^[0-9]\+d[0-9]\+$'\
        --regexp='^[0-9]\+,[0-9]\+d[0-9]\+$'
  done
}

some_diff2(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
```

```
        --regexp='<!-- '\
        --regexp='^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$'
  done
}

some_diff3(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='\(<!-- \|^[0-9]\+d[0-9]\+$\|^[0-9]\+,[0-9]\+d[0-9]\+$\)'
  done
}

some_diff4(){
  for ((i=0; i < 1000; ++i)); do
    # shellcheck disable=SC2312
    diff --suppress-common-lines\
      build_and_checks_variables/phpmd_baseline.xml\
      build_and_checks_variables/temp/phpmd_baseline.xml\
      | grep --invert-match\
        --regexp='\(<!-- \|^[0-9]\+\(,[0-9]\+\)\?d[0-9]\+$\)'
  done
}

#$ time some_diff
# real        0m1,663s
# user        0m0,795s
# sys        0m1,975s
#$ time some_diff2
# real        0m1,597s
# user        0m0,705s
# sys        0m2,009s
#$ time some_diff3
# real        0m1,619s
# user        0m0,758s
# sys        0m1,969s
#$ time some_diff4
# real        0m1,574s
# user        0m0,750s
# sys        0m1,932s
```

3.69 miscellaneous/spampty_data.txt

```


a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

3.70 miscellaneous/string_escaping.html

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
  <meta charset="utf-8">
  <title>Truc</title>
</head>
<body>
<header id="title-block-header">
<h1 class="title">Truc</h1>
</header>
&lt;abc> &amp; &lt;de>
<p title="<&>'
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script>
let a = 'abcd & < > \\ \n  \"\' \u{00003c}/script>';
alert(a);
let b = "abcd & < > \\ \n  \"\'";
alert(b);
</script>

</body>
</html>
```

3.71 miscellaneous/string_escaping.xhtml

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml" xml:lang="fr" lang="fr">
<head>
<meta content="text/html; charset=UTF-8" http-equiv="Content-Type" />
  <title>Truc</title>
</head>
<body>
<h1 class="title">Truc</h1>
&lt;abc> &amp; &lt;de>

<p title="&lt;&amp;>'&#10;
 abcd`\ machin">
This is some text
</p>

<pre>
"&lt;&amp;>'&quot;
abcd`\ machin"
This is some text
</pre>

<script type="text/javascript">
let a = 'abcd &amp; &lt; > \\ \n  \"\'';
alert(a);
let b = "abcd &amp; &lt; > \\ \n  \"\'";
alert(b);
</script>

```

```
</body>
</html>
```

3.72 README.md.tpl

```
# DevOrSysAdminScripts

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[![Linting:ruff:badge]][Ruff-GitHub.com]

|                **Some "small" useful scripts**                |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- But the files "pre-commit" and "post-commit" were not renamed,
  since it is not possible to give them other names to use them as
  pre-commit/post-commit hooks in `git`.
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
```

```
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.
- We used `_?i_` or `_?j_`, etc. in integer variables names in Bash
  scripts.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

[CodeFactor-package-page]: https://www.codefactor.io/repository/\
github/llyaudet/DevOrSysAdminScripts

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3dd6ef85e8db4460991d1ebc404b68dd

[Codacy-package-page]: https://app.codacy.com/gh/LLyaudet/\
DevOrSysAdminScripts/dashboard?utm_source=gh&utm_medium=referral\
&utm_content=&utm_campaign=Badge_grade

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languages/top/llyaudet/DevOrSysAdminScripts

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languages/code-size/llyaudet/DevOrSysAdminScripts

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security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

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[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/

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mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

3.73 string_escaping.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.
```

```
5
6  DevOrSysAdminScripts is free software:
7  you can redistribute it and/or modify it
8  under the terms of the GNU Lesser General Public License
9  as published by the Free Software Foundation,
10 either version 3 of the License,
11 or (at your option) any later version.
12
13 DevOrSysAdminScripts is distributed
14 in the hope that it will be useful,
15 but WITHOUT ANY WARRANTY;
16 without even the implied warranty of
17 MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
18 See the GNU Lesser General Public License for more details.
19
20 You should have received a copy of
21 the GNU Lesser General Public License
22 along with DevOrSysAdminScripts.
23 If not, see <https://www.gnu.org/licenses/>.
24
25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26
27 @category Library
28 @package DevOrSysAdminScripts
29 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
30 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
31 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
32 */
33
34 declare(strict_types=1);
35 declare(encoding='UTF-8');
36
37
38
39 /**
40 The namespace string_escaping is splitted into subnamespaces for each
41 language.
42 This subnamespace is for HTML (surprised?)...
43
44 @package DevOrSysAdminScripts
45 @subpackage string_escaping\HTML
46 */
47 namespace string_escaping\HTML {
48   /**
49   When you want that the string appears as is in HTML text block.
50
51   @param string $s_string The string you want to escape.
52
53   @return string The escaped string.
54   */
55   function escape_text(string $s_string) : string {
56     return str_replace(
57       [
58         '&',
59         '<',
60       ],
61       [
62         '&amp;',
63         '&lt;',
64       ],
65       $s_string,
```

```
66      );
67    }//end escape_text()
68
69
70
71    /**
72    When you want that the string appears as is in an HTML tag attribute
73    that starts and ends with double quotes: my_attribute="something".
74    Example: a title attribute.
75    /!\ in double quotes /!\ my_attribute='something' is not supported.
76
77    @param string $s_string The string you want to escape.
78
79    @return string The escaped string.
80    */
81    function escape_attribute(string $s_string) : string {
82      return str_replace(
83        ['"'],
84        ['&quot;'],
85        $s_string,
86      );
87    }//end escape_attribute()
88
89
90
91    /**
92    When you want that the string appears verbatim inside HTML <pre> tags.
93
94    @param string $s_string The string you want to escape.
95
96    @return string The escaped string.
97    */
98    function escape_pre(string $s_string) : string {
99      return str_replace(
100       [
101         '&',
102         '<',
103       ],
104       [
105         '&amp;',
106         '&lt;',
107       ],
108       $s_string,
109     );
110   }//end escape_pre()
111 }//end namespace string_escaping\HTML
112
113
114
115
116 /**
117 The namespace string_escaping is splitted into subnamespaces for each
118 language.
119 This subnamespace is for XHTML (still surprised?)...
120
121 @package DevOrSysAdminScripts
122 @subpackage string_escaping\XHTML
123 */
124 namespace string_escaping\XHTML {
125   /**
126   When you want that the string appears as is in XHTML text block.
```

```
127
128     @param string $s_string The string you want to escape.
129
130     @return string The escaped string.
131     */
132     function escape_text(string $s_string) : string {
133       return str_replace(
134         [
135           '&',
136           '<',
137         ],
138         [
139           '&amp;',
140           '&lt;',
141         ],
142         $s_string,
143       );
144     }//end escape_text()
145
146
147
148     /**
149     When you want that the string appears as is in an XHTML tag attribute
150     that starts and ends with double quotes: my_attribute="something".
151     Example: a title attribute.
152     /!\ in double quotes /!\ my_attribute='something' is not supported.
153
154     @param string $s_string The string you want to escape.
155
156     @return string The escaped string.
157     */
158     function escape_attribute(string $s_string) : string {
159       return str_replace(
160         [
161           '&',
162           '<',
163           '"',
164           "\n",
165         ],
166         [
167           '&amp;',
168           '&lt;',
169           '&quot;',
170           '&#10;',
171         ],
172         $s_string,
173       );
174     }//end escape_attribute()
175
176
177
178     /**
179     When you want that the string appears verbatim inside XHTML <pre> tags.
180
181     @param string $s_string The string you want to escape.
182
183     @return string The escaped string.
184     */
185     function escape_pre(string $s_string) : string {
186       return str_replace(
187         [
```

```
188           '&',
189           '<',
190         ],
191         [
192           '&amp;',
193           '&lt;',
194         ],
195         $s_string,
196       );
197     }//end escape_pre()
198   }//end namespace string_escaping\XHTML
199
200
201
202
203   /**
204   The namespace string_escaping is splitted into subnamespaces for each
205   language.
206   This subnamespace is for JS (you're hopeless?)...
207
208   @package DevOrSysAdminScripts
209   @subpackage string_escaping\JS
210   */
211   namespace string_escaping\JS {
212
213     use function string_escaping\XHTML\escape_text as XHTML_escape_text;
214
215
216
217     /**
218     When you want that the string can be put in a JS string definition.
219
220     @param string $s_string The string you want to escape.
221
222     @return string The escaped string.
223     */
224     function escape(string $s_string) : string {
225       return str_replace(
226         [
227           '\\',
228           "\n",
229           "'",
230           '"',
231         ],
232         [
233           '\\\\',
234           '\n',
235           '\\'',
236           '\\"',
237         ],
238         $s_string,
239       );
240     }//end escape()
241
242
243
244     /**
245     When you want that the string can be put in a JS string definition
246     to be included inside an HTML file <script> tag.
247
248     @param string $s_string The string you want to escape.
```

```
249
250   @return string The escaped string.
251   */
252   function escape_for_string_definition_inside_HTML(string $s_string)
253   : string {
254     return str_replace(
255       ['</script>'],
256       ['\u{00003c}/script>'],
257       escape($s_string),
258     );
259   }//end escape_for_string_definition_inside_HTML()
260
261
262
263   /**
264   When you want that the string can be put in a JS string definition
265   to be included inside an XHTML file <script> tag.
266
267   @param string $s_string The string you want to escape.
268
269   @return string The escaped string.
270   */
271   function escape_for_string_definition_inside_XHTML(string $s_string)
272   : string {
273     return XHTML_escape_text(escape($s_string));
274   }//end escape_for_string_definition_inside_XHTML()
275 }//end namespace string_escaping\JS
276
277
278
279
280 /**
281 The namespace string_escaping is splitted into subnamespaces for each
282 language.
283 This subnamespace is for TeX.
284
285 @package DevOrSysAdminScripts
286 @subpackage string_escaping\TeX
287 */
288 namespace string_escaping\TeX {
289   /**
290   When you want that the string can be written to a TeX file inside a
291   part where TeX is in text mode.
292
293   @param string $s_string The string you want to escape.
294
295   @return string The escaped string.
296   */
297   function escape_text(string $s_string) : string {
298     // https://tex.stackexchange.com/a/34586/270015
299     return str_replace(
300       [
301         '\\',
302         '&',
303         '%',
304         '$',
305         '#',
306         '_',
307         '{',
308         '}',
309         '~',
```

```
        '^',
      ],
      [
        '\\textbackslash',
        '\\&',
        '\\%',
        '\\$',
        '\\#',
        '\\_',
        '\\{',
        '\\}',
        '\\textasciitilde',
        '\\textasciicircum',
      ],
      $s_string,
    );
  }//end escape_text()
}//end namespace string_escaping\TeX

/*
$ php
<?php
require_once 'string_escaping.libr.php';
var_dump(string_escaping\HTML\escape_text("<&\"\n'truc\\>"));
var_dump(string_escaping\HTML\escape_attribute("<&\"\n'truc\\>"));
var_dump(string_escaping\HTML\escape_pre("<&\"\n'truc\\>"));
var_dump(string_escaping\XHTML\escape_text("<&\"\n'truc\\>"));
var_dump(string_escaping\XHTML\escape_attribute("<&\"\n'truc\\>"));
var_dump(string_escaping\XHTML\escape_pre("<&\"\n'truc\\>"));
var_dump(string_escaping\JS\escape("abcd & < > \\ \n  \"'"));
var_dump(string_escaping\JS\escape_for_string_definition_inside_HTML(
  "abcd & < > \\ \n  \"' </script>"
));
var_dump(string_escaping\JS\escape_for_string_definition_inside_XHTML(
  "abcd & < > \\ \n  \"'"
));
*/
?>
```

3.74 unicode.libr.php

```
<?php

/**
This file is part of DevOrSysAdminScripts.

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either version 3 of the License,
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```

```
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This file was renamed from "unicode.php" to "unicode.libr.php".

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/

declare(strict_types=1);
declare(encoding='UTF-8');

namespace DOSAS_unicode;

use Exception;

/*
Max value for Unicode is 10FFFF != 16*255*255 = 16*65025 = 1040400... XD
Max value for Unicode is 10FFFF = (16*256+255)*256+255
  = (2560+1536+255)*256+255 = (4096+255)*256+255 = 4351*256+255
  = 4351*1000/4 + 4351*6 + 255
  4351000 2175500 1087750
  = 1087750 + 26106 + 255 = 1114111
so there is 1114112 possible values in the "uncarved" range.
244,143,191,191 for UTF-8 yields max code point is ((4*64+15)*64+63)*64+63
  = ((256+15)*64+63)*64+63 = (271*64+63)*64+63 = (16260+1084+63)*64+63
  = 17407*64+63 = 1044420 + 69628 + 63 = 1114111
*/
const MAX_UNICODE_CODE_POINT = 1114111;



/**
Converts an int to the byte-string corresponding to the UTF-8 encoding
of the Unicode code point of same integer value.
Note that not all integers between 0 and 1114111 are valid code points
for UTF-8.

@param int<0, 1114111> $i_unicode_code_point The code point.

@throws \Exception When the code point is outside of Unicode range.

@return string The corresponding UTF-8 character.
*/
function unicode_code_point_to_UTF8(int $i_unicode_code_point) : string {
  // var_dump($i_unicode_code_point);
  if($i_unicode_code_point < 0){
    throw new Exception('A unicode code point must not be negative.');
  }
  // 0xxxxxxx ASCII
  if($i_unicode_code_point < 128){
    return chr($i_unicode_code_point);
  }
  $i_continuation_base_value = 128;
  // 110xxxxx 10xxxxxx
  if($i_unicode_code_point < 2048){  // 2**11
```

```
82      $i_first_byte_base_value = 192;
83      /*
84      $i_last_byte_significant_bits = $i_unicode_code_point % 64  // 2**6;
85      $i_first_byte_significant_bits = intdiv(
86        $i_unicode_code_point - $i_last_byte_significant_bits,
87        64,  // 2**6
88      );
89      $s_result = chr(
90        $i_first_byte_base_value
91        + $i_first_byte_significant_bits
92      );
93      $s_result .= chr(
94        $i_continuation_base_value
95        + $i_last_byte_significant_bits
96      );
97      return $s_result;
98      */
99      return (
100       chr($i_first_byte_base_value + ($i_unicode_code_point >> 6))
101       .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
102     );
103   }//end if($i_unicode_code_point < 2048)
104   // 1110xxxx 10xxxxxx 10xxxxxx
105   if($i_unicode_code_point < 65536){  // 2**16
106     $i_first_byte_base_value = 224;
107     /*
108     $arr_arr_data_per_byte_reverse = [
109       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
110       ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
111       ['i_bits' => 4, 'i_base_value' => $i_first_byte_base_value],
112     ];
113     $i_rest = $i_unicode_code_point;
114     $s_result = '';
115     foreach($arr_arr_data_per_byte_reverse as $arr_data){
116       // _Pragma unroll
117       // J'aurais pu "roller" les 4 cas des ifs.
118       // Mais pour les perfs, il faudrait tout unroller vu le nombre
119       // d'itérations. Et j'aimerais bien comparer du code unrollé
120       // aux 2 niveaux à la main avec celui unrollé par le compilateur
121       // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
122       // l'intérieur d'abord (pour ne pas refaire le même boulot)
123       // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
124       // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
125       $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
126       $i_rest = intdiv(
127         $i_rest - $i_significant_bits,
128         2**$arr_data['i_bits'],
129       );
130       // .= or =.
131       // Below could have been a case of "=.".
132       $s_result = (
133         chr($i_significant_bits + $arr_data['i_base_value'])
134         .$s_result
135       );
136     }//end foreach($arr_arr_data_per_byte_reverse as $arr_data)
137     return $s_result;
138     */
139     return (
140       chr($i_first_byte_base_value + ($i_unicode_code_point >> 12))
141       .chr(
142         $i_continuation_base_value
```

```
        + (($i_unicode_code_point >> 6) & 63)
      )
      .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
    );
  }//end if($i_unicode_code_point < 65536)
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  // if($i_unicode_code_point < 2097152){  // 2**21
  if($i_unicode_code_point < MAX_UNICODE_CODE_POINT){
    $i_first_byte_base_value = 240;
    /*
    $arr_arr_data_per_byte_reverse = [
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 6, 'i_base_value' => $i_continuation_base_value],
      ['i_bits' => 3, 'i_base_value' => $i_first_byte_base_value],
    ];
    $i_rest = $i_unicode_code_point;
    $s_result = '';
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data['i_bits']);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data['i_bits'],
      );
      // .= or =.
      // Below could have been a case of "=.".
      $s_result = (
        chr($i_significant_bits + $arr_data['i_base_value'])
        .$s_result
      );
    }
    return $s_result;
    */
    return (
      chr($i_first_byte_base_value + ($i_unicode_code_point >> 18))
      .chr(
        $i_continuation_base_value
        + (($i_unicode_code_point >> 12) & 63)
      )
      .chr(
        $i_continuation_base_value
        + (($i_unicode_code_point >> 6) & 63)
      )
      .chr($i_continuation_base_value + ($i_unicode_code_point & 63))
    );
  }//end if($i_unicode_code_point < MAX_UNICODE_CODE_POINT)
  // phpcs:disable Squiz.Strings.DoubleQuoteUsage.NotRequired
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
    ." Du coup, ça peut donner quelques \"surprises\"."
    ." Sinon, y'a aussi ASCII dans MySQL."
    ." Et comme ça joue sur la mémoire et les performances,"
    ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
    ." :)"
  );
  // phpcs:enable Squiz.Strings.DoubleQuoteUsage.NotRequired
}//end unicode_code_point_to_UTF8()

```

```
204
205
206 /**
207 Converts a string of hexadecimal digits to the byte-string corresponding
208 to the UTF-8 encoding of the Unicode code point of same integer value
209 that is denoted by the hexadecimal digits.
210
211 @param string $s_code_point_in_hexadecimal_notation The code point.
212
213 @throws \Exception When the code point is outside of Unicode range.
214
215 @return string The corresponding UTF-8 character.
216 */
217 function hexa_code_point_to_UTF8(
218   string $s_code_point_in_hexadecimal_notation
219 ) : string {
220   $s_code_point_in_hexadecimal_notation = strtoupper(
221     $s_code_point_in_hexadecimal_notation
222   );
223   $i_unicode_code_point = 0;
224   $i_ord_0 = ord('0');
225   $i_ord_A = ord('A');
226   $i_digit_offset = -$i_ord_0;
227   $i_letter_offset = 10 - $i_ord_A;
228   for(
229     $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
230     $i < $i_max;
231     ++$i
232   ){
233     $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
234     $i_unicode_code_point *= 16;
235     if(ctype_digit($s_hexa_digit)){
236       $i_unicode_code_point += (
237         ord($s_hexa_digit) + $i_digit_offset
238       );
239       continue;
240     }
241     if(ctype_xdigit($s_hexa_digit)){
242       $i_unicode_code_point += (
243         ord($s_hexa_digit) + $i_letter_offset
244       );
245       continue;
246     }
247     throw new Exception('Not an hexadecimal digit.');
248   }
249   if($i_unicode_code_point > MAX_UNICODE_CODE_POINT){
250     throw new Exception(
251       'Hexadecimal integer is greater than UTF-8 maximum code point.'
252     );
253   }
254   return unicode_code_point_to_UTF8($i_unicode_code_point);
255 }//end hexa_code_point_to_UTF8()
256
257
258
259 /**
260 Converts a string of decimal digits to the byte-string corresponding
261 to the UTF-8 encoding of the Unicode code point of same integer value
262 that is denoted by the decimal digits.
263
264 @param string $s_code_point_in_decimal_notation The code point.
```

```
265
266 @throws \Exception When the code point is outside of Unicode range.
267
268 @return string The corresponding UTF-8 character.
269 */
270 function decimal_code_point_to_UTF8(
271   string $s_code_point_in_decimal_notation
272 ) : string {
273   $i_unicode_code_point = 0;
274   $i_ord_0 = ord('0');
275   $i_digit_offset = -$i_ord_0;
276   for(
277     $i = 0, $i_max = strlen($s_code_point_in_decimal_notation);
278     $i < $i_max;
279     ++$i
280   ){
281     $s_digit = $s_code_point_in_decimal_notation[$i];
282     $i_unicode_code_point *= 10;
283     if(ctype_digit($s_digit)){
284       $i_unicode_code_point += (ord($s_digit) + $i_digit_offset);
285       continue;
286     }
287     throw new Exception('Not a decimal digit.');
288   }
289   if($i_unicode_code_point > MAX_UNICODE_CODE_POINT){
290     throw new Exception(
291       'Decimal integer is greater than UTF-8 maximum code point.'
292     );
293   }
294   return unicode_code_point_to_UTF8($i_unicode_code_point);
295 }//end decimal_code_point_to_UTF8()
296
297
298
299 /**
300 This function throws an exception with extended debug informations
301 if the input string is not valid ASCII.
302 Otherwise, it returns true.
303
304 @param string $s_string The input string.
305
306 @throws \DOSAS_unicode\InvalidEncodingException
307         When the input string is not valid ASCII.
308
309 @return true
310 */
311 function check_string_is_valid_ASCII(string $s_string) : bool {
312   //Fast-path
313   if(mb_check_encoding($s_string, 'ASCII')){
314     return true;
315   }
316
317   $i_offset_in_octets_from_string_start = 0;
318   $i_current_line_number = 1;
319   $i_offset_in_octets_from_line_start = -1;
320   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
321     $i_current_octet = ord($s_string[$i]);
322     $i_offset_in_octets_from_string_start = $i;
323     ++$i_offset_in_octets_from_line_start;
324
325     if($i_current_octet > 127){
```

```
      throw new InvalidEncodingException(
        'Non-ASCII character found on line '
        .$i_current_line_number
        .'; the octet/character '
        .($i_offset_in_octets_from_line_start + 1)
        .' has value '
        .$i_current_octet
        .' which is not ASCII.'
        .' (Sequential position without line splitting:'
        .' This is at octet/character '
        .($i_offset_in_octets_from_string_start + 1)
        .'.)',
        [
          'current_octet' => $i_current_octet,
          'offset_from_string_start'
            => $i_offset_in_octets_from_string_start,
          'line' => $i_current_line_number,
          'offset_from_line_start'
            => $i_offset_in_octets_from_line_start,
        ],
      );
    }//end if($i_current_octet > 127)

    // if($s_string[$i] === "\n"){
    if($i_current_octet === 10){
      ++$i_current_line_number;
      $i_offset_in_octets_from_line_start = -1;
    }
  }//for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
  return true;
}//end check_string_is_valid_ASCII()



/**
This function throws an exception with extended debug informations
if the content of the file with given file_path is not valid ASCII.
Otherwise, it returns true.

@param string $s_file_path The input file_path.

@throws \Exception When the file is not found.
@throws \DOSAS_unicode\InvalidEncodingException
        When the input string is not valid ASCII.

@return true
*/
function check_file_is_valid_ASCII(string $s_file_path) : bool {
  $s_string = file_get_contents($s_file_path);
  if($s_string === false){
    throw new Exception('File '.$s_file_path.' not found.');
  }
  return check_string_is_valid_ASCII($s_string);
}//end check_file_is_valid_ASCII()



/**
This function returns the message string and data array for
DOSAS_unicode\InvalidEncodingException inside check_string_is_valid_UTF8().

```

```
387 @param string $s_custom_message The custom part of the message.
388 @param int $i_current_octet The current octet in the string.
389 @param int $i_continuation_octet_needed The current number of continuation
390                                         octets needed.
391 @param int $i_offset_in_octets_from_string_start
392             The offset in octets from string start.
393 @param int $i_offset_in_characters_from_string_start
394             The offset in characters from string start.
395 @param int $i_character_start_position_from_string_start
396             The offset in octets from string start at which the current
397             character starts.
398 @param int $i_current_line_number The current line number.
399 @param int $i_offset_in_octets_from_line_start
400             The offset in octets from line start.
401 @param int $i_offset_in_characters_from_line_start
402             The offset in characters from line start.
403 @param int $i_character_start_position_from_line_start
404             The offset in octets from line start at which the current
405             character starts.
406 @param int $i_current_continuation_octet_minimum
407             The minimum value of the second continuation octet as specified
408             by the grammar.
409 @param int $i_current_continuation_octet_maximum
410             The maximum value of the second continuation octet as specified
411             by the grammar.
412
413 @return array{'message': string, 'data': array<mixed>}
414 */
415 function get_message_and_data_array(
416   string $s_custom_message,
417   int $i_current_octet,
418   int $i_continuation_octet_needed,
419   int $i_offset_in_octets_from_string_start,
420   int $i_offset_in_characters_from_string_start,
421   int $i_character_start_position_from_string_start,
422   int $i_current_line_number,
423   int $i_offset_in_octets_from_line_start,
424   int $i_offset_in_characters_from_line_start,
425   int $i_character_start_position_from_line_start,
426   int $i_current_continuation_octet_minimum,
427   int $i_current_continuation_octet_maximum,
428 ) : array {
429   return [
430     'message' => (
431       'Non-UTF8 character found on line '
432       .$i_current_line_number
433       .'; the octet '
434       .($i_offset_in_octets_from_line_start + 1)
435       .', part of the character '
436       .($i_offset_in_characters_from_line_start + 1)
437       .', has value '
438       .$i_current_octet
439       .$s_custom_message
440       .' This character starts at octet '
441       .($i_character_start_position_from_line_start + 1)
442       .' of the current line.'
443       .' (Sequential positions without line splitting:'
444       .' This is at character '
445       .($i_offset_in_characters_from_string_start + 1)
446       .' and octet '
447       .($i_offset_in_octets_from_string_start + 1)
```

```
      .'.'
      .' This character starts at octet '
      .($i_character_start_position_from_string_start + 1)
      .'.)'
    ),
    'data' => [
      'current_octet' => $i_current_octet,
      'continuation_octet_needed' => $i_continuation_octet_needed,
      'offset_in_octets_from_string_start'
        => $i_offset_in_octets_from_string_start,
      'offset_in_characters_from_string_start'
        => $i_offset_in_characters_from_string_start,
      'character_start_position_from_string_start'
        => $i_character_start_position_from_string_start,
      'line' => $i_current_line_number,
      'offset_in_octets_from_line_start'
        => $i_offset_in_octets_from_line_start,
      'offset_in_characters_from_line_start'
        => $i_offset_in_characters_from_line_start,
      'character_start_position_from_line_start'
        => $i_character_start_position_from_line_start,
      'current_continuation_octet_minimum'
        => $i_current_continuation_octet_minimum,
      'current_continuation_octet_maximum'
        => $i_current_continuation_octet_maximum,
    ],
  ];
}//end get_message_and_data_array()



/**
Checks that the version of PHP allows using the fast-path inside
check_string_is_valid_UTF8.

@return bool
*/
function get_use_fast_path() : bool {
  // Before PHP 5.4, Unicode support has bugs,
  // both in mb_check_encoding and preg_match.
  // See https://github.com/php/php-src/issues/22279
  return PHP_VERSION_ID >= 50400;
}//end get_use_fast_path()



/**
This function throws an exception with extended debug informations
if the input string is not valid UTF-8.
Otherwise, it returns true.

See https://datatracker.ietf.org/doc/html/rfc3629
See https://github.com/Seldaek/jsonlint/pull/98
for why it may be a good idea to have the last two parameters.
Note that a very clever compiler would merge both booleans into
a single integer parameter using binary flags.
If you want to use this function with fast-path and fallback on slow-path
in case of error, then set both booleans to true.
If, for testing/benchmarking purpose, you want to compare the results
of fast-path and slow-path, call this function twice with a single boolean
set, and store/compare the results.
```

```
509
510 @param string $s_string The input string.
511 @param bool $b_fast_path Use the fast-path or not to check quickly that
512                          there is no error.
513 @param bool $b_slow_path Use the slow-path or not to check that
514                          there is no error or give detailed error message.
515
516 @throws \Exception When called with $b_fast_path set to true but PHP
517                    version is incompatible.
518 @throws \DOSAS_unicode\InvalidEncodingException
519         When the input string is not valid UTF-8.
520
521 @return bool
522 */
523 function check_string_is_valid_UTF8(
524   string $s_string,
525   bool $b_fast_path = true,
526   bool $b_slow_path = true,
527 ) : bool {
528   // Fast-path
529   // But before PHP 5.4 Unicode support has bugs.
530   // See https://github.com/php/php-src/issues/22279
531   if($b_fast_path && !get_use_fast_path()){
532     throw new Exception(
533       'Fast-path is not available with version '.PHP_VERSION_ID.' of PHP.'
534     );
535   }
536   if($b_fast_path){
537     if(function_exists('mb_check_encoding')){
538       if(mb_check_encoding($s_string, 'UTF-8')){
539         return true;
540       }
541     }
542     else{
543       if(preg_match('//u', $s_string) === 1){
544         return true;
545       }
546     }
547   }
548   else{
549     if(!$b_slow_path){
550       throw new Exception(
551         "Don't call check_string_is_valid_UTF8 without at least one of"
552         .' fast-path or slow-path required.'
553       );
554     }
555   }
556
557   if(!$b_slow_path){
558     return false;
559   }
560
561   $i_current_octet = null;
562   $i_continuation_octet_needed = 0;
563   $i_offset_in_octets_from_string_start = 0;
564   $i_offset_in_characters_from_string_start = -1;
565   $i_character_start_position_from_string_start = 0;
566   $i_current_line_number = 1;
567   $i_offset_in_octets_from_line_start = -1;
568   $i_offset_in_characters_from_line_start = -1;
569   $i_character_start_position_from_line_start = -1;
```

```
570
571     // For the second octet of a character, hence first continuation octet,
572     // further restriction may apply.
573     $I_CONTINUATION_OCTET_MINIMUM = 128;
574     $I_CONTINUATION_OCTET_MAXIMUM = 191;
575     $i_current_continuation_octet_minimum = $I_CONTINUATION_OCTET_MINIMUM;
576     $i_current_continuation_octet_maximum = $I_CONTINUATION_OCTET_MAXIMUM;
577     $s_message_for_continuation_octet_above_maximum = null;
578
579     for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
580       $i_current_octet = ord($s_string[$i]);
581       $i_offset_in_octets_from_string_start = $i;
582       ++$i_offset_in_octets_from_line_start;
583
584       /*
585       The octet values C0, C1, F5 to FF never appear.
586       C0 = 12*16      = 192 = 11000000
587       C1 = 12*16 + 1  = 193 = 11000001
588       F5 = 15*16 + 5  = 245 = 11110101
589       FF = 15*16 + 15 = 255 = 11111111
590       */
591       if(
592         $i_current_octet === 192
593         || $i_current_octet === 193
594         || $i_current_octet === 245
595         || $i_current_octet === 255
596       ){
597         [
598           'message' => $s_message,
599           'data' => $arr_data,
600         ] = get_message_and_data_array(
601           ' which is one of the four forbidden values (C0, C1, F5, FF).',
602           $i_current_octet,
603           $i_continuation_octet_needed,
604           $i_offset_in_octets_from_string_start,
605           $i_offset_in_characters_from_string_start,
606           $i_character_start_position_from_string_start,
607           $i_current_line_number,
608           $i_offset_in_octets_from_line_start,
609           $i_offset_in_characters_from_line_start,
610           $i_character_start_position_from_line_start,
611           $i_current_continuation_octet_minimum,
612           $i_current_continuation_octet_maximum,
613         );
614         throw new InvalidEncodingException($s_message, $arr_data);
615       }
616
617       /*
618       UTF8-octets = *( UTF8-char )
619       UTF8-char   = UTF8-1 / UTF8-2 / UTF8-3 / UTF8-4
620       UTF8-1      = %x00-7F
621       UTF8-2      = %xC2-DF UTF8-tail
622         Notice that values C0 and C1 are forbidden, hence %xC2
623       UTF8-3      = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2( UTF8-tail ) /
624                     %xED %x80-9F UTF8-tail / %xEE-EF 2( UTF8-tail )
625         Notice that E0 = 224 adds a restriction on second octet.
626         Notice that ED = 237 adds a restriction on second octet.
627       UTF8-4      = %xF0 %x90-BF 2( UTF8-tail ) / %xF1-F3 3( UTF8-tail ) /
628                     %xF4 %x80-8F 2( UTF8-tail )
629         Notice that F0 = 240 adds a restriction on second octet.
630         Notice that F4 = 244 adds a restriction on second octet.
```

```
631       UTF8-tail   = %x80-BF
632       */
633       if($i_continuation_octet_needed > 0){
634         if(
635           $i_current_octet < $i_current_continuation_octet_minimum
636           || $i_current_octet > $i_current_continuation_octet_maximum
637         ){
638           [
639             'message' => $s_message,
640             'data' => $arr_data,
641           ] = get_message_and_data_array(
642             $s_message_for_continuation_octet_above_maximum !== null
643             && $i_current_octet > $i_current_continuation_octet_maximum
644             ? $s_message_for_continuation_octet_above_maximum
645             : ' which is not a continuation octet.',
646             $i_current_octet,
647             $i_continuation_octet_needed,
648             $i_offset_in_octets_from_string_start,
649             $i_offset_in_characters_from_string_start,
650             $i_character_start_position_from_string_start,
651             $i_current_line_number,
652             $i_offset_in_octets_from_line_start,
653             $i_offset_in_characters_from_line_start,
654             $i_character_start_position_from_line_start,
655             $i_current_continuation_octet_minimum,
656             $i_current_continuation_octet_maximum,
657           );
658           throw new InvalidEncodingException($s_message, $arr_data);
659         }//end if($i_current_octet < 128 || $i_current_octet >= 192)
660         --$i_continuation_octet_needed;
661         $i_current_continuation_octet_minimum = (
662           $I_CONTINUATION_OCTET_MINIMUM
663         );
664         $i_current_continuation_octet_maximum = (
665           $I_CONTINUATION_OCTET_MAXIMUM
666         );
667         continue;
668       }//end if($i_continuation_octet_needed > 0)
669
670       ++$i_offset_in_characters_from_string_start;
671       ++$i_offset_in_characters_from_line_start;
672       $i_character_start_position_from_string_start = (
673         $i_offset_in_octets_from_string_start
674       );
675       $i_character_start_position_from_line_start = (
676         $i_offset_in_octets_from_line_start
677       );
678
679       if($i_current_octet < 128){ // 0xxxxxxx ASCII
680         // if($s_string[$i] === "\n"){
681         if($i_current_octet === 10){
682           ++$i_current_line_number;
683           $i_offset_in_octets_from_line_start = -1;
684           $i_offset_in_characters_from_line_start = -1;
685         }
686         continue;
687       }
688
689       if(/*$i_current_octet >= 128 &&*/ $i_current_octet < 192){
690         [
691           'message' => $s_message,
```

```
692           'data' => $arr_data,
693         ] = get_message_and_data_array(
694           ' which is a continuation octet.',
695           $i_current_octet,
696           $i_continuation_octet_needed,
697           $i_offset_in_octets_from_string_start,
698           $i_offset_in_characters_from_string_start,
699           $i_character_start_position_from_string_start,
700           $i_current_line_number,
701           $i_offset_in_octets_from_line_start,
702           $i_offset_in_characters_from_line_start,
703           $i_character_start_position_from_line_start,
704           $i_current_continuation_octet_minimum,
705           $i_current_continuation_octet_maximum,
706         );
707         throw new InvalidEncodingException($s_message, $arr_data);
708       }//end if($i_current_octet >= 128 && $i_current_octet < 192)
709
710       if(/*$i_current_octet >= 192 &&*/ $i_current_octet < 224){
711         // 110xxxxx 10xxxxxx
712         $i_continuation_octet_needed = 1;
713         continue;
714       }
715
716       if(/*$i_current_octet >= 224 &&*/ $i_current_octet < 240){
717         // 1110xxxx 10xxxxxx 10xxxxxx
718         $i_continuation_octet_needed = 2;
719         /*
720         UTF8-3 = %xE0 %xA0-BF UTF8-tail / %xE1-EC 2( UTF8-tail ) /
721                  %xED %x80-9F UTF8-tail / %xEE-EF 2( UTF8-tail )
722         Notice that E0 = 224 adds a restriction on second octet.
723         Notice that ED = 237 adds a restriction on second octet.
724
725         The definition of UTF-8 prohibits encoding character numbers between
726         U+D800 and U+DFFF.
727         D8 = 13*16 + 8  = 216 = 11010100
728         DF = 13*16 + 15 = 223 = 11010101
729         216*256 = 55296       = 1101 1000 0000 0000 = D800
730         to
731         223*256 + 255 = 57343 = 1101 1111 1111 1111 = DFFF
732
733         1110xxxx
734         11101101 = 237 = ED
735
736         237,160,128
737         237,191,191
738         Thus, the whole "continuation range" is forbidden if start
739         is 237 and second octet, first continuation octet, is >= 160.
740         */
741         if($i_current_octet === 224){
742           $i_current_continuation_octet_minimum = 160;
743           $i_current_continuation_octet_maximum = 191;  // Normal value
744         }
745         if($i_current_octet === 237){
746           $i_current_continuation_octet_minimum = 128;  // Normal value
747           $i_current_continuation_octet_maximum = 159;
748           $s_message_for_continuation_octet_above_maximum = (
749             ' which is into the forbidden range of surrogate pairs.'
750           );
751         }
752         continue;
```

```
753       }//end if(/*$i_current_octet >= 224 &&*/ $i_current_octet < 240)
754
755       if(/*$i_current_octet >= 240 &&*/ $i_current_octet < /*248*/ 245){
756         // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
757         $i_continuation_octet_needed = 3;
758         /*
759         UTF8-4 = %xF0 %x90-BF 2( UTF8-tail ) / %xF1-F3 3( UTF8-tail ) /
760                  %xF4 %x80-8F 2( UTF8-tail )
761         Notice that F0 = 240 adds a restriction on second octet.
762         Notice that F4 = 244 adds a restriction on second octet.
763         */
764         if($i_current_octet === 240){
765           $i_current_continuation_octet_minimum = 144;
766           $i_current_continuation_octet_maximum = 191;  // Normal value
767         }
768         if($i_current_octet === 244){
769           $i_current_continuation_octet_minimum = 128;  // Normal value
770           $i_current_continuation_octet_maximum = 143;
771         }
772         continue;
773       }
774
775       [
776         'message' => $s_message,
777         'data' => $arr_data,
778       ] = get_message_and_data_array(
779         ' which is invalid.',
780         $i_current_octet,
781         $i_continuation_octet_needed,
782         $i_offset_in_octets_from_string_start,
783         $i_offset_in_characters_from_string_start,
784         $i_character_start_position_from_string_start,
785         $i_current_line_number,
786         $i_offset_in_octets_from_line_start,
787         $i_offset_in_characters_from_line_start,
788         $i_character_start_position_from_line_start,
789         $i_current_continuation_octet_minimum,
790         $i_current_continuation_octet_maximum,
791       );
792       throw new InvalidEncodingException($s_message, $arr_data);
793     }//end for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i)
794
795     if($i_continuation_octet_needed > 0){
796       [
797         'message' => $s_message,
798         'data' => $arr_data,
799       ] = get_message_and_data_array(
800         '; end of string was found instead of a continuation octet.',
801         $i_current_octet,
802         $i_continuation_octet_needed,
803         $i_offset_in_octets_from_string_start,
804         $i_offset_in_characters_from_string_start,
805         $i_character_start_position_from_string_start,
806         $i_current_line_number,
807         $i_offset_in_octets_from_line_start,
808         $i_offset_in_characters_from_line_start,
809         $i_character_start_position_from_line_start,
810         $i_current_continuation_octet_minimum,
811         $i_current_continuation_octet_maximum,
812       );
813       throw new InvalidEncodingException($s_message, $arr_data);
```

```
814     }
815
816     if($b_fast_path){
817       throw new InvalidEncodingException(
818         "Fast-path detected an error that this function couldn't found."
819         .'Please create an issue at '
820         .'"https://github.com/LLyaudet/DevOrSysAdminScripts/issues".',
821         [],
822       );
823     }
824
825     return true;
826   }//end check_string_is_valid_UTF8()
827
828
829
830   /**
831   This function throws an exception with extended debug informations
832   if the content of the file with given file_path is not valid UTF-8.
833   Otherwise, it returns true.
834
835   @param string $s_file_path The input file_path.
836   @param bool $b_fast_path Use the fast-path or not to check quickly that
837                            there is no error.
838
839   @throws \Exception When the file is not found OR when called with
840                     $b_fast_path set to true but PHP version is
841                     incompatible.
842   @throws \DOSAS_unicode\InvalidEncodingException
843           When the input string is not valid UTF-8.
844
845   @return bool
846   */
847   function check_file_is_valid_UTF8(
848     string $s_file_path,
849     bool $b_fast_path = true,
850   ) : bool {
851     $s_string = file_get_contents($s_file_path);
852     if($s_string === false){
853       throw new Exception('File '.$s_file_path.' not found.');
854     }
855     return check_string_is_valid_UTF8($s_string, $b_fast_path);
856   }//end check_file_is_valid_UTF8()
857
858
859
860   /*
861   ?>
862   <?php
863   require_once "./unicode_exceptions.libr.php";
864   require_once "./unicode.libr.php";
865   var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("002B"));
866   var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("00E6"));
867   var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("1400"));
868   var_dump(DOSAS_unicode\hexa_code_point_to_UTF8("10111"));
869   // https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
870   // Some invisible unicode characters that are a problem:
871   // 2000-200F 2028-202F 205F 2060-206F
872   $arr = [
873     '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
874     '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
```

```
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
echo("sed\\\n");
foreach($arr as $s_hexa){
  $s = DOSAS_unicode\hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
  $arr_s2 []= $s2;
  $s3 = "  --expression='s/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}

?>
<?php
require_once "./unicode_exceptions.libr.php";
require_once "./unicode.libr.php";
// var_dump(DOSAS_unicode\check_file_is_valid_UTF8("validutf8.json"));
// var_dump(DOSAS_unicode\check_file_is_valid_UTF8("nonvalidutf8.json"));
var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé"));
var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé", true, false));
var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcdé", false, true));
var_dump(
  DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233), true, false)
);
if(true){
  var_dump(DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233)));
}
else{
  var_dump(
    DOSAS_unicode\check_string_is_valid_UTF8("abcd".chr(233), false, true)
  );
}
//*/
?>
```

3.75 unicode_exceptions.libr.php

```
<?php

/**
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```

```
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25 ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
26 This file was renamed from "unicode.php" to "unicode.libr.php".
27
28 @category Library
29 @package DevOrSysAdminScripts
30 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
31 @copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
32 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
33 */
34
35 declare(strict_types=1);
36 declare(encoding='UTF-8');
37
38 namespace DOSAS_unicode;
39
40 /**
41 A class to enhance standard PHP Exception with an array of data.
42
43 @category Library
44 @package DevOrSysAdminScripts
45 @author Laurent Lyaudet <laurent.lyaudet@gmail.com>
46 @license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
47 */
48 class WithDataArrayException extends \Exception {
49   /**
50   A data array to store details related to an exception.
51
52   @var array<mixed> $arr_data
53   */
54   protected $arr_data;
55
56
57
58   /**
59   Constructor for WithDataArrayException.
60
61   @param string $message
62   @param array<mixed> $arr_data
63
64   @return void
65   */
66   public function __construct(string $message, array $arr_data = []) {
67     $this->arr_data = $arr_data;
68     parent::__construct($message);
69   }//end __construct() WithDataArrayException
70
71
72
```

```
  /**
  Getter to obtain the data array with details of the exception.

  @return array<mixed>
  */
  public function get_data() : array {
    return $this->arr_data;
  }//end get_data()
}//end class WithDataArrayException




/**
A class to specialize WithDataArrayException for use with invalid
encoding detection (ASCII and UTF-8 validation) in unicode.libr.php.

@category Library
@package DevOrSysAdminScripts
@author Laurent Lyaudet <laurent.lyaudet@gmail.com>
@license https://www.gnu.org/licenses/lgpl-3.0.html LGPLv3+
*/
class InvalidEncodingException extends WithDataArrayException {
}//end class InvalidEncodingException

?>
```

3.76 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts.
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#
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  # Options:
  #   --add-test-files
  #   --check-download=y|keep (=keep to keep the downloaded files)
  #   --fixed-point-build
  #   --verbose
```

```
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
  readonly LFBFL_fixed_point_build
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="--check-download=keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="--check-download=y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_add_test_files=0
  if [[ "$*" == *--add-test-files* ]]; then
    LFBFL_add_test_files=1
  fi
  readonly LFBFL_add_test_files
  declare -i LFBFL_remove_test_files=0
  if [[ "$*" == *--remove-test-files* ]]; then
    LFBFL_remove_test_files=1
  fi
  readonly LFBFL_remove_test_files
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )

  declare -i LFBFL_i
  local LFBFL_file_path="miscellaneous/12345678901234567890123456789012345678901234567890"
  declare -i LFBFL_digit
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    mkdir --parents -- --a/
    {
      printf "<?php\nfunction my_function(:;){\n\n}\n"
      # shellcheck disable=SC2016
      printf '$my_variable = "aaaa
```

```
  printf "    \"\"\"Some comment.\"\"\"\n"
  printf "\n"
  printf "    print(my_argument)\n"
  printf "\n"
  printf "def my_function4(my_argument : int):\n"
  printf "    \"\"\"Some comment.\"\"\"\n"
  printf "    def my_sub_function():\n"
  printf "        print(my_argument)\n"
  printf "    return my_sub_function\n"
  printf "\n"
  printf "def my_function5(my_argument : int):\n"
  printf "    \"\"\"Some comment.\"\"\"\n"
  printf "\n"
  printf "    def my_sub_function():\n"
  printf "        print(my_argument)\n"
  printf "    return my_sub_function\n"
  printf "\n"
} > --a.py
cat -- --a.py
cp -- --a.py --a/
{
  printf "def my_function(:;):\n\n\n"
  printf 'my_variable = "aaaa
```

```
    cp -- --a.sh --a/

    # We add a file with space in it for tests.
    touch "a b.sh"

    # The following files cannot be commited,
    # because the backslash in them breaks CodeFactor.io.
    for ((LFBFL_i = 1; LFBFL_i < 90; ++LFBFL_i)); do
      LFBFL_digit=$((LFBFL_i % 10))
      LFBFL_file_path+="${LFBFL_digit}"
      touch "${LFBFL_file_path}"
      touch "${LFBFL_file_path}\\"
    done
  fi
  ./build_and_checks.exec.sh "."\
    "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"\
    "${LFBFL_check_download}"
  if [[ LFBFL_add_test_files -eq 1 ]]; then
    cat -- --a.py
    cat -- --a2.py
  fi
  if [[ LFBFL_remove_test_files -eq 1 ]]; then
    rm --force --recursive -- --a/
    rm --force -- --a.php --a.py --a2.py --a.sh
    rm "a b.sh"
    for ((LFBFL_i = 1; LFBFL_i < 90; ++LFBFL_i)); do
      LFBFL_digit=$((LFBFL_i % 10))
      LFBFL_file_path+="${LFBFL_digit}"
      rm --force "${LFBFL_file_path}"
      rm --force "${LFBFL_file_path}\\"
    done
  fi
}

upgrade_build_and_checks "$@"
```

3.77 wget_sha512.libr.sh

```
#!/usr/bin/env bash
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#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1=file_path
  # $2=download_URL
  # $3=correct_sha512
  # Options:
  #   --check-download=y|keep (=keep to keep the downloaded file)
  #   --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  local LFBFL_check_download=""
  if [[ "$*" == *--check-download=keep* ]]; then
    LFBFL_check_download="keep"
  elif [[ "$*" == *--check-download=y* ]]; then
    LFBFL_check_download="y"
  fi
  readonly LFBFL_check_download
  declare -i LFBFL_i_error=0

  if [[ ! -f "$1" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"} --output-document="$1"\
      -- "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum -- "$1"\
    | cut --fields=1 --delimiter=' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    LFBFL_i_error=1
  fi
  if [[ -n "${LFBFL_check_download}" ]]; then
    wget ${LFBFL_verbose:+"${LFBFL_verbose}"}\
      --output-document="$1.wget_sha512.temp"\
      -- "$2"
    declare -r LFBFL_present_sha512_2=$(
      sha512sum -- "$1.wget_sha512.temp"\
      | cut --fields=1 --delimiter=' '
    )
    if [[ "${LFBFL_check_download}" != "keep" ]]; then
      rm -- "$1.wget_sha512.temp"
    fi
    if [[ "${LFBFL_present_sha512_2}" != "$3" ]]; then
      printf\
        "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
        "Online version of $1" "$3" "${LFBFL_present_sha512_2}"
      LFBFL_i_error=1
    fi
  fi
  # shellcheck disable=SC2248
  return ${LFBFL_i_error}
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !